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United States Patent	12384837
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Cady; Roger K. et al.

Treatment of most bothersome symptom (MBS) associated with migraine using anti-CGRP antibodies

Abstract

Methods for treatment of most bothersome symptom (MBS) associated with migraine are provided. Exemplary methods provide improvement in MBS associated with migraine within 1 month of administration of anti-CGRP antibodies of the invention. Also provided are methods for improvement of patient impression of change (PGIC) associated with migraine. Exemplary methods comprise administration of an anti-CGRP antagonistic antibody to a patient in need thereof.

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Appl. No.: 18/182822

Filed: March 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240101653 A1	Mar. 28, 2024

Related U.S. Application Data

continuation parent-doc US 16860239 20200428 ABANDONED child-doc US 18182822
us-provisional-application US 63005950 20200406

Publication Classification

Int. Cl.: C07K16/18 (20060101); A61K9/00 (20060101); A61K39/00 (20060101); A61P25/06 (20060101)

U.S. Cl.:

CPC C07K16/18 (20130101); A61K9/0019 (20130101); A61P25/06 (20180101); A61K2039/545 (20130101)

Field of Classification Search

CPC: C07K (16/18); C07K (2317/565); C07K (2317/76); A61K (9/0019); A61K (2039/545); A61K (39/39591); A61K (2039/505); A61P (25/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5116964	12/1991	Capon et al.	N/A	N/A
5266561	12/1992	Cooper et al.	N/A	N/A
5364841	12/1993	Cooper et al.	N/A	N/A
5585089	12/1995	Queen et al.	N/A	N/A
5624821	12/1996	Winter et al.	N/A	N/A
5648260	12/1996	Winter et al.	N/A	N/A
5942227	12/1998	Cooper et al.	N/A	N/A
6180370	12/2000	Queen et al.	N/A	N/A
6313097	12/2000	Eberlein et al.	N/A	N/A
6509014	12/2002	De Lacharriere et al.	N/A	N/A
6521609	12/2002	Doods et al.	N/A	N/A
6737056	12/2003	Presta	N/A	N/A
6956107	12/2004	Fung et al.	N/A	N/A
7279471	12/2006	Mueller et al.	N/A	N/A
7479488	12/2008	Mueller et al.	N/A	N/A
7696209	12/2009	Mueller et al.	N/A	N/A
7700735	12/2009	Young et al.	N/A	N/A
7879991	12/2010	Vater et al.	N/A	N/A
7927863	12/2010	Cregg et al.	N/A	N/A
7935340	12/2010	Garcia-Martinez et al.	N/A	N/A
8007794	12/2010	Zeller et al.	N/A	N/A
8293239	12/2011	Poulsen et al.	N/A	N/A
8298536	12/2011	Poulsen et al.	N/A	N/A
8586045	12/2012	Zeller et al.	N/A	N/A
8597649	12/2012	Zeller et al.	N/A	N/A
8623366	12/2013	Pios et al.	N/A	N/A
8734802	12/2013	Zeller et al.	N/A	N/A
9073991	12/2014	Allan et al.	N/A	N/A

9708393	12/2016	Russo et al.	N/A	N/A
9745373	12/2016	Kovacevich et al.	N/A	N/A
9855332	12/2017	Russo et al.	N/A	N/A
10066009	12/2017	Kovacevich et al.	N/A	N/A
10179809	12/2018	Kovacevich et al.	N/A	N/A
10189895	12/2018	Kovacevich et al.	N/A	N/A
10208112	12/2018	Kovacevich et al.	N/A	N/A
10214582	12/2018	Kovacevich et al.	N/A	N/A
10266587	12/2018	Russo et al.	N/A	N/A
10533048	12/2019	Kovacevich et al.	N/A	N/A
11639380	12/2022	Cady	424/130.1	A61K 9/0019
11639381	12/2022	Cady	424/133.1	A61K 47/26
2001/0036647	12/2000	Choudary et al.	N/A	N/A
2002/0162125	12/2001	Salmon et al.	N/A	N/A
2002/0164707	12/2001	Adamou et al.	N/A	N/A
2003/0027213	12/2002	Zhu et al.	N/A	N/A
2003/0181462	12/2002	Doods et al.	N/A	N/A
2003/0194404	12/2002	Greenfeder et al.	N/A	N/A
2004/0110170	12/2003	Pisegna et al.	N/A	N/A
2004/0132824	12/2003	Gil et al.	N/A	N/A
2005/0234054	12/2004	Mueller et al.	N/A	N/A
2006/0183700	12/2005	Vater et al.	N/A	N/A
2006/0270045	12/2005	Cregg et al.	N/A	N/A
2009/0023644	12/2008	Southard et al.	N/A	N/A
2009/0028784	12/2008	Garcia-Martinez et al.	N/A	N/A
2009/0220489	12/2008	Zeller et al.	N/A	N/A
2010/0152171	12/2009	Rudolf et al.	N/A	N/A
2011/0054150	12/2010	Poulsen et al.	N/A	N/A
2011/0257371	12/2010	Poulsen et al.	N/A	N/A
2011/0305711	12/2010	Allan et al.	N/A	N/A
2012/0000192	12/2011	Zeller et al.	N/A	N/A
2012/0114741	12/2011	Aung-Din	N/A	N/A
2012/0225075	12/2011	Pios et al.	N/A	N/A
2012/0294797	12/2011	Kovacevich et al.	N/A	N/A
2012/0294802	12/2011	Russo et al.	N/A	N/A
2012/0294822	12/2011	Russo et al.	N/A	N/A
2013/0216535	12/2012	Zeller et al.	N/A	N/A
2013/0295087	12/2012	Poulsen et al.	N/A	N/A
2013/0295088	12/2012	Poulsen et al.	N/A	N/A

2015/0266948	12/2014	Bigal et al.	N/A	N/A
2017/0088612	12/2016	Bigal	N/A	N/A
2017/0174754	12/2016	Kovacevich et al.	N/A	N/A
2018/0127490	12/2017	Bigal et al.	N/A	N/A
2018/0142029	12/2017	Boone et al.	N/A	N/A
2018/0161434	12/2017	Russo et al.	N/A	N/A
2019/0211085	12/2018	Kovacevich et al.	N/A	N/A
2019/0240331	12/2018	Russo et al.	N/A	N/A
2019/0367590	12/2018	Russo et al.	N/A	N/A
2020/0010537	12/2019	Baker et al.	N/A	N/A
2020/0216524	12/2019	Cady et al.	N/A	N/A
2020/0216525	12/2019	Cady et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2006313434	12/2006	AU	N/A
2611433	12/2005	CA	N/A
2626120	12/2011	CA	N/A
101309704	12/2007	CN	N/A
101979650	12/2010	CN	N/A
103421114	12/2012	CN	N/A
015526	12/2007	EA	N/A
0212432	12/1986	EP	N/A
1031350	12/1999	EP	N/A
1770091	12/2006	EP	N/A
1556020	12/2008	EP	N/A
1957106	12/2012	EP	N/A
Hei6-87890	12/1993	JP	N/A
08-268874	12/1995	JP	N/A
2005523418	12/2004	JP	N/A
2007517911	12/2006	JP	N/A
2009-515942	12/2008	JP	N/A
2011046710	12/2010	JP	N/A
2011513386	12/2010	JP	N/A
2011513387	12/2010	JP	N/A
5123197	12/2012	JP	N/A
2014-517699	12/2013	JP	N/A
2017-515579	12/2016	JP	N/A
10-1250049	12/2012	KR	N/A
2329062	12/2007	RU	N/A
WO 1996/0004928	12/1995	WO	N/A
WO 97/09046	12/1996	WO	N/A
WO 98/09630	12/1997	WO	N/A
WO 98/11128	12/1997	WO	N/A
WO 98/56779	12/1997	WO	N/A
WO 00/18764	12/1999	WO	N/A
WO 2001/022972	12/2000	WO	N/A

WO 2003/045424	12/2002	WO	N/A
WO 2003/093472	12/2002	WO	N/A
WO 03/104236	12/2002	WO	N/A
WO 2004/003019	12/2003	WO	N/A
WO 2004/014351	12/2003	WO	N/A
WO 2004/050683	12/2003	WO	N/A
WO 2004058184	12/2003	WO	N/A
WO 2004/082602	12/2003	WO	N/A
WO 2004/082605	12/2003	WO	N/A
WO 2004/082678	12/2003	WO	N/A
WO 2004/083187	12/2003	WO	N/A
WO 2004/087649	12/2003	WO	N/A
WO 2004/091514	12/2003	WO	N/A
WO 2004/092166	12/2003	WO	N/A
WO 2004/092168	12/2003	WO	N/A
WO 2004096122	12/2003	WO	N/A
WO 2004097421	12/2003	WO	N/A
WO 2005/009962	12/2004	WO	N/A
WO 2005/040395	12/2004	WO	N/A
WO 2005041757	12/2004	WO	N/A
WO 2005070444	12/2004	WO	N/A
WO 2005/100360	12/2004	WO	N/A
WO 2006/077212	12/2005	WO	N/A
WO 2007/025212	12/2006	WO	N/A
WO 2007/048026	12/2006	WO	N/A
WO 2007/054800	12/2006	WO	N/A
WO 2007/054809	12/2006	WO	N/A
WO 2007/061676	12/2006	WO	N/A
WO 2007/076336	12/2006	WO	N/A
WO 2007/141285	12/2006	WO	N/A
WO 2008/011190	12/2007	WO	N/A
2008144757	12/2007	WO	N/A
WO 2009/109908	12/2008	WO	N/A
WO 2009/109911	12/2008	WO	N/A
WO 2010075238	12/2009	WO	N/A
WO 2011/024113	12/2010	WO	N/A
WO 2011/156324	12/2010	WO	N/A
2012162243	12/2011	WO	N/A
2015143409	12/2014	WO	N/A
2015173539	12/2014	WO	N/A
2016171742	12/2015	WO	N/A
2016205037	12/2015	WO	N/A
2017186928	12/2016	WO	N/A
2018/055574	12/2017	WO	N/A
2020146527	12/2019	WO	N/A

OTHER PUBLICATIONS

Alder Biopharmaceuticals. “Alder BioPharmaceuticals announces positive eptinezumab Phase 3 results for prevention of frequent episodic migraine.” (2017). cited by applicant

Dodick, David W., et al. “Eptinezumab demonstrated efficacy in sustained prevention of episodic

and chronic migraine beginning on day 1 after dosing.” Headache: The Journal of Head and Face Pain 60.10 (2020): 2220-2231. cited by applicant

Dodick, David W., et al. “Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial.” The lancet neurology 13.11 (2014): 1100-1107. cited by applicant

Edvinsson, L. “The Trigeminovascular pathway: role of CGRP and CGRP receptors in migraine. Headache. 57 (Suppl 2): 47-55.” (2017). cited by applicant

George, Judy. “Eptinezumab Effective in Chronic Migraine: Intravenous CGRP blocker shows rapid treatment effect,” MedPage Today, Apr. 27, 2018. cited by applicant

Lee, Mi Ji, et al. “Feasibility of serum CGRP measurement as a biomarker of chronic migraine: a critical reappraisal.” The journal of headache and pain 19.1 (2018): 1-8. cited by applicant

Maasumi, Kasra, Rebecca L. Michael, and Alan M. Rapoport. “CGRP and migraine: the role of blocking calcitonin gene-related peptide ligand and receptor in the management of migraine.” Drugs 78 (2018): 913-928. cited by applicant

Marmura MJ, et al. Preventive migraine treatment with eptinezumab reduced acute headache medication and headache frequency to below diagnostic thresholds in patients with chronic migraine and medication-overuse headache. Headache: The Journal of Head and Face Pain. Oct. 2021;61(9): 1421-31. cited by applicant

Peters, Golden L. “Migraine overview and summary of current and emerging treatment options.” Am J Manag Care 25.2 Suppl (2019): S23-S34. cited by applicant

Raffaelli, Bianca, and Uwe Reuter. “The biology of monoclonal antibodies: focus on calcitonin gene-related peptide for prophylactic migraine therapy.” Neurotherapeutics 15.2 (2018): 324-335. cited by applicant

Silberstein, S. D., et al. “Eptinezumab results for the prevention of episodic migraine over one year in the PROMISE-1 (PRevention of migraine via intravenous Eptinezumab safety and efficacy-1) trial.” Headache. vol. 58. No. 8. 111 River St, Hoboken 07030-5774, NJ USA: Wiley, 2018. p. 1298. cited by applicant

Singh SR, Zhang J, O'Dell C, Hsieh MC, Goldstein J, Liu J, Srivastava A. Effect of polysorbate 80 quality on photostability of a monoclonal antibody. Aaps Pharmscitech. Jun. 2012;13:422-30. cited by applicant

Tepper, Stewart J. “CGRP and headache: a brief review.” Neurological Sciences 40 (2019): 99-105. cited by applicant

The Department of Health and Human Services U.S. Food and Drug Administration, The Pediatric Exclusivity Provision, Jan. 2001 Status Report to Congress (Year 2001). cited by applicant

The International Classification of Headache Disorders, second edition, Cephalalgia, 24(Suppl 1) 2004 (Year: 2004). cited by applicant

Warne, Nicholas W. “Development of high concentration protein biopharmaceuticals: the use of platform approaches in formulation development.” European journal of pharmaceuticals and biopharmaceuticals 78.2 (2011): 208-212. cited by applicant

Dodick, David, et al. “A single intravenous administration of ALD403 (eptinezumab) reduces use of triptans among patients with chronic migraine.” Cephalalgia. vol. 37. 1 Olivers Yard, 55 City Road, London EC1Y 1SP, England: Sage Publications Ltd, 2017. cited by applicant

Lipton, Richard B et al. “Patient-identified most bothersome symptom in preventive migraine treatment with eptinezumab: A novel patient-centered outcome.” Headache vol. 61,5 (2021): 766-776. doi:10.1111/head.14120. cited by applicant

Lipton, Richard B et al. “Evaluating the clinical utility of the patient-identified most bothersome symptom measure from PROMISE-2 for research in migraine prevention.” Headache vol. 62,6 (2022): 690-699. doi:10.1111/head.14295. cited by applicant

Alstadhaug, Karl B et al. “Preventing and treating medication overuse headache.” Pain reports vol.

2,4 e612. Jul. 26, 2017, doi:10.1097/PR9.0000000000000612. cited by applicant

Androulakis, X. Michelle, et al. "Central executive and default mode network intranet work functional connectivity patterns in chronic migraine." *Journal of neurological disorders*. cited by applicant

Burstein et al. "The neurobiology of photophobia." *Journal of neuro-ophthalmology: the official journal of the North American Neuro-Ophthalmology Society*. Mar. 2019;39(1):94. cited by applicant

Carlsen, Louise Ninett, et al. "Complete detoxification is the most effective treatment of medication-overuse headache: a randomized controlled open-label trial." *Cephalalgia* 38.2 (2018): 225-236. cited by applicant

Cevoli, Sabina, et al. "Family history for chronic headache and drug overuse as a risk factor for headache chronification." *Headache: The Journal of Head and Face Pain* 49.3 (2009): 412-418. cited by applicant

Chen, Zhiye, et al. "Altered functional connectivity architecture of the brain in medication overuse headache using resting state fMRI." *The Journal of Headache and Pain* 18.1 (2017): 1-9. cited by applicant

Christensen et al. "Migraine induction with calcitonin gene-related peptide in patients from erenumab trials." *The Journal of Headache and Pain*. Dec. 2018; 19(1): 1-9. cited by applicant

Iranian Office Action dated Apr. 15, 2022, for Pat. Appl. No. 140050140003002468, filed Jun. 15, 2021 entitled "Treatment of Medication Overuse Headache Using Anticgrp or Anti-CGRP-R Antibodies". cited by applicant

Iranian Office Action dated Feb. 7, 2022, for Pat. Appl. No. 140050140003002305, filed Jun. 9, 2021 entitled "Acute Treatment and Rapid Treatment of Headache Using Anti-CGRP Antibodies." cited by applicant

Covasala et al. "Calcitonin gene-related peptide receptors in rat trigeminal ganglion do not control spinal trigeminal activity." *Journal of neurophysiology*. Jul. 15, 2012; 108(2):431-40. cited by applicant

Ferrari, Anna, et al. "Need for analgesics/drugs of abuse: a comparison between headache patients and addicts by the Leeds Dependence Questionnaire (LDQ)." *Cephalalgia* 26.2 (2006): 187-193. cited by applicant

Ferraro, Stefania et al. "In medication-overuse headache, fMRI shows long-lasting dysfunction in midbrain areas." *Headache* vol. 52, 10 (2012): 1520-34. doi:10.1111/j.1526-4610.2012.02276.x. cited by applicant

Find, Ninette Louise et al. "Medication overuse headache in Europe and Latin America: general demographic and clinical characteristics, referral pathways and national distribution of painkillers in a descriptive, multinational, multicenter study." *The journal of headache and pain* 17.1 (2016): 1-12. cited by applicant

Fuh, Jong-Ling et al. "Does medication overuse headache represent a behavior of dependence?." *Pain* vol. 119,1-3 (2005): 49-55. doi:10.1016/j.pain.2005.09.034. cited by applicant

Fumal, Arnaud, et al. "Orbitofrontal cortex involvement in chronic analgesic-overuse headache evolving from episodic migraine." *Brain* 129.2 (2006): 543-550. cited by applicant

Goadsby et al. *Pathophysiology of Migraine: A Disorder of Sensory Processing*. Physiological reviews. Apr;97 (2):553-622. cited by applicant

Grande, Ragnhild Berling, et al. "The Severity of Dependence Scale detects people with medication overuse: the Akershus study of chronic headache." *Journal of Neurology, Neurosurgery & Psychiatry* 80.7 (2009): 784-789. cited by applicant

"Headache Classification Committee of the International Headache Society (IHS) The International Classification of Headache Disorders, 3rd edition." *Cephalalgia : an international journal of headache* vol. 38,1 (2018): 1-211. doi:10.1177/0333102417738202. cited by applicant

Kelman L. "Pain characteristics of the acute migraine attack." *Headache: The Journal of Head and*

Face Pain. Jun. 2006;46(6):942-53. cited by applicant

Kopruszinski et al. "Prevention of stress-or nitric oxide donor-induced medication overuse headache by a calcitonin gene-related peptide antibody in rodents." *Cephalalgia*. May 2017;37(6):560-70. cited by applicant

Kumar et al. "Protective role of a-calcitonin gene-related peptide in cardiovascular diseases." *Frontiers in physiology*. Jul. 2, 2019;10:821. cited by applicant

Lai, Tzu-Hsien, et al. "Gray matter changes related to medication overuse in patients with chronic migraine." *Cephalalgia* 36.14 (2016): 1324-1333. cited by applicant

Lundqvist, C., et al. "An adapted Severity of Dependence Scale is valid for the detection of medication overuse: the Akershus study of chronic headache." *European Journal of Neurology* 18.3 (2011): 512-518. cited by applicant

Lundqvist, Christofer, et al. "The severity of dependence score correlates with medication overuse in persons with secondary chronic headaches. The Akershus study of chronic headache." *Pain®* 148.3 (2010): 487-491. cited by applicant

Messlinger et al. "The Big CGRP Flood-sources, Sinks and Signalling Sites in the Trigemino-vascular System." *The Journal of Headache and Pain*. Dec. 2018;19(1):1-7. cited by applicant

Newman-Norlund, Roger D., et al. "Cortical and subcortical changes following sphenopalatine ganglion blocks in chronic migraine with medication overuse headache: a preliminary longitudinal study." *Women's midlife health* 6.1 (2020): 1-8. cited by applicant

Riederer, Franz, et al. "Decrease of gray matter volume in the midbrain is associated with treatment response in medication-overuse headache: possible influence of orbitofrontal cortex." *Journal of Neuroscience* 33.39 (2013): 15343-15349. cited by applicant

Riederer, Franz, et al. "Grey matter changes associated with medication-overuse headache: correlations with disease related disability and anxiety." *The world journal of biological psychiatry* 13.7 (2012): 517-525. cited by applicant

Storer et al. "Calcitonin gene-related peptide (CGRP) modulates nociceptive trigemino-vascular transmission in the cat." *British journal of pharmacology*. Aug. 2004;142(7):1171-81. cited by applicant

Torta, D. M., et al. "Nucleus accumbens functional connectivity discriminates medication-overuse headache." *NeuroImage: Clinical* 11 (2016): 686-693. cited by applicant

Van Dongen et al. "Migraine biomarkers in cerebrospinal fluid : A systematic review and meta-analysis." *Cephalalgia*. Jan. 2017;37(1):49-63. cited by applicant

Wang et al. "Monoclonal antibody exposure in rat and cynomolgus monkey cerebrospinal fluid following systemic administration." *Fluids and Barriers of the CNS*. Dec. 2018;15(1):1-0. cited by applicant

Winner et al. "Effects of Intravenous Eptinezumab vs Placebo on Headache Pain and Most Bothersome Symptom When Initiated During a Migraine Attack: A Randomized Clinical Trial." *JAMA*. Jun. 15, 2021;325(23):2348-56. cited by applicant

Scuteri et al. "New trends in migraine pharmacology: targeting calcitonin gene-related peptide (CGRP) with monoclonal antibodies." *Frontiers in pharmacology*. Apr. 9, 2019;10:363. cited by applicant

Clinical Trial No. LY2951742, started Mar. 2015, "A Study of LY2951742 in Participants With Episodic Cluster Headache," [online] retrieved from [ClinicalTrials.gov](https://clinicaltrials.gov) at <https://clinicaltrials.gov/ct2/show/study/NCT02397473?term=LY2951742&rank=9>, [retrieved Sep. 3, 2016]. 6 pages. cited by applicant

[Machine translated from website] "Dysfunction of the temporomandibular joint," [online] as published on the Colgate-Palmolive Company website and retrieved from <http://www.colgate.ru/ru/oc/oral-health/conditions/temporomandibular-disorder> 2017 [retrieved Jun. 1, 2017]; 7 pages. cited by applicant

“Cluster Headache,” Wolff’s Headache 1974, p. 348. cited by applicant

“Highlights of Prescribing Information” BLA STN 103000/5215—FDA Approved Labeling Text, Botox Package Insert, Oct. 2010, 25 pages. cited by applicant

“Teva to Acquire Labrys Biologics, Inc.: Novel Migraine Prophylaxis Treatment Adds Significant New Dimension to Teva’s Growing Pain Care Franchise” “Business Wire Jun. 3, 2014.” 4 pages. cited by applicant

“TMJ Disorders,” National Institute of Dental and Craniofacial Research, NIH Publication No. 15-3487, Apr. 2015. 20 pages. cited by applicant

Abdiche YN, et al. “Probing the binding mechanism and affinity of tanezumab, a recombinant humanized anti-NGF monoclonal antibody, using a repertoire of biosensors,” Protein Sci. Aug. 2008;17(8):1326-35. cited by applicant

Adwanikar H, et al. Spinal CGRP1 receptors contribute to supraspinally organized pain behavior and pain-related sensitization of amygdala neurons. Pain. Nov. 2007;132(1-2):53-66. Epub Mar. 1, 2007. cited by applicant

Akerman S, et al. “Nitric oxide synthase inhibitors can antagonize neurogenic and calcitonin gene-related peptide induced dilation of dural meningeal vessels,” Br J Pharmacol. Sep. 2002;137(1):62-8. cited by applicant

Akerman, S., et al. “Pearls and pitfalls in experimental in vivo models of migraine: dural trigeminovascular nociception,” Cephalalgia. Jun. 2013;33(8):577-92. cited by applicant

Alder Biopharmaceuticals Inc., “Alder Presents Positive ALD403 Clinical Data at European Headache and Migraine Trust International Congress,” Press Release, Sep. 15, 2016. cited by applicant

Alder Biopharmaceuticals Inc., “Alder Presents Positive Clinical Data for ALD403 at the 17th Congress of the International Headache Society” Press Release, May 15, 2015. (3 pages). cited by applicant

Alder Biopharmaceuticals Inc., “Alder Reports Phase 2b Trial of ALD403 Meets Primary and Secondary Endpoints Demonstrating Migraine Prevention in Patients with Chronic Migraine,” Press Release, Mar. 28, 2016. (4 pages). cited by applicant

Alder Biopharmaceuticals Inc., “Alder Reports Positive Top-Line 24-Week Data Demonstrating Persistent Migraine Prevention in Phase 2b Study of ALD403 in Patients with Chronic Migraine” Press Release, Jul. 25, 2016. (3 pages). cited by applicant

Alder Biopharmaceuticals Inc., “Data From Proof-of-Concept Clinical Trial of ALD403, a Monoclonal Antibody Against CGRP for the Prevention of Migraine, to be Presented at 56th Annual Scientific Meeting of the American Headache Society,” Press Release, Jun. 26, 2014. 2 pages. cited by applicant

Almagro JC et al. “Chapter 13 Antibody Engineering: Humanization, Affinity Maturation, and Selection Techniques.” Therapeutic Monoclonal Antibodies: From Bench to Clinic (Zhiqiang An (Editor)) Oct. 2009: 311-34. cited by applicant

Amara SG, et al. “Expression in brain of a messenger RNA encoding a novel neuropeptide homologous to calcitonin gene-related peptide.” Science. Sep. 13, 1985;229(4718):1094-7. cited by applicant

Ambalavanar R., et al. “Deep tissue inflammation upregulates neuropeptides and evokes nociceptive behaviors which are modulated by a neuropeptide antagonist.” Pain. Jan. 2006;120(1-2):53-68. Epub Dec. 13, 2005. cited by applicant

Amrutkar DV. “Calcitonin gene-related peptide (CGRP) uptake and release in rat dura mater, trigeminal ganglion and trigeminal nucleus caudalis,” PhD thesis, Faculty of Health and Medical Sciences University of Copenhagen, Academic advisor: Inger Jansen-Olesen and Jes Olesen, Submitted: Feb. 20, 2013. cited by applicant

An Z. “Therapeutic Monoclonal Antibodies: From Bench to Clinic.” Wiley & Sons, Inc., 2009 Chapter 31, 711-62. cited by applicant

Andersen DC, et al. "Production technologies for monoclonal antibodies and their fragments," *Curr Opin Biotechnol.* Oct. 2004;15(5):456-62. cited by applicant

Andrew DP, et al. "Monoclonal antibodies distinguishing alpha and beta forms of calcitonin gene-related peptide." *J Immunol Methods.* Nov. 6, 1990;134(1):87-94. cited by applicant

Antibody Structure and Function, Chapter 4 of Elgert's *Immunology: Understanding the Immune System*, pp. 58-78. Wiley 1998. cited by applicant

Aoki KR. "Review of a proposed mechanism for the antinociceptive action of botulinum toxin type A," *Neurotoxicology.* Oct. 2005;26(5):785-93. cited by applicant

Aoki-Nagase T, et al. "Attenuation of antigen-induced airway hyperresponsiveness in CGRP-deficient mice," *Am J Physiol Lung Cell Mol Physiol.* Nov. 2002;283(5):L963-70. cited by applicant

Armour KL, et al. "Recombinant human IgG molecules lacking Fcγ receptor I binding and monocyte triggering activities," *Eur J Immunol.* Aug. 1999;29(8):2613-24. cited by applicant

Arulmani U, et al. "Calcitonin gene-related peptide and its role in migraine pathophysiology." *Eur J Pharmacol.* Oct. 1, 2004;500(1-3):315-30. cited by applicant

Arulmani U, et al. "Experimental migraine models and their relevance in migraine therapy," *Cephalalgia.* Jun. 2006;26(6):642-59. cited by applicant

Arulmozhi DK, et al., "Migraine: current concepts and emerging therapies." *Vascul Pharmacol.* Sep. 2005;43(3):176-87. cited by applicant

Asghar, MS, et al. "Evidence for a vascular factor in migraine," *Ann Neurol.* Apr. 2011;69(4):635-45. cited by applicant

Ashina M, "Vascular changes have a primary role in migraine," *Cephalalgia.* Apr. 2012;32(5):428-30. cited by applicant

Ashina M, et al. "Evidence for increased plasma levels of calcitonin gene-related peptide in migraine outside of attacks." *Pain.* May 2000;86(1-2):133-8. cited by applicant

Ashina M, et al. "Pearls and pitfalls in human pharmacological models of migraine: 30 years' experience," *Cephalalgia.* Jun. 2013;33(8):540-53. cited by applicant

Ashina M, et al. "Plasma levels of calcitonin gene-related peptide in chronic tension-type headache," *Neurology.* Nov. 14, 2000;55(9):1335-40. cited by applicant

Ashina M. "Calcitonin gene-related peptide in tension-type headache," *ScientificWorldJournal.* Jun. 7, 2002;2:1527-31. cited by applicant

Aziz Q., "Visceral hypersensitivity: fact or fiction." *Gastroenterology.* Aug. 2006;131(2):661-4. cited by applicant

Bagdy, G, et al. "Headache-type adverse effects of NO donors: vasodilation and beyond," *Br J Pharmacol.* May 2010;160(1):20-35. cited by applicant

Balint RF, et al. "Antibody engineering by parsimonious mutagenesis." *Gene.* Dec. 27, 1993;137(1):109-18. cited by applicant

Barker JN, et al. "Progress in psoriasis. Psoriasis: from gene to clinic. London, UK, Dec. 5-7, 1996," *Mol Med Today.* May 1997;3(5):193-4. cited by applicant

Batra SK, et al. "Pharmacokinetics and biodistribution of genetically engineered antibodies," *Curr Opin Biotechnol.* Dec. 2002;13(6):603-8. cited by applicant

Baxter LT, et al. "Biodistribution of monoclonal antibodies: scale-up from mouse to human using a physiologically based pharmacokinetic model," *Cancer Res.* Oct. 15, 1995;55(20):4611-22. cited by applicant

Bell RD, et al. "Breaching the blood-brain barrier for drug delivery," *Neuron.* Jan. 8, 2014;81(1):1-3. cited by applicant

Benarroch EE. "CGRP: sensory neuropeptide with multiple neurologic implications," *Neurology.* Jul. 19, 2011;77(3):281-7. cited by applicant

Benemei S, et al. "CGRP receptors in the control of pain and inflammation," *Curr Opin Pharmacol.* Feb. 2009;9(1):9-14. cited by applicant

Benemei S, et al. "Migraine," *Handb Exp Pharmacol*. 2009;(194):75-89. cited by applicant

Benemei S, et al. "Pain pharmacology in migraine: focus on CGRP and CGRP receptors," *Neurol Sci*. May 2007;28 Suppl 2:S89-93. cited by applicant

Benincosa LJ, et al. "Pharmacokinetics and Pharmacodynamics of a Humanized Monoclonal Antibody to Factor IX in Cynomolgus Monkeys," *J Pharmacol Exp Ther*. Feb. 2000;292(2):810-6. cited by applicant

Bennett AD, et al. "Alleviation of mechanical and thermal allodynia by CGRP(8-37) in a rodent model of chronic central pain." *Pain*. May 2000;86(1-2):163-75. cited by applicant

Benschop U.S. Appl. No. 60/753,044, filed Dec. 22, 2005. File History. 48 pages. cited by applicant

Biacore 3000 Instrument Handbook, Mar. 1999. 201 pages. cited by applicant

Bigal and Krymchantowski, "Emerging drugs for migraine prophylaxis and treatment," *Med. Gen. Med*. 2006;8(2):31. cited by applicant

Bigal M. "Clinical Trials Update—2012: Year in Review—A Comment" *Headache*. Jun. 2013;53(6):1003-4. cited by applicant

Bigal ME, et al. "Emerging drugs for migraine prophylaxis and treatment," *MedGenMed*. May 4, 2006;8(2):31. cited by applicant

Bigal ME, et al. "Ergotamine and dihydroergotamine: a review," *Curr Pain Headache Rep*. Feb. 2003;7(1):55-62. cited by applicant

Bigal ME, et al. "Headache prevention outcome and body mass index," *Cephalalgia*. Apr. 2006;26(4):445-50. cited by applicant

Bigal ME, et al. "Migraine in the Triptan Era: Lessons From Epidemiology, Pathophysiology, and Clinical Science," *Headache*. Feb. 2009;49 Suppl 1:S21-33. cited by applicant

Bigal ME, et al. "Migraine in the triptan era: progresses achieved, lessons learned and future developments," *Arq Neuropsiquiatr*. Jun. 2009;67(2B):559-69. cited by applicant

Bigal ME, et al. "Modifiable risk factors for migraine progression," *Headache*. Oct. 2006;46(9):1334-43. cited by applicant

Bigal ME, et al. "Monoclonal Antibodies for Migraine: Preventing Calcitonin Gene-Related Peptide Activity," *CNS Drugs*. May 2014;28(5):389-99. cited by applicant

Bigal ME, et al. "New developments in migraine prophylaxis," *Expert Opin Pharmacother*. Apr. 2003;4(4):433-43. cited by applicant

Bigal ME, et al. "New migraine preventive options: an update with pathophysiological considerations," *Rev Hosp Clin Fac Med Sao Paulo*. Nov.-Dec. 2002;57(6):293-8. cited by applicant

Bigal ME, et al. "Obesity and migraine: a population study," *Neurology*. Feb. 28, 2006;66(4):545-50. cited by applicant

Bigal ME, et al. "Obesity is a risk factor for transformed migraine but not chronic tension-type headache," *Neurology*. Jul. 25, 2006;67(2):252-7. cited by applicant

Bigal ME, et al. "Prophylactic migraine therapy: emerging treatment options," *Curr Pain Headache Rep*. Jun. 2004;8(3):178-84. cited by applicant

Bigal ME, et al. "Safety and tolerability of LBR-101, a humanized monoclonal antibody that blocks the binding of CGRP to its receptor: Results of the Phase 1 program," *Cephalalgia*. Dec. 23, 2013;34(7):483-492. cited by applicant

Bigal ME, et al. "Safety, tolerability, and efficacy of TEV-48125 for preventive treatment of high-frequency episodic migraine: a multicentre, randomised, double-blind, placebo-controlled, phase 2b study," *Lancet Neurol*. Nov. 2015;14(11):1081-90. cited by applicant

Bigal ME, et al. "The preventive treatment of migraine," *Neurologist*. Jul. 2006;12(4):204-13. cited by applicant

Bigal ME, et al. "The triptans," *Expert Rev Neurother*. May 2009;9(5):649-59. cited by applicant

Bigal, ME "Glutamate Receptor Antagonists," *Headache Currents*, 1:20-21. Jul. 2004. cited by

applicant

Birder L, et al. "Neural control of the lower urinary tract: peripheral and spinal mechanisms," *Neurourol Urodyn*. 2010;29(1):128-39. cited by applicant

Boeckh M, et al. "Phase 1 Evaluation of the Respiratory Syncytial Virus-Specific Monoclonal Antibody Palivizumab in Recipients of Hematopoietic Stem Cell Transplants," *J Infect Dis*. Aug. 1, 2001;184(3):350-4. cited by applicant

Bolay H, et al. "Intrinsic brain activity triggers trigeminal meningeal afferents in a migraine model," *Nat Med*. Feb. 2002;8(2):136-42. cited by applicant

Brain SD, et al. "CGRP receptors: a headache to study, but will antagonists prove therapeutic in migraine?" *Trends Pharmacol Sci*. Feb. 2002;23(2):51-3. cited by applicant

Brain SD, et al. "Vascular actions of calcitonin gene-related peptide and adrenomedullin." *Physiol Rev*. Jul. 2004;84(3):903-34. cited by applicant

Brekke OH, et al. "Therapeutic Antibodies For Human Diseases At The Dawn Of The Twenty-First Century," *Nat Rev Drug Discov*. Jan. 2003;2(1):52-62. cited by applicant

Brorson K, et al. "Mutational analysis of avidity and fine specificity of anti-levan antibodies." *J Immunol*. Dec. 15, 1999;163(12):6694-701. cited by applicant

Brüggemann M, et al. "The Immunogenicity Of Chimeric Antibodies," *J Exp Med*. Dec. 1, 1989;170(6):2153-7. cited by applicant

Brummell DA, et al. "Probing the combining site of an anti-carbohydrate antibody by saturation-mutagenesis: role of the heavy-chain CDR3 residues." *Biochemistry*. Feb. 2, 1993;32(4):1180-7. cited by applicant

Buckley TL, et al. "The partial inhibition of inflammatory responses induced by capsaicin using the Fab fragment of a selective calcitonin gene-related peptide antiserum in rabbit skin." *Neuroscience*. Jun. 1992;48(4):963-8. cited by applicant

Burks EA, "In vitro scanning saturation mutagenesis of an antibody binding pocket." *Proc Natl Acad Sci U S A*. Jan. 21, 1997;94(2):412-7. cited by applicant

Buzzi MG, et al. "The antimigraine drug, sumatriptan (GR43175), selectively blocks neurogenic plasma extravasation from blood vessels in dura mater," *Br J Pharmacol*. Jan. 1990;99(1):202-6. cited by applicant

Carter PJ. "Potent antibody therapeutics by design," *Nat Rev Immunol*. May 2006;6(5):343-57. cited by applicant

Casset F, et al. "A peptide mimetic of an anti-CD4 monoclonal antibody by rational design." *Biochem Biophys Res Commun*. Jul. 18, 2003;307(1):198-205. cited by applicant

Castaño A, et al. "Headache in symptomatic intracranial hypertension secondary to leptospirosis: a case report," *Cephalalgia*. Apr. 2005;25(4):309-11. cited by applicant

Cernuda-Morollón E, et al. "CGRP and VIP levels as predictors of efficacy of Onabotulinumtoxin type A in chronic migraine," *Headache*. Jun. 2014;54(6):987-95. cited by applicant

Chancellor MB, et al. "Neurophysiology of stress urinary incontinence," *Rev Urol*. 2004;6 Suppl 3:S19-28. cited by applicant

Charbit, A et al. "Dopamine: what's new in migraine?" *Curr Opin Neurol*. Jun. 2010;23(3):275-81. cited by applicant

Charles A, "Migraine is not primarily a vascular disorder," *Cephalalgia*. Apr. 2012;32(5):431-2. cited by applicant

Chauhan M, et al. "Studies on the effects of the N-terminal domain antibodies of calcitonin receptor-like receptor and receptor activity-modifying protein 1 on calcitonin gene-related peptide-induced vasorelaxation in rat uterine artery," *Biol Reprod*. Jun. 2004;70(6):1658-63. cited by applicant

Chen JT, et al. "Menopausal flushes and calcitonin-gene-related peptide," *Lancet*. Jul. 3, 1993;342(8862):49. cited by applicant

Chen Y, et al. "Selection and analysis of an optimized anti-VEGF antibody: crystal structure of an

affinity-matured Fab in complex with antigen.” *J Mol Biol.* Nov. 5, 1999;293(4):865-81. cited by applicant

Cheung B et al. “Adrenomedullin: Its Role in the Cardiovascular System,” *Semin Vasc Med.* May 2004;4(2):129-34. cited by applicant

Chowdhury PS, et al. “Tailor-made antibody therapeutics,” *Methods.* May 2005;36(1):11-24. cited by applicant

Chuang YC, et al. “Intraprostatic botulinum toxin a injection inhibits cyclooxygenase-2 expression and suppresses prostatic pain on capsaicin induced prostatitis model in rat,” *J Urol.* Aug. 2008;180(2):742-8. cited by applicant

Chuang YC, et al. “Urodynamic and immunohistochemical evaluation of intravesical botulinum toxin A delivery using liposomes,” *J Urol.* Aug. 2009;182(2):786-92. cited by applicant

Cianchetti C. “The role of the neurovascular scalp structures in migraine,” *Cephalalgia.* Jul. 2012;32(10):778-84. cited by applicant

Clinical Trial No. LY2951742, started Mar. 2015,
<https://clinicaltrials.gov/ct2/show/study/NCT02397473?term=LY2951742&rank=9>, retrieved Sep. 3, 2016. cited by applicant

Colcher D, et al. “Pharmacokinetics and biodistribution of genetically-engineered antibodies,” *Q J Nucl Med.* Dec. 1998;42(4):225-41. cited by applicant

Colgate.ru Website on Temporomandibular Joint Disorders, 2017;
<http://www.colgate.ru/ru/ru/oc/oral-health/conditions/temporomandibular-disorder> 1 page. cited by applicant

Colman PM. “Effects of amino acid sequence changes on antibody-antigen interactions.” *Res Immunol.* Jan. 1994;145(1):33-6. cited by applicant

Conner AC, et al. “Interaction of calcitonin-gene-related peptide with its receptors.” *Biochem Soc Trans.* Aug. 2002;30(4):451-5. cited by applicant

Conner AC, et al. “Ligand binding and activation of the CGRP receptor,” *Biochem Soc Trans.* Aug. 2007;35(Pt 4):729-32. cited by applicant

Connor K M et al: “Randomized, controlled trial of telcagepant for the acute treatment of migraine.”, *Neurology* Sep. 22, 2009, vol. 73, No. 12, Sep. 22, 2009 (Sep. 22, 2009), pp. 970-977, XP002732737, ISSN: 1526-632X. cited by applicant

Correia IR. “Stability of IgG isotypes in serum,” *MAbs.* May-Jun. 2010;2(3):221-32. cited by applicant

Cottrell GS, et al. “Localization of calcitonin receptor-like receptor (CLR) and receptor activity-modifying protein 1 (RAMP1) in human gastrointestinal tract,” *Peptides.* Jun. 2012;35(2):202-11. cited by applicant

Covell DG, et al. “Pharmacokinetics of monoclonal immunoglobulin G1, F(ab')₂, and Fab' in mice.” *Cancer Res.* Aug. 1986;46(8):3969-78. cited by applicant

Cutrer F. “Pathophysiology of Migraine,” *Semin Neurol.* Apr. 2006;26(2):171-80. cited by applicant

Cutrer F. “Pathophysiology of Migraine,” *Semin Neurol.* Apr. 2010;30(2):120-30. cited by applicant

Dakhama A, et al. “Calcitonin gene-related peptide: role in airway homeostasis,” *Curr Opin Pharmacol.* Jun. 2004;4(3):215-20. cited by applicant

Davies J, et al. “Affinity improvement of single antibody VH domains: residues in all three hypervariable regions affect antigen binding.” *Immunotechnology.* Sep. 1996;2(3):169-79. cited by applicant

Davis CD et al. “The Tortuous Road to an Ideal CGRP Function Blocker for the Treatment of Migraine,” *Curr Top Med Chem.* 2008;8(16):1468-79. cited by applicant

Davletov B, et al. “Beyond BOTOX: advantages and limitations of individual botulinum neurotoxins,” *Trends Neurosci.* Aug. 2005;28(8):446-52. cited by applicant

De Pascalis R, et al. "Grafting of "abbreviated" complementarity-determining regions containing specificity-determining residues essential for ligand contact to engineer a less immunogenic humanized monoclonal antibody." *J Immunol.* Sep. 15, 2002;169(6):3076-84. cited by applicant

Delafoy L, et al. "Interactive involvement of brain derived neurotrophic factor, nerve growth factor, and calcitonin gene related peptide in colonic hypersensitivity in the rat." *Gut.* Jul. 2006;55(7):940-5. Epub Jan. 9, 2006. cited by applicant

Denekas T, et al. "Inhibition of stimulated meningeal blood flow by a calcitonin gene-related peptide binding mirror-image RNA oligonucleotide," *Br J Pharmacol.* Jun. 2006;148(4):536-43. cited by applicant

Deng R et al. "Projecting human pharmacokinetics of therapeutic antibodies from nonclinical data," *MAbs.* Jan.-Feb. 2011;3(1):61-6. cited by applicant

Derosa G, et al. "Optimizing combination treatment in the management of type 2 diabetes," *Vasc Health Risk Manag.* 2007;3(5):665-71. cited by applicant

Diamond S, et al. "Patterns of diagnosis and acute and preventive treatment for migraine in the United States: results from the American Migraine Prevalence and Prevention study," *Headache.* Mar. 2007;47(3):355-63. cited by applicant

Diener HC, et al. "Utility of topiramate for the treatment of patients with chronic migraine in the presence or absence of acute medication overuse," *Cephalalgia.* Oct. 2009;29(10):1021-7. cited by applicant

Dockray et al., "Immunoneutralization studies with calcitonin gene-related peptide," *Ann. NY Acad Sci.* 1992;657:258-67. cited by applicant

Dodick D, et al. "Cluster Headache: Diagnosis, Management and Treatment," *Wolff's Headache* 2001, p. 283. cited by applicant

Dodick DW, et al. "Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial," *Lancet Neurol.* Nov. 2014;13(11):1100-7. cited by applicant

Doggrell S. "Migraine and beyond: cardiovascular therapeutic potential for CGRP modulators," *Expert Opin Investig Drugs.* Jun. 2001;10(6):1131-8. cited by applicant

Dolgin E. "Antibody drugs set to revive flagging migraine target," *Nat Rev Drug Discov.* Apr. 2013;12(4):249-50. cited by applicant

Doods H, et al. "Pharmacological profile of BIBN4096BS, the first selective small molecule CGRP antagonist." *Br J Pharmacol.* Feb. 2000;129(3):420-3. cited by applicant

Doods, H et al. "CGRP antagonists: unravelling the role of CGRP in migraine," *Trends Pharmacol Sci.* Nov. 2007;28(11):580-7. cited by applicant

Dooley JS, et al. "Antibiotics in the treatment of biliary infection," *Gut.* Sep. 1984;25(9):988-98. cited by applicant

Drake AW, et al. "Characterizing high-affinity antigen/antibody complexes by kinetic- and equilibrium-based methods," *Anal Biochem.* May 1, 2004;328(1):35-43. cited by applicant

Dressler and Saberi, "Botulinum toxin: mechanisms of action," *Eur. Neurol.* 2005;53:3-9. cited by applicant

Dressler D, et al. "Botulinum toxin: mechanisms of action," *Arq Neuropsiquiatr.* Mar. 2005;63(1):180-5. cited by applicant

Dufner P, et al. "Harnessing phage and ribosome display for antibody optimisation." *Trends Biotechnol.* Nov. 2006;24(11):523-9. Epub Sep. 26, 2006. cited by applicant

Durham P. "CGRP-receptor antagonists—a fresh approach to migraine therapy?" *N Engl J Med.* Mar. 11, 2004;350(11):1073-5. cited by applicant

Durham Paul L et al: "Calcitonin Gene-Related Peptide (CGRP) Receptor Antagonists in the Treatment of Migraine", *CNS Drugs*, vol. 24, No. 7, 2010, pp. 539-548. cited by applicant

Durham PL et al. "New insights into the molecular actions of serotonergic antimigraine drugs," *Pharmacol Ther.* Apr.-May 2002;94(1-2):77-92. cited by applicant

Durham PL, et al. "Regulation of calcitonin gene-related peptide secretion from trigeminal nerve cells by botulinum toxin type A: implications for migraine therapy," *Headache*. Jan. 2004;44(1):35-42; discussion 42-3. cited by applicant

Durham PL. "Calcitonin Gene-Related Peptide (CGRP) and Migraine," *Headache*. Jun. 2006;46 Suppl 1:S3-8. cited by applicant

Durham PL. "Inhibition of calcitonin gene-related peptide function: a promising strategy for treating migraine," *Headache*. Sep. 2008;48(8):1269-75. cited by applicant

Edvinsson L et al. "Blockade of CGRP receptors in the intracranial vasculature: a new target in the treatment of headache," *Cephalalgia*. Aug. 2004;24(8):611-22. cited by applicant

Edvinsson L et al. "CGRP Receptor Antagonism and Migraine," *Neurotherapeutics*. Apr. 2010;7(2):164-75. cited by applicant

Edvinsson L et al. "Extracerebral manifestations in migraine. A peptidergic involvement?" *J Intern Med*. Oct. 1990;228(4):299-304. cited by applicant

Edvinsson L et al. "Neurobiology in primary headaches," *Brain Res Brain Res Rev*. Jun. 2005;48(3):438-56. cited by applicant

Edvinsson L et al. "Perivascular neuropeptides (NPY, VIP, CGRP and SP) in human brain vessels after subarachnoid haemorrhage," *Acta Neurol Scand*. Nov. 1994;90(5):324-30. cited by applicant

Edvinsson L et al. "The blood-brain barrier in migraine treatment," *Cephalalgia*. Dec. 2008;28(12):1245-58. cited by applicant

Edvinsson L et al: "New drugs in migraine treatment and prophylaxis: telcagepant and topiramate", *The Lancet*, the Lancet Publishing Group, GB, vol. 376, No. 9741, Aug. 21, 2010 (Aug. 21, 2010), pp. 645-655. cited by applicant

Edvinsson L, et al. "Calcitonin gene-related peptide and cerebral blood vessels: distribution and vasomotor effects," *J Cereb Blood Flow Metab*. Dec. 1987;7(6):720-8. cited by applicant

Edvinsson L, et al. "Inhibitory effect of BIBN4096BS, CGRP(8-37), a CGRP antibody and an RNA-Spiegelmer on CGRP induced vasodilatation in the perfused and non-perfused rat middle cerebral artery." *Br J Pharmacol*. Mar. 2007;150(5):633-40. Epub Jan. 22, 2007. cited by applicant

Edvinsson L, et al. "Innervation of the human middle meningeal artery: immunohistochemistry, ultrastructure, and role of endothelium for vasomotility," *Peptides*. 1998;19(7):1213-25. cited by applicant

Edvinsson L, et al. "Neuropeptides in migraine and cluster headache," *Cephalalgia*. Oct. 1994;14(5):320-7. cited by applicant

Edvinsson L. "Aspects on the Pathophysiology of Migraine and Cluster Headache," *Pharmacol Toxicol*. Aug. 2001;89(2):65-73. cited by applicant

Edvinsson L. "Calcitonin Gene-Related Peptide (CGRP) and the Pathophysiology of Headache Therapeutic Implications," *CNS Drugs*. 2001;15(10):745-53. cited by applicant

Edvinsson L. "CGRP blockers in migraine therapy: where do they act?" *Br J Pharmacol*. Dec. 2008;155(7):967-9. cited by applicant

Edvinsson L. "CGRP-receptor antagonism in migraine treatment," *Lancet*. Dec. 20, 2008;372(9656):2089-90. cited by applicant

Edvinsson L. "Clinical Data on the CGRP Antagonist BIBN4096BS for Treatment of Migraine Attacks," *CNS Drug Rev*. 2005 Spring;11(1):69-76. cited by applicant

Edvinsson L. "Innervation and effects of dilatory neuropeptides on cerebral vessels. New aspects," *Blood Vessels*. 1991;28(1-3):35-45. cited by applicant

Edvinsson L. "Neuronal Signal Substances as Biomarkers of Migraine," *Headache*. Jul.-Aug. 2006;46(7):1088-94. cited by applicant

Edvinsson L. "New therapeutic target in primary headaches—blocking the CGRP receptor," *Expert Opin Ther Targets*. Jun. 2003;7(3):377-83. cited by applicant

Edvinsson L. "Novel migraine therapy with calcitonin gene-regulated peptide receptor antagonists," *Expert Opin Ther Targets*. Sep. 2007;11(9):1179-88. cited by applicant

Edvinsson L: "CGRP blockers in migraine therapy: where do they act?", *British Journal of Pharmacology*, vol. 155, No. 7, Dec. 2008 (Dec. 2008), pp. 967-969. cited by applicant

Edvinsson Lars: "CGRP-receptor antagonism in migraine treatment.", *Lancet* Dec. 20, 2008, vol. 372, No. 9656, Dec. 20, 2008 (Dec. 20, 2008), pp. 2089-2090. cited by applicant

Eftekhari S et al. "Differentiation of Nerve Fibers Storing CGRP and CGRP Receptors in the Peripheral Trigemino-vascular System," *J Pain*. Nov. 2013;14(11):1289-303. cited by applicant

Elshourbagy NA, et al. "Molecular cloning and characterization of the porcine calcitonin gene-related peptide receptor." *Endocrinology*. Apr. 1998;139(4):1678-83. cited by applicant

Emerick GT. "Migraines in the Presence of Glaucoma, Recent advances in diagnosis and management," *Glaucoma Today*, Sep./Oct. 2008, 21-23. cited by applicant

Escott et al., "Effect of a calcitonin gene-related peptide antagonist (CGRP8-37) on skin vasodilatation and oedema induced by stimulation of the rat saphenous nerve," *Br. J. Pharmacol.* 1993;110:772-6. cited by applicant

Escott KJ, et al. "Trigeminal ganglion stimulation increases facial skin blood flow in the rat: a major role for calcitonin gene-related peptide." *Brain Res*. Jan. 9, 1995;669(1):93-9. cited by applicant

Esfandiyari T. "The Role Of Calcitonin Gene-Related Peptide (CGRP) In Colonic Inflammation, And Secretion In The Rat Distal Colon," Thesis, University of Calgary, Department of Neuroscience and Gastrointestinal Sciences. 1999. 145 pages. cited by applicant

Evans BN, et al. "CGRP-RCP, a novel protein required for signal transduction at calcitonin gene-related peptide and adrenomedullin receptors," *J Biol Chem*. Oct. 6, 2000;275(40):31438-43. cited by applicant

Evans RW, et al. "Target doses and titration schedules for migraine preventive medications," *Headache*. Jan. 2006;46(1):160-4. cited by applicant

Evans RW. "Exploding head syndrome followed by sleep paralysis: a rare migraine aura," *Headache*. Apr. 2006;46(4):682-3. cited by applicant

Everitt DE et al. "The Pharmacokinetics, Antigenicity, and Fusion-Inhibition Activity of RSHZ19, a Humanized Monoclonal Antibody to Respiratory Syncytial Virus, in Healthy Volunteers," *J Infect Dis*. Sep. 1996;174(3):463-9. cited by applicant

Faraci FM, et al. "Vascular responses of dura mater," *Am J Physiol*. Jul. 1989;257(1 Pt 2):H157-61. cited by applicant

Farinelli, I et al. "Future drugs for migraine," *Intern Emerg Med*. Oct. 2009;4(5):367-73. cited by applicant

Feuerstein G et al. "Clinical perspectives of calcitonin gene related peptide pharmacology," *Can J Physiol Pharmacol*. Jul. 1995;73(7):1070-4. cited by applicant

File History U.S. Appl. No. 60/736,623, filed Nov. 14, 2005, Zeller, et al. *Antagonist Antibodies Directed Against Calcitonin Gene-Related Peptide and Methods Using Same*. 110 pages. cited by applicant

Fischer MJ et al. "The Nonpeptide Calcitonin Gene-Related Peptide Receptor Antagonist BIBN4096BS Lowers the Activity of Neurons with Meningeal Input in the Rat Spinal Trigeminal Nucleus," *J Neurosci*. Jun. 22, 2005;25(25):5877-83. cited by applicant

Fischer MJ. "Calcitonin gene-related peptide receptor antagonists for migraine," *Expert Opin Investig Drugs*. Jul. 2010;19(7):815-23. cited by applicant

Forssman B, et al. "Atenolol for migraine prophylaxis," *Headache*. Jul. 1983;23(4):188-90. cited by applicant

Forster ER, et al. "The role of calcitonin gene-related peptide in gastric mucosal protection in the rat," *Exp Physiol*. Jul. 1991;76(4):623-6. cited by applicant

Friend PJ, et al. "Phase I study of an engineered aglycosylated humanized CD3 antibody in renal transplant rejection," *Transplantation*. Dec. 15, 1999;68(11):1632-7. cited by applicant

Frobert Y, et al. "A sensitive sandwich enzyme immunoassay for calcitonin gene-related peptide

(CGRP): characterization and application.” *Peptides*. 1999;20(2):275-84. cited by applicant

Galitsky BA, et al. “Predicting amino acid sequences of the antibody human VH chains from its first several residues,” *Proc Natl Acad Sci U S A*. Apr. 28, 1998;95(9):5193-8. cited by applicant

Gallai V, et al. “Vasoactive peptide levels in the plasma of young migraine patients with and without aura assessed both interictally and ictally.” *Cephalalgia*. Oct. 1995;15(5):384-90. cited by applicant

Gangula PR, et al. “Increased blood pressure in alpha-calcitonin gene-related peptide/calcitonin gene knockout mice,” *Hypertension*. Jan. 2000;35(1 Pt 2):470-5. cited by applicant

Gearing D, et al. “A fully caninised anti-NGF monoclonal antibody for pain relief in dogs,” *BMC Vet Res*. Nov. 9, 2013;9:226. cited by applicant

Geppetti P et al. “Antidromic vasodilatation and the migraine mechanism,” *J Headache Pain*. Mar. 2012;13(2):103-11. cited by applicant

Geppetti P et al. “CGRP and migraine: neurogenic inflammation revisited,” *J Headache Pain*. Apr. 2005;6(2):61-70. cited by applicant

Geppetti P et al. “Novel therapeutic targets,” *Neurol Sci*. May 2006;27 Suppl 2:S111-4. cited by applicant

Giamberardino MA, et al. “Emerging drugs for migraine treatment,” *Expert Opin Emerg Drugs*. Mar. 2015;20(1):137-47. cited by applicant

Gillies S et al. “Improving the efficacy of antibody-interleukin 2 fusion proteins by reducing their interaction with Fc receptors,” *Cancer Res*. May 1, 1999;59(9):2159-66. cited by applicant

Giniatullin R et al. “Molecular Mechanisms of Sensitization of Pain-transducing P2X3 Receptors by the Migraine Mediators CGRP and NGF,” *Mol Neurobiol*. Feb. 2008;37(1):83-90. cited by applicant

Glennie MJ, et al. “Clinical trials of antibody therapy,” *Immunol Today*. Aug. 2000;21(8):403-10. cited by applicant

Glover V, et al. “Can the vascular and neurogenic theories of migraine finally be reconciled?” *Trends Pharmacol Sci*. Jan. 1989;10(1):1-3. cited by applicant

Goadsby PJ et al. “Release of vasoactive peptides in the extracerebral circulation of humans and the cat during activation of the trigeminovascular system,” *Ann Neurol*. Feb. 1988;23(2):193-6. cited by applicant

Goadsby PJ, et al. “Migraine—current understanding and treatment.” *N Engl J Med*. Jan. 24, 2002;346(4):257-70. cited by applicant

Goadsby PJ, et al. “Vasoactive peptide release in the extracerebral circulation of humans during migraine headache.” *Ann Neurol*. Aug. 1990;28(2):183-7. cited by applicant

Goadsby PJ. “Advances in the understanding of headache,” *Br Med Bull*. Oct. 5, 2005;73-74:83-92. Print 2005. cited by applicant

Goadsby PJ. “Calcitonin gene-related peptide antagonists as treatments of migraine and other primary headaches,” *Drugs*. 2005;65(18):2557-67. cited by applicant

Goadsby PJ. “Can we develop neurally acting drugs for the treatment of migraine?” *Nat Rev Drug Discov*. Sep. 2005;4(9):741-50. cited by applicant

Goadsby PJ. “Headache: a good year for research,” *Lancet Neurol*. Jan. 2006;5(1):5-6. cited by applicant

Goadsby PJ. “Migraine Pathophysiology,” *Headache*. Apr. 2005;45 Suppl 1:S14-24. cited by applicant

Goadsby PJ. “New targets in the acute treatment of headache,” *Curr Opin Neurol*. Jun. 2005;18(3):283-8. cited by applicant

Goadsby PJ. “The vascular theory of migraine—a great story wrecked by the facts,” *Brain*. Jan. 2009;132(Pt 1):6-7. cited by applicant

Goadsby, PJ, et al. “Randomized, double-blind, placebo-controlled trial of ALD403, an anti-CGRP antibody in the prevention of frequent episodic migraine.” 56th Annual Scientific Meeting of the

American Headache Society, Jun. 2014. 4 pages. cited by applicant

Gómez-Foix AM, et al., "Anti-insulin effects of amylin and calcitonin-gene-related peptide on hepatic glycogen metabolism," *Biochem J.* Jun. 15, 1991;276 (Pt 3):607-10. cited by applicant

Green LL, et al. "Antigen-specific human monoclonal antibodies from mice engineered with human Ig heavy and light chain YACs," *Nat Genet.* May 1994;7(1):13-21. cited by applicant

Grunenberger F. "[Calcitonin gene-related peptide (CGRP): a vasodilator neuropeptide with many potential applications]" *Pathol Biol (Paris)*. Dec. 1993;41(10):936-42. cited by applicant

Gupta S et al. "Evidence for CGRP re-uptake in rat dura mater encephali," *Br J Pharmacol.* Dec. 2010;161(8):1885-98. cited by applicant

Gupta S et al. "Intravital microscopy on a closed cranial window in mice: a model to study trigeminovascular mechanisms involved in migraine," *Cephalalgia.* Nov. 2006;26(11):1294-303. cited by applicant

Gupta S et al. "Potential role of female sex hormones in the pathophysiology of migraine," *Pharmacol Ther.* Feb. 2007;113(2):321-40. cited by applicant

Gupta S et al. "The relevance of preclinical research models for the development of antimigraine drugs: focus on 5-HT(1B/1D) and CGRP receptors," *Pharmacol Ther.* Oct. 2010;128(1):170-90. cited by applicant

Hakala JM, et al. "Modelling constrained calcitonin gene-related peptide analogues." *Protein Eng.* Feb. 1996;9(2):143-8. cited by applicant

Halimi S, et al. "Combination treatment in the management of type 2 diabetes: focus on vildagliptin and metformin as a single tablet," *Vasc Health Risk Manag.* 2008;4(3):481-92. cited by applicant

Hanes J et al. "Picomolar affinity antibodies from a fully synthetic naive library selected and evolved by ribosome display," *Nat Biotechnol.* Dec. 2000;18(12):1287-92. cited by applicant

Hansen JM, et al. "Calcitonin gene-related peptide triggers migraine-like attacks in patients with migraine with aura," *Cephalalgia.* Oct. 2010;30(10):1179-86. cited by applicant

Hargreaves R. "New Migraine and Pain Research," *Headache.* Apr. 2007;47 Suppl 1:S26-43. cited by applicant

Hatcher JP, et al. "Biologics: the next-generation therapeutics for analgesia?" *Expert Rev Neurother.* Nov. 2011;11(11):1653-8. cited by applicant

Hay D et al. "A comparison of the actions of BIBN4096BS and CGRP(8-37) on CGRP and adrenomedullin receptors expressed on SK-N-MC, L6, Col 29 and Rat 2 cells," *Br J Pharmacol.* Sep. 2002;137(1):80-6. cited by applicant

Hay D et al. "International Union of Pharmacology. LXIX. Status of the Calcitonin Gene-Related Peptide Subtype 2 Receptor," *Pharmacol Rev.* Jun. 2008;60(2):143-5. cited by applicant

Hay D et al. "The pharmacology of CGRP-responsive receptors in cultured and transfected cells," *Peptides.* Nov. 2004;25(11):2019-26. cited by applicant

Hay D et al. "The Preclinical Pharmacology of BIBN4096BS, a CGRP Antagonist," *Cardiovasc Drug Rev.* 2005 Spring;23(1):31-42. cited by applicant

Hay D. "What Makes a CGRP2 Receptor?" *Clin Exp Pharmacol Physiol.* Oct. 2007;34(10):963-71. cited by applicant

Hay DL, et al. "CL/RAMP2 and CL/RAMP3 produce pharmacologically distinct adrenomedullin receptors: a comparison of effects of adrenomedullin22-52, CGRP8-37 and BIBN4096BS," *Br J Pharmacol.* Oct. 2003;140(3):477-86. Epub Aug. 26, 2003. cited by applicant

Hershey JC, et al. "Investigation of the species selectivity of a nonpeptide CGRP receptor antagonist using a novel pharmacodynamic assay," *Regul Pept.* Apr. 15, 2005;127(1-3):71-7. cited by applicant

Hill RG et al. "Neuropeptide and Kinin Antagonists," *Handb Exp Pharmacol.* 2007;(177):181-216. cited by applicant

Hillmen P, et al. "Effect of eculizumab on hemolysis and transfusion requirements in patients with

paroxysmal nocturnal hemoglobinuria,” *N Engl J Med.* Feb. 5, 2004;350(6):552-9. cited by applicant

Hinton PR, et al. “Engineered human IgG antibodies with longer serum half-lives in primates,” *J Biol Chem.* Feb. 20, 2004;279(8):6213-6. cited by applicant

Hirsch S et al. “The CGRP receptor antagonist BIBN4096BS peripherally alleviates inflammatory pain in rats,” *Pain.* May 2013;154(5):700-7. cited by applicant

Ho TW et al. “CGRP and its receptors provide new insights into migraine pathophysiology,” *Nat Rev Neurol.* Oct. 2010;6(10):573-82. cited by applicant

Ho TW, et al. “Impact of recent prior opioid use on rizatriptan efficacy. A post hoc pooled analysis,” *Headache.* Mar. 2009;49(3):395-403. cited by applicant

Ho TW, et al. “Randomized controlled trial of the CGRP receptor antagonist telcagepant for migraine prevention,” *Neurology.* Sep. 9, 2014;83(11):958-66. cited by applicant

Ho TW, et al. “Efficacy and tolerability of MK-0974 (telcagepant), a new oral antagonist of calcitonin gene-related peptide receptor, compared with zolmitriptan for acute migraine: a randomised, placebo-controlled, parallel-treatment trial,” *Lancet.* Dec. 20, 2008;372(9656):2115-23. cited by applicant

Hoff AO et al. “Increased bone mass is an unexpected phenotype associated with deletion of the calcitonin gene,” *J Clin Invest.* Dec. 2002;110(12):1849-57. cited by applicant

Hoffmann J, et al. “New Agents for Acute Treatment of Migraine: CGRP Receptor Antagonists, iNOS Inhibitors,” *Curr Treat Options Neurol.* Feb. 2012;14(1):50-9. cited by applicant

Holland J et al. “Calcitonin Gene-Related Peptide Reduces Brain Injury in a Rat Model of Focal Cerebral Ischemia,” *Stroke.* Oct. 1994;25(10):2055-8; discussion 2058-9. cited by applicant

Holliger P, et al. “Engineered antibody fragments and the rise of single domains,” *Nat Biotechnol.* Sep. 2005;23(9):1126-36. cited by applicant

Holm P, et al. “Functional mapping and single chain construction of the anti-cytokeratin 8 monoclonal antibody TS1.” *Mol Immunol.* Feb. 2007;44(6):1075-84. Epub Sep. 20, 2006. cited by applicant

Holman JJ, et al. “Human alpha- and beta-CGRP and rat alpha-CGRP are coronary vasodilators in the rat.” *Peptides.* Mar.-Apr. 1986;7(2):231-5. cited by applicant

Holt LJ, et al. “Domain antibodies: proteins for therapy.” *Trends Biotechnol.* Nov. 2003;21(11):484-90. cited by applicant

Holzer P et al. “Afferent Nerve-Mediated Protection Against Deep Mucosal Damage in the Rat Stomach,” *Gastroenterology.* Apr. 1990;98(4):838-48. cited by applicant

Holzer P et al. “Sensory neurons mediate protective vasodilatation in rat gastric mucosa,” *Am J Physiol.* Mar. 1991;260(3 Pt 1):G363-70. cited by applicant

Holzer P et al. “Stimulation Of Afferent Nerve Endings By Intragastric Capsaicin Protects Against Ethanol-Induced Damage Of Gastric Mucosa,” *Neuroscience.* Dec. 1988;27(3):981-7. cited by applicant

Holzer P. “Implications of tachykinins and calcitonin gene-related peptide in inflammatory bowel disease,” *Digestion.* Jul.-Aug. 1998;59(4):269-83. cited by applicant

Holzer P. “Capsaicin: Cellular Targets, Mechanisms of Action, and Selectivity for Thin Sensory Neurons,” *Pharmacol Rev.* Jun. 1991;43(2):143-201. cited by applicant

Hong KW, et al. “Effect of omega-conotoxin GVIA and omega-agatoxin IVA on the capsaicin-sensitive calcitonin gene-related peptide release and autoregulatory vasodilation in rat pial arteries,” *J Cereb Blood Flow Metab.* Jan. 1999;19(1):53-60. cited by applicant

Hong KW, et al. “Pharmacological coupling and functional role for CGRP receptors in the vasodilation of rat pial arterioles,” *Am J Physiol.* Jan. 1996;270(1 Pt 2):H317-23. cited by applicant

Hong KW, et al. “Pharmacological evidence that calcitonin gene-related peptide is implicated in cerebral autoregulation,” *Am J Physiol.* Jan. 1994;266(1 Pt 2):H11-6. cited by applicant

Hoogenboom HR, et al. “Multi-subunit proteins on the surface of filamentous phage:

methodologies for displaying antibody (Fab) heavy and light chains,” *Nucleic Acids Res.* Aug. 11, 1991;19(15):4133-7. cited by applicant

Hoogenboom HR. “Selecting and screening recombinant antibody libraries,” *Nat Biotechnol.* Sep. 2005;23(9):1105-16. cited by applicant

Hopkins, CR. “ACS Chemical Neuroscience Molecule Spotlight on Telcagepant (MK-0974),” *ACS Chem Neurosci.* Jul. 20, 2011;2(7):334-5. cited by applicant

Hu H, et al. “Acute migraine treatment with rizatriptan in real world settings—focusing on treatment strategy, effectiveness, and behavior,” *Headache.* Feb. 2009;49 Suppl 1:S34-42. cited by applicant

Hubbard JA, et al. “Identification of the epitopes of calcitonin gene-related peptide (CGRP) for two anti CGRP monoclonal antibodies by 2D NMR,” *Protein Sci.* Sep. 1997;6(9):1945-52. cited by applicant

Hudson PJ, et al. “Engineered antibodies,” *Nat Med.* Jan. 2003;9(1):129-34. cited by applicant

Hughes SR et al. “A calcitonin gene-related peptide (CGRP) antagonist (CGRP8-37) inhibits microvascular responses induced by CGRP and capsaicin in skin,” *Br J Pharmacol.* Nov. 1991;104(3):738-42. cited by applicant

Hurley D. “CGRP Drug Improves Wellness on Headache-Free Days, Study Finds,” *Neurology Today*, p. 31, Jul. 2016. cited by applicant

Hwang WY, et al. “Immunogenicity of engineered antibodies,” *Methods.* May 2005;36(1):3-10. cited by applicant

Ibrahimi K, et al. “Development of an experimental model to study trigeminal nerve-mediated vasodilation on the human forehead,” *Cephalalgia.* Jan. 3, 2014;34(7):514-522. cited by applicant

Idusogie EE, et al. “Mapping of the C1q Binding Site on Rituxan, a Chimeric Antibody with a Human IgG1 Fc,” *J Immunol.* Apr. 15, 2000;164(8):4178-84. cited by applicant

Iovino M, et al. “Safety, tolerability and pharmacokinetics of BIBN 4096 BS, the first selective small molecule calcitonin gene-related peptide receptor antagonist, following single intravenous administration in healthy volunteers,” *Cephalalgia.* Aug. 2004;24(8):645-56. cited by applicant

Janeway CA et al. “Immuno Biology: The Immune System in Health and Disease.” *Current Biology Ltd./Garland Publishing Inc.* 1994 Glossary page G:2. cited by applicant

Jang YJ, et al. “The structural basis for DNA binding by an anti-DNA autoantibody.” *Mol Immunol.* Dec. 1998;35(18):1207-17. cited by applicant

Jansen-Olesen I, et al. “In-depth characterization of CGRP receptors in human intracranial arteries,” *Eur J Pharmacol.* Nov. 28, 2003;481(2-3):207-16. cited by applicant

Jones PT, et al. “Replacing the complementarity-determining regions in a human antibody with those from a mouse,” *Nature.* May 29-Jun. 4, 1986;321(6069):522-5. cited by applicant

Juaneda C, et al. “The molecular pharmacology of CGRP and related peptide receptor subtypes,” *Trends Pharmacol Sci.* Nov. 2000;21(11):432-8. cited by applicant

Juhasz G, et al. “NO-induced migraine attack: strong increase in plasma calcitonin gene-related peptide (CGRP) concentration and negative correlation with platelet serotonin release.” *Pain.* Dec. 2003;106(3):461-70. cited by applicant

Juhl L, et al. “Effect of two novel CGRP-binding compounds in a closed cranial window rat model,” *Eur J Pharmacol.* Jul. 12, 2007;567(1-2):117-24. cited by applicant

Julia V, et al. “Tachykininergic mediation of viscerosensitive responses to acute inflammation in rats: role of CGRP.” *Am J Physiol.* Jan. 1997;272(1 Pt 1):G141-6. cited by applicant

Jung ST, et al. “Bypassing glycosylation: engineering aglycosylated full-length IgG antibodies for human therapy,” *Curr Opin Biotechnol.* Dec. 2011;22(6):858-67. cited by applicant

Kaiser EA, et al. “CGRP and migraine: could PACAP play a role too?” *Neuropeptides.* Dec. 2013;47(6):451-61. cited by applicant

Kapoor K, et al. “Effects of BIBN4096BS on cardiac output distribution and on CGRP-induced carotid haemodynamic responses in the pig,” *Eur J Pharmacol.* Aug. 15, 2003;475(1-3):69-77. cited

by applicant

Kapoor K, et al. "Effects of the CGRP receptor antagonist BIBN4096BS on capsaicin-induced carotid haemodynamic changes in anaesthetised pigs," Br J Pharmacol. Sep. 2003;140(2):329-38. cited by applicant

Kapoor, K. "Novel Potential Antimigraine Compounds: Carotid and Systemic Haemodynamic Effects in a Porcine Model of Migraine," Thesis, Erasmus University, Rotterdam. With summary in Dutch. 2003. 157 pages. cited by applicant

Karasek C., et al. "Characterization of the intrinsic binding features of three anti-CGRP therapeutic antibodies effective in preventing migraine: a comparative pre-clinical case study of ALD403, LY-2951742, TEV-48125." 5th European Headache and Migraine Trust International Congress, Sep. 2016. 4 pages. cited by applicant

Kato K, et al. "CGRP antagonists enhance gastric acid secretion in 2-h pylorus-ligated rats," Peptides. 1995;16(7):1257-62. cited by applicant

Kawamura M, et al. "Antinociceptive effect of intrathecally administered antiserum against calcitonin gene-related peptide on thermal and mechanical noxious stimuli in experimental hyperalgesic rats." Brain Res. Sep. 11, 1989;497(1):199-203. cited by applicant

Kaymakcalan Z, et al. "Comparisons of affinities, avidities, and complement activation of adalimumab, infliximab, and etanercept in binding to soluble and membrane tumor necrosis factor," Clin Immunol. May 2009;131(2):308-16. cited by applicant

Keates AC, et al. "CGRP upregulation in dorsal root ganglia and ileal mucosa during Clostridium difficile toxin A-induced enteritis," Am J Physiol. Jan. 1998;274(1 Pt 1):G196-202. cited by applicant

Kennel SJ, et al. "Direct Binding of Radioiodinated Monoclonal Antibody to Tumor Cells: Significance of Antibody Purity and Affinity for Drug Targeting or Tumor Imaging," Hybridoma. 1983;2(3):297-310. cited by applicant

Kim SJ, et al. "Antibody Engineering for the Development of Therapeutic Antibodies," Mol Cells. Aug. 31, 2005;20(1):17-29. cited by applicant

Kipriyanov S, et al. "Generation and Production of Engineered Antibodies," Mol Biotechnol. Jan. 2004;26(1):39-60. cited by applicant

Kipriyanov S. "Generation of Antibody Molecules Through Antibody Engineering" from Methods in Molecular Biology, vol. 207: Recombinant Antibodies for Cancer Therapy Methods and Protocols, 2003 pp. 3-25. cited by applicant

Knotkova H, et al. "Imaging intracranial plasma extravasation in a migraine patient: a case report," Pain Med. May-Jun. 2007;8(4):383-7. cited by applicant

Kobayashi D, et al. "Calcitonin Gene-Related Peptide Mediated Neurogenic Vasorelaxation in the Isolated Canine Lingual Artery," Jpn J Pharmacol. Apr. 1995;67(4):329-39. cited by applicant

Kobayashi H, et al. "Tryptophan H33 plays an important role in pyrimidine (6-4) pyrimidone photoproduct binding by a high-affinity antibody." Protein Eng. Oct. 1999;12(10):879-84. cited by applicant

Krymchantowski AV, et al. "New and emerging prophylactic agents for migraine," CNS Drugs. 2002;16(9):611-34. cited by applicant

Krymchantowski AV, et al. "Rizatriptan in migraine," Expert Rev Neurother. Sep. 2005;5(5):597-603. cited by applicant

Krymchantowski AV, et al. "Rizatriptan vs. rizatriptan plus trimebutine for the acute treatment of migraine: a double-blind, randomized, cross-over, placebo-controlled study," Cephalalgia. Jul. 2006;26(7):871-4. cited by applicant

Krymchantowski AV, et al. "Topiramate plus nortriptyline in the preventive treatment of migraine: a controlled study for nonresponders," J Headache Pain. Jan. 2012;13(1):53-9. cited by applicant

Kumar S, et al. "Molecular cloning and expression of the Fabs of human autoantibodies in *Escherichia coli*. Determination of the heavy or light chain contribution to the anti-DNA/-

cardiolipin activity of the Fab.” *J Biol Chem*. Nov. 10, 2000;275(45):35129-36. cited by applicant

Kunkel RS, et al. “Surgical treatment of chronic migrainous neuralgia,” *Cleve Clin Q*. 1974 Winter;41(4):189-92. cited by applicant

Kuraishi Y, et al. “Antinociception induced in rats by intrathecal administration of antiserum against calcitonin gene-related peptide.” *Neurosci Lett*. Oct. 17, 1988;92(3):325-9. cited by applicant

Kurosawa M, et al. “Increase of meningeal blood flow after electrical stimulation of rat dura mater encephali: mediation by calcitonin gene-related peptide,” *Br J Pharmacol*. Apr. 1995;114(7):1397-402. cited by applicant

Kuus-Reichel K, et al. “Will Immunogenicity Limit the Use, Efficacy, and Future Development of Therapeutic Monoclonal Antibodies?” *Clin Diagn Lab Immunol*. Jul. 1994;1(4):365-72. cited by applicant

Lambrecht N, et al. “Role of calcitonin gene-related peptide and nitric oxide in the gastroprotective effect of capsaicin in the rat,” *Gastroenterology*. May 1993;104(5):1371-80. cited by applicant

Lance J. “Migraine Pain Originates from Blood Vessels,” *Headache Pathogenesis: Monoamines, Neuropeptides, Purines, and Nitric Oxide*, edited by J. Olesen and L. Edvinsson, Lippincott-Raven Publishers, Philadelphia, 1997. Chapter 1, pp. 3-9. cited by applicant

Lassen LH, et al. “CGRP may play a causative role in migraine.” *Cephalalgia*. Feb. 2002;22(1):54-61. cited by applicant

Lassen LH, et al. “Involvement of calcitonin gene-related peptide in migraine: regional cerebral blood flow and blood flow velocity in migraine patients,” *J Headache Pain*. Jun. 2008;9(3):151-7. cited by applicant

Lazzeria M, et al. “The Challenge of the Overactive Bladder: From Laboratory to New Drugs,” *European Association of Urology*, vol. 5, Issue 6, Dec. 2007, pp. 250-258. cited by applicant

Lee CV, et al. “High-affinity human antibodies from phage-displayed synthetic Fab libraries with a single framework scaffold,” *J Mol Biol*. Jul. 23, 2004;340(5):1073-93. cited by applicant

Leighton B, et al. “Pancreatic amylin and calcitonin gene-related peptide cause resistance to insulin in skeletal muscle in vitro,” *Nature*. Oct. 13, 1988;335(6191):632-5. cited by applicant

Levêque D, et al. “Pharmacokinetics of therapeutic monoclonal antibodies used in oncology,” *Anticancer Res*. May-Jun. 2005;25(3c):2327-43. cited by applicant

Levy D, et al. “A critical view on the role of migraine triggers in the genesis of migraine pain,” *Headache*. Jun. 2009;49(6):953-7. cited by applicant

Levy D, et al. “Calcitonin gene-related peptide does not excite or sensitize meningeal nociceptors: implications for the pathophysiology of migraine,” *Ann Neurol*. Nov. 2005;58(5):698-705. cited by applicant

Levy D, et al. “Migraine pain and nociceptor activation—where do we stand?” *Headache*. May 2010;50(5):909-16. cited by applicant

Levy D, et al. “The vascular theory of migraine: leave it or love it?” *Ann Neurol*. Apr. 2011;69(4):600-1. cited by applicant

Li DS, et al. “Role of calcitonin gene-related peptide in gastric hyperemic response to intragastric capsaicin,” *Am J Physiol*. Oct. 1991;261(4 Pt 1):G657-61. cited by applicant

Lin HC, et al. “Immunoneutralization of Calcitonin Gene-Related Peptide (CGRP) During Inhibition of Intestinal Transit by Fat,” *Gastroenterology* vol. 114, No. 4, 1998. 1 page. Abstract No. G3253. cited by applicant

Lin YS, et al. “Preclinical pharmacokinetics, interspecies scaling, and tissue distribution of a humanized monoclonal antibody against vascular endothelial growth factor,” *J Pharmacol Exp Ther*. Jan. 1999;288(1):371-8. cited by applicant

Link AS, et al. “Treatment of migraine attacks based on the interaction with the trigemino-cerebrovascular system,” *J Headache Pain*. Feb. 2008;9(1):5-12. cited by applicant

Lipton RB, et al. “CGRP antagonists in the acute treatment of migraine,” *Lancet Neurol*. Jun.

2004;3(6):332. cited by applicant

Lipton RB, et al. "Headache: triumphs in translational research," *Lancet Neurol.* Jan. 2005;4(1):11-2. cited by applicant

Lipton RB, et al. "Moving forward—essential questions for the next 10 years," *Headache.* Feb. 2009;49 Suppl 1:S43-6. cited by applicant

Little M, et al. "Of mice and men: hybridoma and recombinant antibodies." *Immunol Today.* Aug. 2000;21(8):364-70. cited by applicant

Lonberg N, et al. "Antigen-specific human antibodies from mice comprising four distinct genetic modifications," *Nature.* Apr. 28, 1994;368(6474):856-9. cited by applicant

Lonberg N, et al. "Human antibodies from transgenic animals," *Nat Biotechnol.* Sep. 2005;23(9):1117-25. cited by applicant

Longoni M, et al. "Inflammation and excitotoxicity: role in migraine pathogenesis," *Neurol Sci.* May 2006;27 Suppl 2:S107-10. cited by applicant

Louis SM, et al. "Antibodies to calcitonin-gene related peptide reduce inflammation induced by topical mustard oil but not that due to carrageenin in the rat." *Neurosci Lett.* Jul. 31, 1989;102(2-3):257-60. cited by applicant

Louis SM, et al. "Immunization with calcitonin gene-related peptide reduces the inflammatory response to adjuvant arthritis in the rat," *Neuroscience.* 1990;39(3):727-31. cited by applicant

Louis SM, et al. "The role of substance P and calcitonin gene-related peptide in neurogenic plasma extravasation and vasodilatation in the rat." *Neuroscience.* 1989;32(3):581-6. cited by applicant

MacCallum RM, et al. "Antibody-antigen interactions: contact analysis and binding site topography." *J Mol Biol.* Oct. 11, 1996;262(5):732-45. cited by applicant

MacGregor EA. "Migraine in pregnancy and lactation: a clinical review," *J Fam Plann Reprod Health Care.* Apr. 2007;33(2):83-93. cited by applicant

Majima, M, et al. "Roles of calcitonin gene-related peptide in enhancement of angiogenesis," *Inflammation and Regeneration* vol. 31 No. 2 Mar. 2011, 146-150. cited by applicant

Mallee JJ, et al. "Receptor activity-modifying protein 1 determines the species selectivity of non-peptide CGRP receptor antagonists." *J Biol Chem.* Apr. 19, 2002;277(16):14294-8. cited by applicant

Marcelo E. Bigal et al: "Calcitonin Gene-Related Peptide (CGRP) and Migraine Current Understanding and State of Development", *Headache*, vol. 53, No. 8, Sep. 12, 2013 (Sep. 12, 2013), pp. 1230-1244. cited by applicant

Mareska M, et al. "Lambert-Eaton myasthenic syndrome," *Semin Neurol.* Jun. 2004;24(2):149-53. cited by applicant

Marquez de Prado B and Russo AF, "CGRP receptor antagonists: A new frontier of anti- migraine medications," *Drug Discov Today Ther Strateg.* 2006 Winter;3(4):593-597. cited by applicant

Marshall I, et al. "Human and rat alpha-CGRP but not calcitonin cause mesenteric vasodilatation in rats." *Eur J Pharmacol.* Apr. 16, 1986;123(2):217-22. cited by applicant

Martínez-Sáenz A, et al. "Role of calcitonin gene-related peptide in inhibitory neurotransmission to the pig bladder neck," *J Urol.* Aug. 2011;186(2):728-35. cited by applicant

Maynard JA, et al. "Protection against anthrax toxin by recombinant antibody fragments correlates with antigen affinity," *Nat Biotechnol.* Jun. 2002;20(6):597-601. cited by applicant

McCafferty J, et al. "Phage antibodies: filamentous phage displaying antibody variable domains," *Nature.* Dec. 6, 1990;348(6301):552-4. cited by applicant

McCulloch J, et al. "Calcitonin gene-related peptide: functional role in cerebrovascular regulation," *Proc Natl Acad Sci U S A.* Aug. 1986;83(15):5731-5. cited by applicant

McLatchie LM, et al. "RAMPs regulate the transport and ligand specificity of the calcitonin-receptor-like receptor," *Nature.* May 28, 1998;393(6683):333-9. cited by applicant

Mehrotra S, et al. "Current and prospective pharmacological targets in relation to antimigraine action," *Naunyn Schmiedebergs Arch Pharmacol.* Oct. 2008;378(4):371-94. cited by applicant

Mense S. "Pathophysiology of low back pain and the transition to the chronic state—experimental data and new concepts." *Schmerz*. Dec. 2001;15(6):413-7. cited by applicant

Messlinger K, et al. "Neuropeptide effects in the trigeminal system: pathophysiology and clinical relevance in migraine," *Keio J Med*. 2011;60(3):82-9. cited by applicant

Messlinger K. "Migraine: where and how does the pain originate?" *Exp Brain Res*. Jun. 2009;196(1):179-93. cited by applicant

Messlinger, et al. "Inhibition of neurogenic blood flow increases in the rat cranial dura matter by a CGRP-binding Spiegelmer," *Cephalalgia*, No. F022 2004. cited by applicant

Middlemiss DN. "Direct evidence for an interaction of beta-adrenergic blockers with the 5-HT receptor," *Nature*. May 19, 1977;267(5608):289-90. cited by applicant

Middlemiss DN. "Stereoselective blockade at [3H]5-HT binding sites and at the 5-HT autoreceptor by propranolol," *Eur J Pharmacol*. Jun. 1, 1984;101(3-4):289-93. cited by applicant

Mirick GR, et al. "A review of human anti-globulin antibody (HAGA, HAMA, HACA, HAHA) responses to monoclonal antibodies. Not four letter words," *Q J Nucl Med Mol Imaging*. Dec. 2004;48(4):251-7. cited by applicant

Molina JM, et al. "Induction of insulin resistance in vivo by amylin and calcitonin gene-related peptide," *Diabetes*. Feb. 1990;39(2):260-5. cited by applicant

Moore CK, et al. "Urological Applications of Botulinum Toxin," *Female Urology: A Practical Clinical Guide*. 2007 Chapter 14:213-217. cited by applicant

Moore EL, et al. "Targeting a family B GPCR/RAMP receptor complex: CGRP receptor antagonists and migraine," *Br J Pharmacol*. May 2012;166(1):66-78. cited by applicant

Morara S, et al. "Monoclonal antibodies reveal expression of the CGRP receptor in Purkinje cells, interneurons and astrocytes of rat cerebellar cortex," *Neuroreport*. Nov. 16, 1998;9(16):3755-9. cited by applicant

Morell A, et al. "Metabolic properties of IgG subclasses in man." *J Clin Invest*. Apr. 1970;49(4):673-80. cited by applicant

Morrison SL, et al. "Chimeric human antibody molecules: mouse antigen-binding domains with human constant region domains," *Proc Natl Acad Sci U S A*. Nov. 1984;81(21):6851-5. cited by applicant

Moskowitz MA, "Neurogenic inflammation in the pathophysiology and treatment of migraine," *Neurology*. Jun. 1993;43(6 Suppl 3):S16-20. cited by applicant

Moskowitz MA, et al. "CGRP: blood flow and more?" *Cephalalgia*. Aug. 1996;16(5):287. cited by applicant

Moskowitz MA. "Pathophysiology of headache—past and present," *Headache*. Apr. 2007;47 Suppl 1:S58-63. cited by applicant

Mould DR, et al. "A population pharmacokinetic-pharmacodynamic analysis of single doses of clenoliximab in patients with rheumatoid arthritis," *Clin Pharmacol Ther*. Sep. 1999;66(3):246-57. cited by applicant

Mountain A, et al. "Engineering antibodies for therapy," *Biotechnol Genet Eng Rev*. 1992;10:1-142. cited by applicant

Muff R, et al. "Calcitonin, calcitonin gene-related peptide, adrenomedullin and amylin: homologous peptides, separate receptors and overlapping biological actions," *Eur J Endocrinol*. Jul. 1995;133(1):17-20. cited by applicant

Mulderry PK, et al. "Differential expression of alpha-CGRP and beta-CGRP by primary sensory neurons and enteric autonomic neurons of the rat." *Neuroscience*. Apr. 1988;25(1):195-205. cited by applicant

Mullins MW, et al. "Characterization of a calcitonin gene-related peptide (CGRP) receptor on mouse bone marrow cells." *Regul Pept*. Nov. 19, 1993;49(1):65-72. cited by applicant

Nakamura-Craig M, et al. "Effect of neurokinin A, substance P and calcitonin gene related peptide in peripheral hyperalgesia in the rat paw." *Neurosci Lett*. Mar. 11, 1991;124(1):49-51. cited by

applicant

Naot D, et al. "The role of peptides and receptors of the calcitonin family in the regulation of bone metabolism," Bone. Nov. 2008;43(5):813-8. cited by applicant

Negro A, et al. "CGRP receptor antagonists: an expanding drug class for acute migraine?" Expert Opin Investig Drugs. Jun. 2012;21(6):807-18. cited by applicant

Newman R, et al. "Modification of the Fc region of a primatized IgG antibody to human CD4 retains its ability to modulate CD4 receptors but does not deplete CD4(+) T cells in chimpanzees," Clin Immunol. Feb. 2001;98(2):164-74. cited by applicant

Ng-Mak DS, et al. "Migraine treatment with rizatriptan and almotriptan: a crossover study," Headache. May 2009;49(5):655-62. cited by applicant

Nippon Rinsho, "Recent Development of Calcitonin Gene-related Peptide (CGRP) receptor antagonist," 2005, vol. 63, Suppl. 10, pp. 263-266 [Original With English Translation]. cited by applicant

Nishimoto N, et al. "Anti-interleukin-6 receptor antibody therapy in rheumatic diseases," Endocr Metab Immune Disord Drug Targets. Dec. 2006;6(4):373-81. cited by applicant

Oates PJ, et al. "Studies on the mechanism of ethanol-induced gastric damage in rats," Gastroenterology. Jan. 1988;94(1):10-21. cited by applicant

Ober RJ, et al. "Visualizing the site and dynamics of IgG salvage by the MHC class I-related receptor, FcRn," J Immunol. Feb. 15, 2004;172(4):2021-9. cited by applicant

O'Connell JP, et al. "On the role of the C-terminus of alpha-calcitonin-gene-related peptide (alpha CGRP). The structure of des-phenylalaninamide³⁷-alpha CGRP and its interaction with the CGRP receptor," Biochem J. Apr. 1, 1993;291 (Pt 1):205-10. cited by applicant

Oh-hashii Y, et al. "Elevated sympathetic nervous activity in mice deficient in alphaCGRP," Circ Res. Nov. 23, 2001;89(11):983-90. cited by applicant

Olesen J, et al. "Calcitonin gene-related peptide receptor antagonist BIBN 4096 BS for the acute treatment of migraine." N Engl J Med. Mar. 11, 2004;350(11):1104-10. cited by applicant

Olesen J, et al. "Chapter 16: Calcitonin Gene-Related Peptide and Other Peptides." The Headaches Third Edition. Lippincott Williams & Wilkins 2006 159-164. cited by applicant

Olesen J, et al. "Chapter 31: CGRP Involvement in Migraines." The Headaches Third Edition. Lippincott Williams & Wilkins 2006 289-99. cited by applicant

Olesen J, et al. "Emerging migraine treatments and drug targets," Trends Pharmacol Sci. Jun. 2011;32(6):352-9. cited by applicant

Olesen J, et al. "Finding new drug targets for the treatment of migraine attacks," Cephalalgia. Sep. 2009;29(9):909-20. cited by applicant

Olesen J, et al. "Migraine: a research field matured for the basic neurosciences," Trends Neurosci. Jan. 1991;14(1):3-5. cited by applicant

Olesen J, et al. "Origin of pain in migraine: evidence for peripheral sensitisation," Lancet Neurol. Jul. 2009;8(7):679-90. cited by applicant

Olesen J. "Migraine: A neural pathway for photophobia in migraine," Nat Rev Neurol. May 2010;6(5):241-2. cited by applicant

Ondo WG, et al. "Botulinum toxin A for chronic daily headache: a randomized, placebo-controlled, parallel design study," Cephalalgia. Jan. 2004;24(1):60-5. cited by applicant

O'Sullivan J, et al. "Migraine development, treatments, research advances, and anesthesia implications," AANA J. Feb. 2006;74(1):61-9. cited by applicant

Ottosson A, et al. "Release of histamine from dural mast cells by substance P and calcitonin gene-related peptide," Cephalalgia. May 1997;17(3):166-74. cited by applicant

Pabst MA, et al. "Ablation of capsaicin sensitive afferent nerves impairs defence but not rapid repair of rat gastric mucosa," Gut. Jul. 1993;34(7):897-903. cited by applicant

Panconesi A, et al. "Migraine pain: reflections against vasodilatation," J Headache Pain. Oct. 2009;10(5):317-25. cited by applicant

Panka DJ, et al. "Defining the structural correlates responsible for loss of arsonate affinity in an IDCR antibody isolated from an autoimmune mouse," *Mol Immunol*. Aug. 1993;30(11):1013-20. cited by applicant

Paone DV, et al. "Calcitonin gene-related peptide receptor antagonists for the treatment of migraine: a patent review," *Expert Opin Ther Pat*. Dec. 2009;19(12):1675-713. cited by applicant

Papadopoulos N, et al. "Binding and neutralization of vascular endothelial growth factor (VEGF) and related ligands by VEGF Trap, ranibizumab and bevacizumab." *Angiogenesis*. Jun. 2012;15(2):171-85. cited by applicant

Papp K, et al. "The treatment of moderate to severe psoriasis with a new anti-CD11a monoclonal antibody," *J Am Acad Dermatol*. Nov. 2001;45(5):665-74. cited by applicant

Pavlou AK, et al. "Recombinant protein therapeutics—success rates, market trends and values to 2010," *Nat Biotechnol*. Dec. 2004;22(12):1513-9. cited by applicant

Peroutka SJ, et al. "Neurogenic inflammation and migraine: implications for the therapeutics," *Mol Interv*. Oct. 2005;5(5):304-11. cited by applicant

Peskar BM, et al. "A monoclonal antibody to calcitonin gene-related peptide abolishes capsaicin-induced gastroprotection." *Eur J Pharmacol*. Nov. 30, 1993;250(1):201-3. cited by applicant

Petersen KA, et al. "BIBN4096BS antagonizes human alpha-calcitonin gene related peptide-induced headache and extracerebral artery dilatation." *Clin Pharmacol Ther*. Mar. 2005;77(3):202-13. cited by applicant

Petersen KA, et al. "Effect of hypotension and carbon dioxide changes in an improved genuine closed cranial window rat model," *Cephalalgia*. Jan. 2005;25(1):23-9. cited by applicant

Petersen KA, et al. "Inhibitory effect of BIBN4096BS on cephalic vasodilatation induced by CGRP or transcranial electrical stimulation in the rat." *Br J Pharmacol*. Nov. 2004;143(6):697-704. cited by applicant

Petersen KA, et al. "Presence and function of the calcitonin gene-related peptide receptor on rat pial arteries investigated in vitro and in vivo," *Cephalalgia*. Jun. 2005;25(6):424-32. cited by applicant

Petersen KA, et al. "The effect of nonpeptide CGRP-antagonist, BIBN4096BS on human-alphaCGRP induced headache and hemodynamics in healthy volunteers," *Cephalalgia*, vol. 23, extract from Abstracts of the XI Congress of the International Headache Society, p. 725, 2003. cited by applicant

Petkova SB, et al. "Enhanced half-life of genetically engineered human IgG1 antibodies in a humanized FcRn mouse model: potential application in humorally mediated autoimmune disease," *Int Immunol*. Dec. 2006;18(12):1759-69. cited by applicant

Pietrobon D, et al. "Pathophysiology of migraine," *Annu Rev Physiol*. 2013;75:365-91. cited by applicant

Plessas IN, et al. "Migraine-like episodic pain behavior in a dog: can dogs suffer from migraines?" *J Vet Intern Med*. Sep.-Oct. 2013;27(5):1034-40. cited by applicant

Plourde V, et al. "CGRP antagonists and capsaicin on celiac ganglia partly prevent postoperative gastric ileus." *Peptides*. Nov.-Dec. 1993;14(6):1225-9. cited by applicant

Poyner DR, et al. "International Union of Pharmacology. XXXII. The mammalian calcitonin gene-related peptides, adrenomedullin, amylin, and calcitonin receptors," *Pharmacol Rev*. Jun. 2002;54(2):233-46. cited by applicant

Presta L. "Antibody engineering for therapeutics," *Curr Opin Struct Biol*. Aug. 2003;13(4):519-25. cited by applicant

Presta LG, et al. "Engineering therapeutic antibodies for improved function," *Biochem Soc Trans*. Aug. 2002;30(4):487-90. cited by applicant

Prewett M, et al. "The biologic effects of C225, a chimeric monoclonal antibody to the EGFR, on human prostate carcinoma." *J Immunother Emphasis Tumor Immunol*. Nov. 1996;19(6):419-27. cited by applicant

Qing-Hui Niu, et al. "Expression of mast cell and calcitonin gene related peptides in the mucosa of irritable bowel syndrome," *World Chinese Journal of Digestology*, Jan. 18, 2009 p. 213-217; ISSN 1099-3079. cited by applicant

Raddant AC, et al. "Calcitonin gene-related peptide in migraine: intersection of peripheral inflammation and central modulation," *Expert Rev Mol Med*. Nov. 29, 2011;13:e36. cited by applicant

Ramadan NM, et al. "New and future migraine therapy," *Pharmacol Ther*. Oct. 2006;112(1):199-212. cited by applicant

Ramadan NM. "Acute treatments: future developments," *Curr Med Res Opin*. 2001;17 Suppl 1:s81-6. cited by applicant

Ramos ML, et al. "AMG 334 CGRP antibody for migraine: time to celebrate?" *Lancet Neurol*. Apr. 2016;15(4):347-9. cited by applicant

Rapoport AM, Bigal ME, et al. "Naratriptan in the preventive treatment of refractory chronic migraine." In Olsen J, Silberstein SD, Tfelt-Hansen P, eds. *Preventive Pharmacotherapy of Headache Disorders*. Copenhagen: Oxford University Press, 2004, Chapter 31. cited by applicant

Rapoport AM, et al. "Intranasal medications for the treatment of migraine and cluster headache," *CNS Drugs*. 2004;18(10):671-85. cited by applicant

Rapoport AM, et al. "Levetiracetam in the preventive treatment of transformed migraine: A prospective, open-label, pilot study," *Curr Ther Res Clin Exp*. May 2005;66(3):212-21. cited by applicant

Rapoport AM, et al. "Migraine preventive therapy: current and emerging treatment options," *Neurol Sci*. May 2005;26 Suppl 2:s111-20. cited by applicant

Rapoport AM, et al. "Preventive migraine therapy: what is new," *Neurol Sci*. Oct. 2004;25 Suppl 3:S177-85. cited by applicant

Raybould HE, et al. "Selective ablation of spinal afferent neurons containing CGRP attenuates gastric hyperemic response to acid," *Peptides*. Mar.-Apr. 1992;13(2):249-54. cited by applicant

Reasbeck PG, et al. "Calcitonin gene-related peptide: enteric and cardiovascular effects in the dog," *Gastroenterology*. Oct. 1988;95(4):966-71. cited by applicant

Recober A, et al. "Calcitonin gene-related peptide: A molecular link between obesity and migraine?" *Drug News Perspect*. Mar. 2010;23(2):112-7. cited by applicant

Recober A, et al. "Calcitonin gene-related peptide: an update on the biology," *Curr Opin Neurol*. Jun. 2009;22(3):241-6. cited by applicant

Recober A, et al. "Olcegepant, a non-peptide CGRP1 antagonist for migraine treatment," *IDrugs*. Aug. 2007;10(8):566-74. cited by applicant

Recober A, et al., "Role of calcitonin gene-related peptide in light-averse behavior: implications for migraine," *J Neurosci*. Jul. 8, 2009;29(27):8798-804. cited by applicant

Reddy MP, et al. "Elimination of Fc receptor-dependent effector functions of a modified IgG4 monoclonal antibody to human CD4," *J Immunol*. Feb. 15, 2000;164(4):1925-33. cited by applicant

Reff ME, et al. "A review of modifications to recombinant antibodies: attempt to increase efficacy in oncology applications," *Crit Rev Oncol Hematol*. Oct. 2001;40(1):25-35. cited by applicant

Reff ME, et al. "Depletion of B cells in vivo by a chimeric mouse human monoclonal antibody to CD20," *Blood*. Jan. 15, 1994;83(2):435-45. cited by applicant

Reichert JM, et al. "Monoclonal antibody successes in the clinic," *Nat Biotechnol*. Sep. 2005;23(9):1073-8. cited by applicant

Reinshagen M, et al. "Calcitonin gene-related peptide mediates the protective effect of sensory nerves in a model of colonic injury." *J Pharmacol Exp Ther*. Aug. 1998;286(2):657-61. cited by applicant

Reuter U, et al. "Experimental models of migraine," *Funct Neurol*. 2000;15 Suppl 3:9-18. cited by applicant

Reuter U. "Anti-CGRP antibodies: a new approach to migraine prevention," *Lancet Neurol.* Sep. 2014;13(9):857-9. cited by applicant

Rolston RK, et al., "Intravenous calcitonin gene-related peptide stimulates net water secretion in rat colon in vivo," *Dig Dis Sci.* Apr. 1989;34(4):612-6. cited by applicant

Roon KI, et al. "No acute antimigraine efficacy of CP-122,288, a highly potent inhibitor of neurogenic inflammation: results of two randomized, double-blind, placebo-controlled clinical trials," *Ann Neurol.* Feb. 2000;47(2):238-41. cited by applicant

Roopenian DC, et al. "FcRn: the neonatal Fc receptor comes of age," *Nat Rev Immunol.* Sep. 2007;7(9):715-25. cited by applicant

Roque AC, et al. "Antibodies and genetically engineered related molecules: production and purification," *Biotechnol Prog.* May-Jun. 2004;20(3):639-54. cited by applicant

Roskos LK, et al. "The Clinical Pharmacology of Therapeutic Monoclonal Antibodies," *Drug Development Research* 2004 61:108-120. cited by applicant

Rother RP, et al. "Discovery and development of the complement inhibitor eculizumab for the treatment of paroxysmal nocturnal hemoglobinuria," *Nat Biotechnol.* Nov. 2007;25(11):1256-64. cited by applicant

Rovero P, et al. "CGRP antagonist activity of short C-terminal fragments of human alpha CGRP, CGRP(23-37) and CGRP(19-37)." *Peptides.* Sep.-Oct. 1992;13(5):1025-7. cited by applicant

Rudikoff S, et al. "Single amino acid substitution altering antigen-binding specificity." *Proc Natl Acad Sci U S A.* Mar. 1982;79(6):1979-83. cited by applicant

Ruiz-Gayo M, et al. "Vasodilatory effects of cholecystokinin: new role for an old peptide?" *Regul Pept.* Dec. 10, 2006;137(3):179-84. cited by applicant

Russo AF, et al., "A Potential Preclinical Migraine Model: CGRP-Sensitized Mice," *Mol Cell Pharmacol.* 2009;1(5):264-270. cited by applicant

Russo AF. "Calcitonin gene-related peptide (CGRP): a new target for migraine," *Annu Rev Pharmacol Toxicol.* 2015;55:533-52. cited by applicant

Russo. "CGRP Meeting Abstract Book," *The 4th International Meeting on CGRP, Copenhagen,* Sep. 2001, 71 pages. cited by applicant

Russo. "CGRP Meeting Abstract Book," *Joint International Symposium on Calcitonin Gene-Related Peptide, Amylin and Calcitonin; 4th Symposium on Adrenomedullin and Proadrenomedullin N-20 Peptide, Zurich, Switzerland,* Mar. 2004. 38 pages. cited by applicant

Ryan AM, et al. "Preclinical safety evaluation of rhuMAbVEGF, an antiangiogenic humanized monoclonal antibody," *Toxicol Pathol.* Jan.-Feb. 1999;27(1):78-86. cited by applicant

Ryan S. "Medicines for migraine," *Arch Dis Child Educ Pract Ed.* Apr. 2007;92(2):ep50-5. cited by applicant

Saleh MN, et al. "Phase I trial of the chimeric anti-GD2 monoclonal antibody ch14.18 in patients with malignant melanoma." *Hum Antibodies Hybridomas.* Jan. 1992;3(1):19-24. cited by applicant

Salonen R, et al. "Triptans: do they differ?" *Curr Pain Headache Rep.* Apr. 2002;6(2):133-9. cited by applicant

Salvatore CA, et al. "Pharmacological characterization of MK-0974 [N-[(3R,6S)-6-(2,3-difluorophenyl)-2-oxo-1-(2,2,2-trifluoroethyl)azepan-3-yl]-4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)piperidine-1-carboxamide], a potent and orally active calcitonin gene-related peptide receptor antagonist for the treatment of migraine," *J Pharmacol Exp Ther.* Feb. 2008;324(2):416-21. cited by applicant

Sams-Nielsen A, et al. "Pharmacological evidence for CGRP uptake into perivascular capsaicin sensitive nerve terminals," *Br J Pharmacol.* Mar. 2001;132(5):1145-53. cited by applicant

Saphire EO, et al. "Crystal structure of a neutralizing human IGG against HIV-1: a template for vaccine design," *Science.* Aug. 10, 2001;293(5532):1155-9. cited by applicant

Schaible HG, et al. "Mechanisms of pain in arthritis." *Ann N Y Acad Sci.* Jun. 2002;966:343-54. cited by applicant

Schelstraete C, et al. "CGRP antagonists: hope for a new era in acute migraine treatment," *Acta Neurol Belg*. Dec. 2009;109(4):252-61. cited by applicant

Schier R, et al. "Isolation of high-affinity monomeric human anti-c-erbB-2 single chain Fv using affinity-driven selection," *J Mol Biol*. Jan. 12, 1996;255(1):28-43. cited by applicant

Schier R, et al. "Isolation of picomolar affinity anti-c-erbB-2 single-chain Fv by molecular evolution of the complementarity determining regions in the center of the antibody binding site," *J Mol Biol*. Nov. 8, 1996;263(4):551-67. cited by applicant

Schifter S. "Circulating concentrations of calcitonin gene-related peptide (CGRP) in normal man determined with a new, highly sensitive radioimmunoassay," *Peptides*. Mar.-Apr. 1991;12(2):365-9. cited by applicant

Schindler M, et al. "Binding properties of the novel, non-peptide CGRP receptor antagonist radioligand, [(3)H]BIBN4096BS," *Eur J Pharmacol*. May 10, 2002;442(3):187-93. cited by applicant

Schoenen J, et al. "Almotriptan and its combination with aceclofenac for migraine attacks: a study of efficacy and the influence of auto-evaluated brush allodynia," *Cephalalgia*. Oct. 2008;28(10):1095-105. cited by applicant

Schreiber CP. "The pathophysiology of migraine," *Dis Mon*. Oct. 2006;52(10):385-401. cited by applicant

Schwenger N, et al. "Interaction of calcitonin gene-related peptide, nitric oxide and histamine release in neurogenic blood flow and afferent activation in the rat cranial dura mater," *Cephalalgia*. Jun. 2007;27(6):481-91. cited by applicant

Schytz HW, et al. "What have we learnt from triggering migraine?" *Curr Opin Neurol*. Jun. 2010;23(3):259-65. cited by applicant

Seike M, et al. "Increased synthesis of calcitonin gene-related peptide stimulates keratinocyte proliferation in murine UVB-irradiated skin," *J Dermatol Sci*. Feb. 2002;28(2):135-43. cited by applicant

Selenko N, et al. "CD20 antibody (C2B8)-induced apoptosis of lymphoma cells promotes phagocytosis by dendritic cells and cross-priming of CD8+ cytotoxic T cells," *Leukemia*. Oct. 2001;15(10):1619-26. cited by applicant

Seong J, et al. "Radiation-induced alteration of pain-related signals in an animal model with bone invasion from cancer." *Ann N Y Acad Sci*. Dec. 2004;1030:179-86. cited by applicant

Seybold VS. "The role of peptides in central sensitization," *Handb Exp Pharmacol*. 2009; (194):451-91. cited by applicant

Shaw NE, et al. "The effect of monoclonal antibodies to calcitonin gene-related peptide (CGRP) on CGRP-induced vasodilatation in pig coronary artery rings," *Br J Pharmacol*. May 1992;106(1):196-8. cited by applicant

Sheets MD, et al. "Efficient construction of a large nonimmune phage antibody library: the production of high-affinity human single-chain antibodies to protein antigens," *Proc Natl Acad Sci U S A*. May 26, 1998;95(11):6157-62. cited by applicant

Sheftell FD, et al. "Naratriptan in the preventive treatment of refractory transformed migraine: a prospective pilot study," *Headache*. Nov.-Dec. 2005;45(10):1400-6. cited by applicant

Shen YT, et al. "Functional role of alpha-calcitonin gene-related peptide in the regulation of the cardiovascular system," *J Pharmacol Exp Ther*. Aug. 2001;298(2):551-8. cited by applicant

Shevel E. "The extracranial vascular theory of migraine—a great story confirmed by the facts," *Headache*. Mar. 2011;51(3):409-17. cited by applicant

Shields RL, et al. "High resolution mapping of the binding site on human IgG1 for Fc gamma RI, Fc gamma RII, Fc gamma RIIL, and FcRn and design of IgG1 variants with improved binding to the Fc gamma R," *J Biol Chem*. Mar. 2, 2001;276(9):6591-604. cited by applicant

Shulkes A, et al. "Production of calcitonin gene related peptide, calcitonin and PTH-related protein by a prostatic adenocarcinoma," *Clin Endocrinol (Oxf)*. May 1991;34(5):387-93. cited by applicant

Silberstein S, et al. "Botulinum toxin type A as a migraine preventive treatment. For the BOTOX Migraine Clinical Research Group," *Headache*. Jun. 2000;40(6):445-50. cited by applicant

Silberstein SD, "Practice parameter: evidence-based guidelines for migraine headache (an evidence-based review): report of the Quality Standards Subcommittee of the American Academy of Neurology," *Neurology*. Sep. 26, 2000;55(6):754-62. cited by applicant

Silberstein SD. "Emerging target-based paradigms to prevent and treat migraine," *Clin Pharmacol Ther*. Jan. 2013;93(1):78-85. cited by applicant

Silverman AJ, et al. "Mast cells migrate from blood to brain," *J Neurosci*. Jan. 1, 2000;20(1):401-8. cited by applicant

Simmons LC, et al. "Expression of full-length immunoglobulins in *Escherichia coli*: rapid and efficient production of aglycosylated antibodies," *J Immunol Methods*. May 1, 2002;263(1-2):133-47. cited by applicant

Sixt ML, et al. "Calcitonin gene-related peptide receptor antagonist olcegepant acts in the spinal trigeminal nucleus," *Brain*. Nov. 2009;132(Pt 11):3134-41. cited by applicant

Skofitsch G, et al. "Comparative immunohistochemical distribution of amylin-like and calcitonin gene related peptide like immunoreactivity in the rat central nervous system," *Can J Physiol Pharmacol*. Jul. 1995;73(7):945-56. cited by applicant

Smillie SJ, et al. "Calcitonin gene-related peptide (CGRP) and its role in hypertension," *Neuropeptides*. Apr. 2011;45(2):93-104. cited by applicant

Smith KA, et al. "Demystified . . . recombinant antibodies," *J Clin Pathol*. Sep. 2004;57(9):912-7. cited by applicant

Smith TW, et al. "Reversal of advanced digoxin intoxication with Fab fragments of digoxin-specific antibodies." *N Engl J Med*. Apr. 8, 1976;294(15):797-800. cited by applicant

Smith-Gill SJ, et al. "Contributions of immunoglobulin heavy and light chains to antibody specificity for lysozyme and two haptens." *J Immunol*. Dec. 15, 1987;139(12):4135-44. cited by applicant

Solomon S. "Major therapeutic advances in the past 25 years," *Headache*. Apr. 2007;47 Suppl 1:S20-2. cited by applicant

Song MK, et al. "Light chain of natural antibody plays a dominant role in protein antigen binding." *Biochem Biophys Res Commun*. Feb. 16, 2000;268(2):390-4. cited by applicant

Spetz AC, et al. "Momentary increase in plasma calcitonin gene-related peptide is involved in hot flashes in men treated with castration for carcinoma of the prostate," *J Urol*. Nov. 2001;166(5):1720-3. cited by applicant

Sprenger T, et al. "Migraine pathogenesis and state of pharmacological treatment options," *BMC Med*. Nov. 16, 2009;7:71. cited by applicant

Stensrud P, et al. "Comparative trial of Tenormin (atenolol) and Inderal (propranolol) in migraine," *Headache*. Jul. 1980;20(4):204-7. cited by applicant

Storer RJ, et al. "Calcitonin gene-related peptide (CGRP) modulates nociceptive trigeminovascular transmission in the cat," *Br J Pharmacol*. Aug. 2004;142(7):1171-81. cited by applicant

Stovner LJ, et al. "New drugs for migraine," *J Headache Pain*. Dec. 2009;10(6):395-406. cited by applicant

Strassman AM, et al. "On the origin of headaches," *Endeavour*. 1997;21(3):97-100. cited by applicant

Strassman AM, et al. "Response properties of dural nociceptors in relation to headache," *J Neurophysiol*. Mar. 2006;95(3):1298-306. cited by applicant

Subramanian KN, et al. "Safety, tolerance and pharmacokinetics of a humanized monoclonal antibody to respiratory syncytial virus in premature infants and infants with bronchopulmonary dysplasia," *MEDI-493 Study Group, Pediatr Infect Dis J*. Feb. 1998;17(2):110-5. cited by applicant

Tam SH, et al. "Abciximab (ReoPro, chimeric 7E3 Fab) demonstrates equivalent affinity and functional blockade of glycoprotein IIb/IIIa and alpha(v)beta3 integrins," *Circulation*. Sep. 15,

1998;98(11):1085-91. cited by applicant

Tamura M, et al. "Structural correlates of an anticarcinoma antibody: identification of specificity-determining residues (SDRs) and development of a minimally immunogenic antibody variant by retention of SDRs only." *J Immunol.* Feb. 1, 2000;164(3):1432-41. cited by applicant

Tan et al., "Demonstration of the neurotransmitter role of calcitonin gene-related peptides (CGRP) by immunoblockade with anti-CGRP monoclonal antibodies," *Br J Pharmacol.* Mar. 1994;111(3):703-10. cited by applicant

Tan KK, et al. "Calcitonin gene-related peptide as an endogenous vasodilator: immunoblockade studies in vivo with an anti-calcitonin gene-related peptide monoclonal antibody and its Fab' fragment." *Clin Sci (Lond).* Dec. 1995;89(6):565-73. cited by applicant

Tanaka H, et al. "Inhibition of calcitonin gene-related peptide (CGRP) has the potential to extend first-phase insulin secretion," *Exp Clin Endocrinol Diabetes.* May 2013;121(5):280-5. cited by applicant

Taylor AW, et al. "Suppression of nitric oxide generated by inflammatory macrophages by calcitonin gene-related peptide in aqueous humor," *Invest Ophthalmol Vis Sci.* Jul. 1998;39(8):1372-8. cited by applicant

Tedstone, et al. "The effect of islet amyloid polypeptide (amylin) and calcitonin gene-related peptide on glucose removal in the anaesthetized rat and on insulin secretion from rat pancreatic islets in vitro," *Biosci Rep.* Aug. 1990;10(4):339-45. cited by applicant

Tepper SJ, Bigal ME, et al. "Botulinum toxin type A in the treatment of refractory headache." In Olsen J, Silberstein SD, Tfelt-Hansen P, eds. *Preventive Pharmacotherapy of Headache Disorders.* Copenhagen: Oxford University Press, 2004, Chapter 20. cited by applicant

Tepper SJ, et al. "Botulinum neurotoxin type A in the preventive treatment of refractory headache: a review of 100 consecutive cases," *Headache.* Sep. 2004;44(8):794-800. cited by applicant

Tepper SJ, et al. "Clinical and preclinical rationale for CGRP-receptor antagonists in the treatment of migraine," *Headache.* Sep. 2008;48(8):1259-68. cited by applicant

Tepper SJ, et al. "Mechanisms of action of the 5-HT_{1B/1D} receptor agonists," *Arch Neurol.* Jul. 2002;59(7):1084-8. cited by applicant

Teva Pharmaceutical Industries Ltd., Press Release, "Teva to Acquire Labrys Biologics, Inc.," Jun. 3, 2014. 4 pages. cited by applicant

Tfelt-Hansen P, et al. "Effervescent metoclopramide and aspirin (Migravess) versus effervescent aspirin or placebo for migraine attacks: a double-blind study," *Cephalalgia.* Jun. 1984;4(2):107-11. cited by applicant

Tfelt-Hansen P, et al. "Possible site of action of CGRP antagonists in migraine," *Cephalalgia.* Apr. 2011;31(6):748-50. cited by applicant

Tfelt-Hansen PC. "Verisimilitude (or "truthlikeness") as an alternative to pro and cons: migraine and cluster headache mechanisms," *J Headache Pain.* Oct. 2010;11(5):379-89. cited by applicant

Theoharides TC, et al. "The role of mast cells in migraine pathophysiology," *Brain Res Brain Res Rev.* Jul. 2005;49(1):65-76. cited by applicant

Thomas TC, et al. "Inhibition of complement activity by humanized anti-C5 antibody and single-chain Fv," *Mol Immunol.* Dec. 1996;33(17-18):1389-401. cited by applicant

Tjen-A-Looi S, et al. "CGRP and somatostatin modulate chronic hypoxic pulmonary hypertension," *Am J Physiol.* Sep. 1992;263(3 Pt 2):H681-90. cited by applicant

Toda M, et al. "Neuronal system-dependent facilitation of tumor angiogenesis and tumor growth by calcitonin gene-related peptide," *Proc Natl Acad Sci U S A.* Sep. 9, 2008;105(36):13550-5. cited by applicant

Todd J. Schwedt et al: "14th International Headache Congress: Basic Science Highlights", *Headache*, vol. 50, No. 3, Mar. 1, 2010 (Mar. 1, 2010), pp. 520-526. cited by applicant

Tokuda Y, et al. "Dose escalation and pharmacokinetic study of a humanized anti-HER2 monoclonal antibody in patients with HER2/neu-overexpressing metastatic breast cancer," *Br J*

Cancer. Dec. 1999;81(8):1419-25. cited by applicant

Tsujikawa K, et al. "Hypertension and dysregulated proinflammatory cytokine production in receptor activity-modifying protein 1-deficient mice," *Proc Natl Acad Sci U S A*. Oct. 16, 2007;104(42):16702-7. cited by applicant

Turner LC, et al. "A neural shift theory of migraine," *Neuroepidemiology*. 1993;12(4):249-50. cited by applicant

Tvedskov JF, et al. "No increase of calcitonin gene-related peptide in jugular blood during migraine." *Ann Neurol*. Oct. 2005;58(4):561-8. cited by applicant

Tzabazis AZ, et al. "Antihyperalgesic effect of a recombinant herpes virus encoding antisense for calcitonin gene-related peptide." *Anesthesiology*. Jun. 2007;106(6):1196-203. cited by applicant

Uhr M, et al. "Penetration of endogenous steroid hormones corticosterone, cortisol, aldosterone and progesterone into the brain is enhanced in mice deficient for both *mdr1a* and *mdr1b* P-glycoproteins," *J Neuroendocrinol*. Sep. 2002;14(9):753-9. cited by applicant

Unger J. "Migraine headaches: a historical prospective, a glimpse into the future, and migraine epidemiology," *Dis Mon*. Oct. 2006;52(10):367-84. cited by applicant

Vajdos FF, et al. "Comprehensive functional maps of the antigen-binding site of an anti- ErbB2 antibody obtained with shotgun scanning mutagenesis." *J Mol Biol*. Jul. 5, 2002;320(2):415-28. cited by applicant

Van der Schueren BJ, et al. " Calcitonin gene-related peptide8-37 antagonizes capsaicin-induced vasodilation in the skin: evaluation of a human in vivo pharmacodynamic model," *J Pharmacol Exp Ther*. Apr. 2008;325(1):248-55. cited by applicant

Van Rossum D, et al. "Neuroanatomical localization, pharmacological characterization and functions of CGRP, related peptides and their receptors," *Neurosci Biobehav Rev*. Sep. 1997;21(5):649-78. cited by applicant

Vater A, et al. "Short bioactive Spiegelmers to migraine-associated calcitonin gene-related peptide rapidly identified by a novel approach: tailored-SELEX." *Nucleic Acids Res*. Nov. 1, 2003;31(21):e130. cited by applicant

Vaughan TJ, et al. "Human antibodies with sub-nanomolar affinities isolated from a large non-immunized phage display library," *Nat Biotechnol*. Mar. 1996;14(3):309-14. cited by applicant

Villalón CM, et al. "The role of CGRP in the pathophysiology of migraine and efficacy of CGRP receptor antagonists as acute antimigraine drugs," *Pharmacol Ther*. Dec. 2009;124(3):309-23. cited by applicant

Vincent A, et al. "Molecular targets for autoimmune and genetic disorders of neuromuscular transmission," *Eur J Biochem*. Dec. 2000;267(23):6717-28. cited by applicant

Vogler B, et al. "Role of melatonin in the pathophysiology of migraine: implications for treatment," *CNS Drugs*. 2006;20(5):343-50. cited by applicant

Volcy M, et al. "Botulinum toxin A for the treatment of greater occipital neuralgia and trigeminal neuralgia: a case report with pathophysiological considerations," *Cephalalgia*. Mar. 2006;26(3):336-40. cited by applicant

Von Mehren M, et al. "Monoclonal antibody therapy for cancer," *Annu Rev Med*. 2003;54:343-69. cited by applicant

Wachter C, et al. "Visceral vasodilatation and somatic vasoconstriction evoked by acid challenge of the rat gastric mucosa: diversity of mechanisms," *J Physiol*. Jul. 15, 1995;486 (Pt 2):505-16. cited by applicant

Wacnik PW, et al. "Tumor-induced mechanical hyperalgesia involves CGRP receptors and altered innervation and vascularization of DsRed2 fluorescent hindpaw tumors." *Pain*. May 2005;115(1-2):95-106. cited by applicant

Waeber C, et al. "Migraine as an inflammatory disorder." *Neurology*. May 24, 2005;64(10 Suppl 2):S9-15. cited by applicant

Walker CS, et al. "Mice lacking the neuropeptide alpha-calcitonin gene-related peptide are

protected against diet-induced obesity,” *Endocrinology*. Sep. 2010;151(9):4257-69. cited by applicant

Walker CS, et al. “Regulation of signal transduction by calcitonin gene-related peptide receptors,” *Trends Pharmacol Sci*. Oct. 2010;31(10):476-83. cited by applicant

Ward ES, et al. “Binding activities of a repertoire of single immunoglobulin variable domains secreted from *Escherichia coli*.” *Nature*. Oct. 12, 1989;341(6242):544-6. cited by applicant

Weir AN, et al. “Formatting antibody fragments to mediate specific therapeutic functions,” *Biochem Soc Trans*. Aug. 2002;30(4):512-6. cited by applicant

Welch KM, et al. “Mismatch in how oestrogen modulates molecular and neuronal function may explain menstrual migraine,” *Neurol Sci*. May 2006;27 Suppl 2:S190-2. cited by applicant

Werther WA, et al. “Humanization of an anti-lymphocyte function-associated antigen (LFA)-1 monoclonal antibody and reengineering of the humanized antibody for binding to rhesus LFA-1,” *J Immunol*. Dec. 1, 1996;157(11):4986-95. cited by applicant

Wick EC, et al. “Transient receptor potential vanilloid 1, calcitonin gene-related peptide, and substance P mediate nociception in acute pancreatitis.” *Am J Physiol Gastrointest Liver Physiol*. May 2006;290(5):G959-69. Epub Jan. 6, 2006. cited by applicant

Willats WG. “Phage display: practicalities and prospects,” *Plant Mol Biol*. Dec. 2002;50(6):837-54. cited by applicant

Williamson DJ, et al. “Intravital microscope studies on the effects of neurokinin agonists and calcitonin gene-related peptide on dural vessel diameter in the anaesthetized rat,” *Cephalalgia*. Jun. 1997;17(4):518-24. cited by applicant

Williamson DJ, et al. “Neurogenic inflammation in the context of migraine,” *Microsc Res Tech*. May 1, 2001;53(3):167-78. cited by applicant

Williamson DJ, et al. “Sumatriptan inhibits neurogenic vasodilation of dural blood vessels in the anaesthetized rat—intravital microscope studies,” *Cephalalgia*. Jun. 1997;17(4):525-31. cited by applicant

Williamson DJ, et al. “The anti-migraine 5-HT(1B/1D) agonist rizatriptan inhibits neurogenic dural vasodilation in anaesthetized guinea-pigs,” *Br J Pharmacol*. Aug. 2001;133(7):1029-34. cited by applicant

Williamson DJ, et al. “The novel anti-migraine agent rizatriptan inhibits neurogenic dural vasodilation and extravasation,” *Eur J Pharmacol*. Jun. 5, 1997;328(1):61-4. cited by applicant

Wimalawansa SJ, et al. “Comparative study of distribution and biochemical characterization of brain calcitonin gene-related peptide receptors in five different species,” *Neuroscience*. May 1993;54(2):513-9. cited by applicant

Wimalawansa SJ, et al. “Validation, role in perioperative assessment, and clinical applications of an immunoradiometric assay for human calcitonin,” *Peptides*. 1995;16(2):307-12. cited by applicant

Wimalawansa SJ. “Amylin, calcitonin gene-related peptide, calcitonin, and adrenomedullin: a peptide superfamily,” *Crit Rev Neurobiol*. 1997;11(2-3):167-239. cited by applicant

Wimalawansa SJ. “Calcitonin gene-related peptide and its receptors: molecular genetics, physiology, pathophysiology, and therapeutic potentials,” *Endocr Rev*. Oct. 1996;17(5):533-85. cited by applicant

Wimalawansa SJ. “Effects of in vivo stimulation on molecular forms of circulatory calcitonin and calcitonin gene-related peptide in man,” *Mol Cell Endocrinol*. May 28, 1990;71(1):13-9. cited by applicant

Winkler K, et al. “Changing the antigen binding specificity by single point mutations of an anti-p24 (HIV-1) antibody,” *J Immunol*. Oct. 15, 2000;165(8):4505-14. cited by applicant

Winter G, et al. “Making antibodies by phage display technology,” *Annu Rev Immunol*. 1994;12:433-55. cited by applicant

Wong G, et al. “Safety and tolerability of LBR-101, a humanized monoclonal antibody that blocks the binding of CGRP to its receptor,” *Labrys Biologics Poster*, 1 page, 2013 International Headache

Congress. cited by applicant

Wong HC, et al. "Monoclonal antibody to rat alpha-CGRP: production, characterization, and in vivo immunoneutralization activity." *Hybridoma*. Feb. 1993;12(1):93-106. cited by applicant

Wong HC, et al. "Preparation of a monoclonal antibody to rat alpha-CGRP for in vivo immunoneutralization of peptides." *Ann N Y Acad Sci*. Jun. 30, 1992;657:525-7. cited by applicant

Wu D, et al. "Development and potential of non-peptide antagonists for calcitonin-gene-related peptide (CGRP) receptors: evidence for CGRP receptor heterogeneity," *Biochem Soc Trans*. Aug. 2002;30(4):468-73. cited by applicant

Wu H, et al. "Humanization of a murine monoclonal antibody by simultaneous optimization of framework and CDR residues." *J Mol Biol*. Nov. 19, 1999;294(1):151-62. cited by applicant

Wu H, et al. "Humanized antibodies and their applications," *Methods*. May 2005;36(1):1-2. cited by applicant

Wyon Y, et al. "Postmenopausal women with vasomotor symptoms have increased urinary excretion of calcitonin gene-related peptide," *Maturitas*. Nov. 16, 1998;30(3):289-94. cited by applicant

Xu, F.T. Study on the Mechanism of SP and CGRP in the Chronic Pain and Knee Joint. Master Thesis. Guangxi Medical University. May 2005. (In Chinese with English abstract). cited by applicant

Yallampalli C, et al. "Calcitonin gene-related peptide in pregnancy and its emerging receptor heterogeneity," *Trends Endocrinol Metab*. Aug. 2002;13(6):263-9. cited by applicant

Yoshikawa R, et al. "Suppression of ovalbumin-induced allergic diarrhea by diminished intestinal peristalsis in RAMP1-deficient mice," *Biochem Biophys Res Commun*. Jul. 8, 2011;410(3):389-93. cited by applicant

Yu LC, et al. "Roles of calcitonin gene-related peptide and its receptors in pain-related behavioral responses in the central nervous system," *Neurosci Biobehav Rev*. Sep. 2009;33(8):1185-91. cited by applicant

Zeller J, et al. "CGRP function-blocking antibodies inhibit neurogenic vasodilatation without affecting heart rate or arterial blood pressure in the rat." *Br J Pharmacol*. Dec. 2008;155(7):1093-103. doi: 10.1038/bjp.2008.334. Epub Sep. 8, 2008. cited by applicant

Zhang L, et al. "Arthritic calcitonin/alpha calcitonin gene-related peptide knockout mice have reduced nociceptive hypersensitivity," *Pain*. Jan. 2001;89(2-3):265-73. cited by applicant

Zhang M, et al. "Rheumatoid factor specificity of a VH3-encoded antibody is dependent on the heavy chain CDR3 region and is independent of protein A binding." *J Immunol*. Sep. 1, 1998;161(5):2284-9. cited by applicant

Zhuang X, et al. "Brain mast cell degranulation regulates blood-brain barrier," *J Neurobiol*. Dec. 1996;31(4):393-403. cited by applicant

Zittel et al., "Role of spinal afferents and calcitonin gene-related peptide in the postoperative gastric ileus in anesthetized rats," *Ann Surg*. Jan. 1994;219(1):79-87. cited by applicant

Zittel TT, et al. "Calcitonin gene-related peptide and spinal afferents partly mediate postoperative colonic ileus in the rat," *Surgery*. May 1998;123(5):518-27. cited by applicant

Zuckier LS, et al. "Chimeric human-mouse IgG antibodies with shuffled constant region exons demonstrate that multiple domains contribute to in vivo half-life," *Cancer Res*. Sep. 1, 1998;58(17):3905-8. cited by applicant

Misura, et al. "The Eptinezumab: CGRP Complex Structure and Characterization of the Ligand Binding Interface," poster Presented at the American Headache Society (AHS) 61.SUP.st .Annual Scientific Meeting Jul. 11-14, 2019. cited by applicant

Rita Costa A, Elisa Rodrigues M, Henriques M, Azeredo J, Oliveira R. Guidelines to cell engineering for monoclonal antibody production. *Eur J Pharm Biopharm*. 2010;74(2):127-138. doi:10.1016/j.ejpb.2009.10.002. cited by applicant

Potgieter TI, Cukan M, Drummond JE, et al. Production of monoclonal antibodies by

glycoengineered *Pichia pastoris*. *J Biotechnol*. 2009;139(4):318-325. doi:10.1016/j.jbiotec.2008.12.015. cited by applicant

Trill JJ, Shatzman AR, Ganguly S. Production of monoclonal antibodies in COS and CHO cells. *Curr Opin Biotechnol*. 1995;6(5):553-560. doi:10.1016/0958-1669(95)80092-1. cited by applicant

Brandes, Jan Lewis, et al. "Effects of fremanezumab on the use of acute headache medication and associated symptoms of migraine in patients with episodic migraine," *Cephalalgia* 40.5 (2020): 470-477. cited by applicant

Munjal, Sagar, et al. "Most Bothersome Associated Migraine Symptom: Results from 2017 Migraine in America Symptoms and Treatment (MAST) Study (P3. 10-017)." *Neurology* 92.15_supplement (2019): P3-10. cited by applicant

Silberstein, Stephen D., et al. "Fremanezumab for the preventive treatment of chronic migraine." *New England Journal of Medicine* 377.22 (2017): 2113-2122. cited by applicant

Database Embase [online] Jan. 1, 2018 (Jan. 1, 2018), Silberstein S: "The impact of fremanezumab on medication overuse in patients with chronic migraine2018", Database accession No. EMB-624431011 *. cited by applicant

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Background/Summary

SEQUENCE LISTING DISCLOSURE (1) This application is a continuation of U.S. application Ser. No. 16/860,239 filed Apr. 28, 2020, which claims priority to U.S. Provisional Appl. No. 63/005,950, filed Apr. 6, 2020, which is hereby incorporated by reference in its entirety.

SEQUENCE LISTING DISCLOSURE

(1) This application includes as part of its disclosure an electronic sequence listing text file named "1143257o009402.xml", having a size of 771,211 bytes and created on Mar. 13, 2023, which is hereby incorporated by reference in its entirety.

SEQUENCES NOT PERMITTED TO BE ENTERED IN ST.26 XML FILE DUE TO SEQUENCE LENGTH

(2) Table A below lists sequences present in the U.S. priority application Ser. No. 16/860,239 and 63/244,466 (identified above, which are both herein incorporated by reference in their entirety) but cannot be included in the "1143257o009402.xml" file submitted herewith due to the length of the sequences.

(3) TABLE-US-00001

TABLE A	Previous Sequence SEQ ID NO:	#	Length	Type	Organism	Other Information
Gly Asp Ile 8	3	Protein	Artificial	Engineered antibody sequence	ggggacatc	18 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 48	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 58 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 88	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 98 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 128	3	Protein	Artificial	Engineered antibody sequence	gggacatc 138 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 168	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 178 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 208	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 218 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 248	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 258 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 288	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 298 9 DNA
Artificial Engineered antibody sequence	Gly Asp Ile 328	3	Protein	Artificial	Engineered antibody sequence	ggggacatc 338 9 DNA
Artificial Engineered antibody sequence						

sequence Gly Asp Ile 368 3 Protein Artificial Engineered antibody sequence gggacatc 378 9 DNA Artificial Engineered antibody sequence Gly Asp Ile 408 3 Protein Artificial Engineered antibody sequence ggcgacatc 418 9 DNA Artificial Engineered antibody sequence Gly Asp Ile 448 3 Protein Artificial Engineered antibody sequence ggggacatc 458 9 DNA Artificial Engineered antibody sequence Gly Asp Ile 528 3 Protein Artificial Engineered antibody sequence ggggacatc 538 9 DNA Artificial Engineered antibody sequence

BACKGROUND OF THE INVENTION

Field of the Invention

(4) This invention pertains to methods of treatment of most bothersome symptom associated with migraine, using antibodies and fragments thereof (including Fab fragments) that specifically bind to human Calcitonin Gene Related Peptide (hereinafter “CGRP”).

Description of Related Art

(5) Calcitonin Gene Related Peptide (CGRP) is produced as a multifunctional neuropeptide of 37 amino acids in length. Two forms of CGRP, the CGRP-alpha and CGRP-beta forms, exist in humans and have similar activities. CGRP-alpha and CGRP-beta differ by three amino acids in humans, and are derived from different genes. CGRP is released from numerous tissues such as trigeminal nerves, which when activated release neuropeptides within the meninges, mediating neurogenic inflammation that is characterized by vasodilation, vessel leakage, and mast-cell degradation. Durham, P. L., *New Eng. J. Med.*, 350 (11):1073-75 (2004). Biological effects of CGRP are mediated via the CGRP receptor (CGRP-R), which consists of a seven-transmembrane component, in conjunction with receptor-associated membrane protein (RAMP). CGRP-R further requires the activity of the receptor component protein (RCP), which is essential for an efficient coupling to adenylate cyclase through G proteins and the production of cAMP. Doods, H., *Curr. Op. Invest. Drugs*, 2(9):1261-68 (2001).

(6) Migraines are neurovascular disorder affecting approximately 10% of the adult population in the U.S., and are typically accompanied by intense headaches. CGRP is believed to play a prominent role in the development of migraines. In fact, several companies, i.e., Amgen, Eli Lilly, Teva and Alder Biopharmaceuticals (recently acquired by Lundbeck A/S) have developed anti-CGRP and anti-CGRP-R antibodies for use in treating or preventing migraine headaches. The present assignee has previously filed patent applications related to anti-CGRP antibodies and uses thereof including published PCT Application WO/2012/162243 filed May 21, 2012 entitled “ANTI-CGRP COMPOSITIONS AND USE THEREOF”, published PCT Application WO/2012/162257 filed May 21, 2012, entitled “USE OF ANTI-CGRP ANTIBODIES AND ANTIBODY FRAGMENTS TO PREVENT OR INHIBIT PHOTOPHOBIA OR LIGHT AVERSION IN SUBJECTS IN NEED THEREOF, ESPECIALLY MIGRAINE SUFFERERS” published PCT Application WO/2012/162253, filed May 21, 2012, entitled “USE OF ANTI-CGRP OR ANTI-CGRP-R ANTIBODIES OR ANTIBODY FRAGMENTS TO TREAT OR PREVENT CHRONIC AND ACUTE FORMS OF DIARRHEA” and published PCT Application WO/2015/003122, filed Jul. 3, 2014, entitled “REGULATION OF GLUCOSE METABOLISM USING ANTI-CGRP ANTIBODIES” all of which applications are incorporated by reference in their entirety.

BRIEF SUMMARY

(7) The present disclosure provides methods of treatment of most bothersome symptom (MBS) associated with migraine in patient suffering from chronic migraine, comprising administering to a patient in need an effective amount of at least one anti-CGRP antibody or antibody fragment thereof or an anti-CGRP-R antibody or antibody fragment thereof or one or more formulations comprising said antibody or antibody fragment as disclosed herein. Said antibody treatment may be initiated in the interictal period, i.e. in between migraine attacks or in the ictal phase, i.e. during the migraine episode. Said migraine may comprise e.g. chronic migraine or episodic migraine, in a specific aspect of the present invention the patient suffers from chronic migraine. In the present

invention, said anti-CGRP antibody or antibody fragment is denoted Ab6. Ab6 is an anti-CGRP antibody or antibody fragment thereof having the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208; or having the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively. Said anti-CGRP antibody may comprise the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202. Said anti-CGRP antibody may comprise the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212. Said anti-CGRP antibody may comprise the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. Said anti-CGRP antibody may comprise the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567. Said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells. Said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions. Any of the aforementioned anti-CGRP antibodies or antibody fragments, preferably Ab6, may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The administered dosage of said antibody may be between about 100 mg and about 300 mg, such as about 100 mg, about 300 mg, 100 mg, or 300 mg. The dosage may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL

(8) Said patient may exhibit less than 25 headache days per month, less than 20 headache days per month, less than 15 headache days per month, or less than 10 headache days per month. For example, said patient may exhibit less than 14 headache days, less than 13 headache days, less than 12 headache days, less than headache 11 days, less than 10 headache days, less than 9 headache days, less than 8 headache days, less than 7 headache days, or less than 6 headache days per month. Said patient may exhibit between 2-15 headache days, e.g., 3-14 headache days, 4-13 headache days, 5-12 headache days, 6-11 headache days, or 7-10 headache days/month.

(9) Said patient may exhibit less than 10 migraines per month, such as between 1-9 migraines per month, such as between 2-8 migraines per month, between 3-7 migraine per month, between 4-6 migraine per month, or about 5 migraines per month. Said patient may exhibit fewer than 1 migraine per month on average, e.g., on average one migraine every 2 months, one every 3 months, one every 4 or 6 months, or intermediate values such as 2 every 3 months, etc. Said migraine may be diagnosed in accord with the ICHD-3 guidelines.

(10) In addition to headache and associated symptoms as described in the diagnostic criteria of the International Classification of Headache Disorders (ICHD-3) for migraine with or without aura, migraine patients experience a variety of autonomic, cognitive, sensory and motor symptoms

during migraine, these symptoms are experienced uniquely by individual patients. In the present invention, the patients were allowed to self-identify a specific symptom associated with chronic migraine that they considered to be most bothersome. In the present application these symptoms will be referred to as the most bothersome symptom (MBS) associated with migraine. In the present invention the patient could identify their MBS without limitation, which provides a unique patient-centered approach for identifying and measuring the efficacy of antibodies of the invention as treatment of these most bothersome migraine-associated symptoms and hence is expected to have a meaningful impact on the patient's ability to function during migraine. Although nausea, vomiting, photophobia, and phonophobia are migraine-associated symptoms included in ICDH-3 diagnostic criteria, many other symptoms may be observed to occur prior to, after, and even between days with diagnosable migraine. Over the duration of a migraine attack, these can include cognitive symptoms (e.g. memory, executive function, attention deficit), affective symptoms (e.g. mood changes, depression, anxiety, irritability), other sensory symptoms (e.g. osmophobia, taste abnormalities), as well as blurry vision, nasal congestion, rhinorrhea, lacrimation, sweating, ptosis, yawning, polyuria, abdominal cramps, diarrhea, dizziness, and neck pain. The MBS associated with migraine reported by the patients enrolled in the clinical trial described in Example 2 is summarized in Table 1. Although nausea/vomiting, photophobia, and phonophobia were common in the patient population in Example 2, less than half of these patients named one of these 3 symptoms included in ICDH-3 diagnostic criteria as their patient-identified MBS.

(11) Migraine is a complex disorder of the brain associated with multifaceted symptomatology yet expressed in a personalized unique manner. Often persisting over multiple days, the peri-ictal period of migraine can be classified into four distinct phases—prodrome/premonitory, preictal/aura, ictal/headache, postdrome/postictal—with overlapping symptoms occurring during each phase of migraine. The various types and timing of MBS across the course of the migraine is illustrated in FIG. 15. It is highly relevant to assess MBS in migraine patients during clinical trials, since it is recognized that headache pain alone is not considered sufficient to adequately eliminate the impact of migraine on the patient's daily living and health status. The reduction in mean monthly migraine days (MMDs) or a similar endpoint in clinical trials do not fully capture the burden of migraine and the associated symptoms that are affected by therapeutic intervention. The inventors of the present invention found that in addition to reducing MMDs Ab6, an anti-CGRP antibody, was also effective in improving MBS in migraine patients. Improvements in these symptoms associated with treatment were correlated with improved patients' perception of disease status and indirectly with satisfaction with treatment response. It is known that migraine patients often continue to seek treatment for their migraine because of the burden of their MBS, thus supporting the clinical value of treating both the primary migraine pathology and the MBS associated with said migraine.

(12) The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to improve the MBS associated with migraine in patients suffering from migraine, such as chronic or episodic migraine. The MBS parameter rates the patient's assessment of change (improvement or worsening since the start of the study) in this symptom.

(13) The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to improve the patient's global impression of change (PGIC) associated with migraine treatment in patients impacted by migraine, such as chronic or episodic migraine. The patient's global impression of change (PGIC) associated with migraine parameter comprises a single question concerning the patient's impression of the overall change (improvement or worsening since the start of the study) in their disease status evaluated on a 7 point Likert scale anchored by very much improved and very much worse.

(14) The present invention provides anti-CGRP antibodies or antibody fragments thereof, which are able to reduce MMDs as well as improve the patient's most bothersome symptom (MBS) associated with migraine in a manner that is highly correlated with positive change in the patient's global impression of change (PGIC) of migraine treatment. This dual action constitutes an

improved treatment option for patient suffering from migraine, which goes beyond treating the migraine headache, and provides treatment for the collective migraine burden experienced by the patient comprising both migraine headache as well as MBS associated with migraine.

(15) The present invention provides methods of improving most bothersome symptom (MBS) associated with migraine, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(16) The present invention provides methods of improving patient global impression of change (PGIC), comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(17) In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(18) In another aspect, the invention provides methods of improving patient global impression of change (PGIC) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(19) In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and patient global impression of change (PGIC) associated with migraine, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(20) In another aspect, the invention provides methods of improving most bothersome symptom (MBS) associated with migraine and/or patient global impression of change (PGIC) associated with migraine and simultaneously reduce the MMDs, comprising intravenously administering to a patient in need thereof between about 100 mg and about 300 mg of an anti-CGRP antibody, wherein said anti-CGRP antibody preferably comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or 566.

(21) In some exemplary embodiments the dosage of said anti-CGRP antibody may be 100 mg.

(22) In other exemplary embodiments the dosage of said anti-CGRP antibody may be 300 mg.

(23) The method may further comprise intravenously administering 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

(24) The method may further comprise intravenously administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

(25) The antibody may be provided or administered in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

(26) Prior to first dosage, the patient may exhibit between about 10 and about 22 migraine days per month, such as between about 13 and about 19 migraine days per month, such as about 16 migraine days per month.

(27) Prior to first dosage, the patient may exhibit between about 14 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about

21 headache days per month.

(28) Said patient may have been diagnosed with migraine at least 10 years prior to said first dosage, such as at least 15 years prior to said first dosage, such as at least 18 or at least 19 years prior to said first dosage.

(29) Said patient may have been diagnosed with chronic migraine at least 5 years prior to said first dosage, such as at least 8 years prior to said first dosage, such as at least 11 or at least 12 years prior to said first dosage.

(30) The patient may have a headache when administered the anti-CGRP antibody or fragments thereof of the invention.

(31) The patient may have a migraine, such as a migraine with aura, when administered anti-CGRP antibody or fragments thereof of the invention.

(32) Said patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(33) Said patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(34) Said patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(35) Said patient may have a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(36) Said patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(37) Said patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said first dose relative to the baseline number of migraine days experienced by that patient prior to said first dose.

(38) Said patient may experience an improvement in their MBS associated with migraine in the one month period after being administered said first dose measured as the change from the baseline MBS.

(39) Said patient may experience an improvement in their MBS associated with migraine in the 3 month period after being administered said first dose measured as the change from the baseline MBS.

(40) Said patient may experience an improvement in their MBS associated with migraine in the 6 month period after being administered said first dose measured as the change from the baseline MBS.

(41) Said patient may experience an improvement in their PGIC associated with migraine in the one month period after being administered said first dose measured as the change from the baseline.

(42) Said patient may experience an improvement in their PGIC associated with migraine in the 3 month period after being administered said first dose measured as the change from the baseline.

(43) Said patient may experience an improvement in their PGIC associated with migraine in the 6 month period after being administered said first dose measured as the change from the baseline.

(44) The method may further comprise administering, e.g., intravenously, a second dose of an anti-CGRP antibody of the invention to said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said first dose.

(45) Said first dose may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP

antibody.

(46) Prior to said administration, the patient may exhibit between about 15 and about 30 migraine days per month, such as between about 16 and about 28 migraine days per month, such as between about 17 and about 26 migraine days per month, such as about 16 migraine days per month.

(47) Prior to said administration, the patient may exhibit between about 15 and about 27 headache days per month, such as between about 17 and about 24 headache days per month, such as about 20 or about 21 headache days per month.

(48) Said patient may have been diagnosed with migraine at least 10 years prior to said administration, such as at least 15 years prior to said administration, such as at least 18 or at least 19 years prior to said administration.

(49) Said patient may have been diagnosed with chronic migraine at least 5 years prior to said administration, such as at least 8 years prior to said administration, such as at least 11 or at least 12 years prior to said administration.

(50) Said patient may have a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(51) Said patient may have a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(52) Said patient may have a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(53) Said patient may have a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(54) Said patient may have a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(55) Said patient may have a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to said administration.

(56) The method may further comprise administering, e.g., intravenously, a second dose of said anti-CGRP antibody to said patient within about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months, after said administration.

(57) Said administration may comprise about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

(58) Said anti-CGRP antibody may be aglycosylated or if glycosylated only may contain only mannose residues.

(59) Said anti-CGRP antibody may consist of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566. Said anti-CGRP antibody may consist of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(60) In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain of SEQ ID NO: 222 and/or the variable heavy chain of SEQ ID NO: 202. In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the variable light chain encoded by SEQ ID NO: 232 and/or the variable heavy chain encoded by SEQ ID NO: 212.

(61) In some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain of SEQ ID NO: 221 and/or the heavy chain of SEQ ID NO: 201 or SEQ ID NO: 566. In

some embodiments, said anti-human CGRP antibody or antibody fragment comprises the light chain encoded by SEQ ID NO: 231 and/or the heavy chain encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(62) In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the VL polypeptide of SEQ ID NO: 222 and the VH polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

(63) In some embodiments, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

(64) In some embodiments any of the aforementioned anti-CGRP antibodies or antibody fragments may be comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8. The antibody or fragment may be administered by different means, e.g., intravenously, e.g., in a saline solution such as 0.9% sodium chloride in a suitable volume, such as 100 mL.

(65) In some embodiments, about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

(66) In other embodiments, about 100 mg of said anti-CGRP antibody or antibody fragment is administered.

(67) In other embodiments, about 300 mg of said anti-CGRP antibody or antibody fragment is administered, e.g., intravenously.

(68) In exemplary embodiments, the anti-human CGRP antibody or antibody fragment is administered, e.g., intravenously at a frequency which is at most every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks, wherein the antibody dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 3 months or every 12 weeks. The phrase “the antibody dosage is administered in a single formulation or divided into different formulations” refers to the administration of the recited amount of antibody within a relatively short period of time, e.g., within a period of several hours, e.g., 1 to 8 hours, about one day, within about two days, or within about one week, which may be by the same or different routes (e.g., i.v., i.m., and/or s.c.), sites of administration. The term “different formulations” in this context refers to antibody dosages that are administered at different times and/or at different sites and/or different routes, irrespective of whether the dosages are the same or different with respect to the chemical composition of the pharmaceutical formulation in which each dosage is administered; for example, the concentration, excipients, carriers, pH, and the like may be the same or different between the different administered dosages.

(69) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 8 weeks or every 2 months.

(70) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are

administered at a frequency of approximately every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks or every 3 months.

(71) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 16 weeks or every 4 months.

(72) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 20 weeks or every 5 months.

(73) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 24 weeks or every 6 months.

(74) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 28 weeks or every 7 months.

(75) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 32 weeks or every 8 months.

(76) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 36 weeks or every 9 months.

(77) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 40 weeks or every 8 months.

(78) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 44 weeks or every 9 months.

(79) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 48 weeks or every 10 months.

(80) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 52 weeks or every 11 months.

(81) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 56 weeks or every 12 months.

(82) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 15-18 months.

(83) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment dosage is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 18-21 months.

(84) In other exemplary embodiments, the anti-human CGRP antibody dosage or antibody fragment used in the afore-mentioned methods is administered in a single formulation or divided into different formulations which are administered at a frequency of approximately every 2 years.

(85) In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods is administered systemically.

(86) In other exemplary embodiments, the anti-human CGRP antibody or antibody fragment used in the afore-mentioned methods is administered by a mode of administration is selected from intravenous, intramuscular, intravenous, intrathecal, intracranial, topical, intranasal, and oral. In a preferred embodiment, the anti-human CGRP antibody or antibody fragment used in the afore-

mentioned methods is administered intravenously.

(87) In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 10 days.

(88) In other exemplary embodiments, the anti-human CGRP antibody has an in vivo half-life of at least 15 days.

(89) In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20 days.

(90) In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods has an in vivo half-life of at least 20-30 days.

(91) In other exemplary embodiments, the anti-human CGRP antibody is administered at a dosage of between about 100 mg and about 300 mg has an in vivo half-life of $\pm 20\%$ of at least about (284 $\pm$ 44 hours).

(92) In other exemplary embodiments, the anti-human CGRP antibody used in the afore-mentioned methods binds to human α - and β -CGRP.

(93) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 30 days after antibody administration.

(94) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 60 days after antibody administration.

(95) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in inhibition of vasodilation induced by topically applied capsaicin at least 90 days after antibody administration.

(96) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 120 days after antibody administration.

(97) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 150 days after antibody administration.

(98) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin at least 180 days after antibody administration.

(99) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in the inhibition of vasodilation induced by topically applied capsaicin more than 180 days after antibody administration.

(100) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity, within 5% of the maximal response (I_{max}) (as compared to lower antibody doses).

(101) In other exemplary embodiments, the administered anti-human CGRP antibody dosage results in sustained pharmacodynamic (PK) activity which is maintained for at least 2-3 months after antibody administration, wherein PK analysis of the anti-human CGRP antibody is derived from plasma concentrations.

(102) In other exemplary embodiments, the administered anti-human CGRP antibody dosage is between about 100 mg and about 300 mg or more which is administered no more frequently than every 2 months.

(103) The present invention is additionally directed to the use of specific antibodies and fragments thereof having binding specificity for CGRP, in particular antibodies having desired epitopic specificity, high affinity or avidity and/or functional properties. A preferred embodiment of the invention is directed to usage of chimeric or humanized antibodies and fragments thereof (including Fab fragments) capable of binding to CGRP and/or inhibiting the biological activities

mediated by the binding of CGRP to the CGRP receptor (“CGRP-R”) e.g., wherein such antibodies optionally are derived from recombinant cells engineered to express same, optionally yeast or mammalian cells, further optionally *Pichia pastoris* and CHO cells.

(104) In another preferred embodiment of the invention, full length antibodies and Fab fragments thereof are contemplated that inhibit the CGRP-alpha-, CGRP-beta-, and rat CGRP-driven production of cAMP. In a further preferred embodiment of the invention, full length and Fab fragments thereof are contemplated that reduce vasodilation in a recipient following administration.

(105) The invention also contemplates usage of conjugates of anti-CGRP antibodies and binding fragments thereof conjugated to one or more functional or detectable moieties. The invention also contemplates usage of chimeric or humanized anti-CGRP or anti-CGRP/CGRP-R complex antibodies and binding fragments thereof. In one embodiment, binding fragments include, but are not limited to, Fab, Fab', F(ab').sub.2, Fv, scFv fragments, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 provide the polypeptide sequences of the full-length heavy chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

(2) FIG. 2 provide the polypeptide sequences of the full-length light chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and constant region sequences delimited.

(3) FIGS. 3A and 3B provide exemplary polynucleotide sequences encoding the full-length heavy chain for antibody Ab6 with framework regions (FR), complementarity determining regions (CDRs), and variable region coding sequences delimited.

(4) FIG. 4 provide exemplary polynucleotide sequences encoding the full-length light chain for antibody Ab6 with their framework regions (FR), complementarity determining regions (CDRs), and variable region coding sequences delimited.

(5) FIG. 5 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibodies Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

(6) FIG. 6 provides the polypeptide sequence coordinates within the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

(7) FIG. 7 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

(8) FIG. 8 provides the polypeptide sequence coordinates within the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

(9) FIG. 9 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.

(10) FIG. 10 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length heavy chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.

- (11) FIG. 11 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the variable region and complementarity determining regions (CDRs), and the SEQ ID NO of each individual feature.
- (12) FIG. 12 provides the polynucleotide sequence coordinates within the exemplary polynucleotide sequences encoding the full-length light chain polypeptide sequences of antibody Ab6 of sequence features including the framework regions (FRs) and constant region, and the SEQ ID NO of each individual feature.
- (13) FIG. 13 Study design of the clinical trial protocol as summarized in Example 2.
- (14) FIG. 14 displays the efficacy of Ab6 on Mean Monthly Migraine Days (MMDs) in the clinical trial described in Example 2.
- (15) FIG. 15 Illustrates the types and timing of Most Bothersome Symptoms (MBS) across the course of the migraine
- (16) FIG. 16 Illustrates the MBS change from baseline during the 28 day screening period of the clinical trial described in Example 2—i.e. before the first infusion of Ab6.
- (17) FIG. 17 Illustrates the MBS change from baseline 1 month after the first infusion of Ab6 in the clinical trial described in Example 2.
- (18) FIG. 18 Illustrates the PGIC from baseline 1 month after the first infusion of Ab6 in the clinical trial described in Example 2.
- (19) FIG. 19 Illustrates the MBS change from baseline 3 month after the first infusion of Ab6 in the clinical trial described in Example 2.
- (20) FIG. 20 Illustrates the PGIC from baseline 3 month after the first infusion of Ab6 in the clinical trial described in Example 2.
- (21) FIG. 21 Illustrates the MBS change from baseline 6 month after the first infusion of Ab6 in the clinical trial described in Example 2.
- (22) FIG. 22 Illustrates the PGIC from baseline 6 month after the first infusion of Ab6 in the clinical trial described in Example 2.

DETAILED DESCRIPTION

(23) Use of anti-CGRP antibodies for treatment of MBS and/or PGIC associated with migraine, such as chronic migraine or episodic migraine is described herein. Additionally, anti-CGRP antibodies are demonstrated herein to be effective for treatment of MMDs. The treatment efficacy on both MBS and PGIC are shown to be effective in providing relief of MBS and PGIC at 1 month, 3 months and 6 months following the first infusion of an anti-CGRP antibody or fragments thereof of the invention.

Definitions

- (24) It is to be understood that this invention is not limited to the particular methodology, protocols, cell lines, animal species or genera, and reagents described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims. As used herein the singular forms “a”, “and”, and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a cell” includes a plurality of such cells and reference to “the protein” includes reference to one or more proteins and equivalents thereof known to those skilled in the art, and so forth. All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this invention belongs unless clearly indicated otherwise.
- (25) As used herein, the term “most bothersome symptom associated with migraine” refers to symptoms which is identified by an individual patient to be the most bothersome symptom they associate with their migraine. In the present invention the “most bothersome symptom associated with migraine” is specified in Table 1. The “most bothersome symptom associated with migraine” of the present invention described by the patient to the study investigator who assists in medical

interpretation of the patients symptom. The investigator in the clinical study was able to with the patient consultation selected from the group of known migraine symptoms consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty. A patient's "most bothersome symptom associated with migraine" as used in the present invention refers to the self-identified "most bothersome symptom associated with migraine", which may be one or more of the symptoms described herein above or may be classified as "other"

(26) As used herein, the term "improvement of" or "improving" most bothersome symptom associated with migraine refers the change in the patient's assessment of the MBS compared to baseline (i.e the MBS prior to the first dosing with anti-CGRP antibodies or fragments thereof of the invention). An improvement is characterized as ≥ 1 categorical change in the patients assesment of the MBS compared to baseline on the 7 point Likert scale described in Example 2.

(27) As used herein, the term "improvement of" or "improving" patient global impression of change associated with migraine refers the change in the patient's assessment of their disease status compared to baseline (i.e the disease status prior to the first dosing with anti-CGRP antibodies or fragments thereof of the invention). An improvement is characterized as ≥ 1 categorical change in the patients assesment of the PGIC compared to baseline on the 7-step scale described in Example 2.

(28) As used herein, the term "chronic migraine" refers to a condition wherein a patient exhibits, on average, at least 15 headache per month with a subset of these headache days fulling the ICHD-3 criteria for migraine with or without aura. The term "episodic migraine" refers to a condition wherein a patient exhibits, on average, less than 15 day a month of headache with typically 4-15 being a migraine phenotype meeting the ICHD-3 definition of migraine with or without aura.

(29) As used herein, the term "diagnosed with chronic migraine" refers to a patient meeting the clinical criteria for chronic migraine, whether or not a formal diagnosis of that patient was performed.

(30) As used herein, the term "intravenously administering" refers to a mode of administration wherein a substance, e.g., an antibody, is introduced directly into the circulation of that patient, most typically into the venous circulation. The substance may be introduced in a carrier fluid, such as an aqueous solution, e.g., normal saline. The substance may be administered in a single formulation or in multiple formulations, as long as the administration is completed over a short period of time (e.g., within 1 day, preferably within 12 hours, more preferably within 6 hours, and most preferably within 1-2 hours).

(31) As used herein, the term "the baseline number of migraine days" refers to the number of migraine days exhibited by a patient in a specified time period, e.g., prior to treatment. For example, the baseline number of migraine days may be determined over a period of one month, or longer, e.g., by recording each day whether or not a migraine occurred.

(32) As used herein, the term "migraine days per month" refers to the number of days per month on which a patient has a migraine, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of migraine. The number of migraine days per month may be determined by recording each day whether or not a migraine occurred.

(33) As used herein, the term "headache days per month" refers to the number of days per month on which a patient has a headache, i.e., at any time during that day, the patient has symptoms that meet the clinical definition of a headache. The number of headache days per month may be determined by recording each day whether or not a headache occurred.

(34) Calcitonin Gene Related Peptide (CGRP): As used herein, CGRP encompasses not only the following *Homo sapiens* CGRP-alpha and *Homo sapiens* CGRP-beta amino acid sequences

available from American Peptides (Sunnyvale CA) and Bachem (Torrance, CA):

(35) CGRP-alpha: ACDTATCVTHRLAGLLSRSGGVVKNNFVPTNVGSKAF-NH.sub.2 (SEQ ID NO: 561), wherein the terminal phenylalanine is amidated;

(36) CGRP-beta: ACNTATCVTHRLAGLLSRSGGMVKS NFVPTNVGSKAF-NH.sub.2 (SEQ ID NO: 562), wherein the terminal phenylalanine is amidated; but also any membrane-bound forms of these CGRP amino acid sequences, as well as mutants (mutiens), splice variants, isoforms, orthologs, homologues and variants of this sequence.

(37) Expression Vector: These DNA vectors contain elements that facilitate manipulation for the expression of a foreign protein within the target host cell, e.g., a yeast or mammalian cell such as *Pichia pastoris* or CHO cells. Conveniently, manipulation of sequences and production of DNA for transformation is first performed in a bacterial host, e.g. *E. coli*, and usually vectors will include sequences to facilitate such manipulations, including a bacterial origin of replication and appropriate bacterial selection marker. Selection markers encode proteins necessary for the survival or growth of transformed host cells grown in a selective culture medium. Host cells not transformed with the vector containing the selection gene will not survive in the culture medium. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, (b) complement auxotrophic deficiencies, or (c) supply critical nutrients not available from complex media. Exemplary vectors and methods for transformation of yeast are described, for example, in Burke, D., Dawson, D., & Steams, T. (2000). Methods in yeast genetics: a Cold Spring Harbor Laboratory course manual. Plainview, N.Y.: Cold Spring Harbor Laboratory Press.

(38) Expression vectors for use in yeast or mammalian cells will generally further include yeast or mammalian specific sequences, including a selectable auxotrophic or drug marker for identifying transformed yeast strains or transformed mammalian cells. A drug marker may further be used to amplify copy number of the vector in the host cell.

(39) The polypeptide coding sequence of interest is operably linked to transcriptional and translational regulatory sequences that provide for expression of the polypeptide in host cells, e.g., *Pichia pastoris* or CHO cells. These vector components may include, but are not limited to, one or more of the following: an enhancer element, a promoter, and a transcription termination sequence. Sequences for the secretion of the polypeptide may also be included, e.g. a signal sequence, and the like. A yeast or mammalian origin of replication is optional, as expression vectors are often integrated into the host cell genome. In one embodiment of the invention, the polypeptide of interest is operably linked, or fused, to sequences providing for optimized secretion of the polypeptide from yeast diploid cells.

(40) Nucleic acids are "operably linked" when placed into a functional relationship with another nucleic acid sequence. For example, DNA for a signal sequence is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence. Generally, "operably linked" means that the DNA sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading frame. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites or alternatively via a PCR/recombination method familiar to those skilled in the art (Gateway® Technology; Invitrogen, Carlsbad California). If such sites do not exist, the synthetic oligonucleotide adapters or linkers are used in accordance with conventional practice.

(41) Promoters are untranslated sequences located upstream (5') to the start codon of a structural gene (generally within about 100 to 1000 bp) that control the transcription and translation of particular nucleic acid sequences to which they are operably linked. Such promoters fall into several classes: inducible, constitutive, and repressible promoters (that increase levels of transcription in response to absence of a repressor). Inducible promoters may initiate increased levels of transcription from DNA under their control in response to some change in culture conditions, e.g., the presence or absence of a nutrient or a change in temperature.

(42) The promoter fragment may also serve as the site for homologous recombination and integration of the expression vector into the same site in the host genome; alternatively a selectable marker is used as the site for homologous recombination. Examples of suitable promoters from *Pichia* include the AOX1 and promoter (Cregg et al. (1989) *Mol. Cell. Biol.* 9:1316-1323); ICL1 promoter (Menendez et al. (2003) *Yeast* 20(13):1097-108); glyceraldehyde-3-phosphate dehydrogenase promoter (GAP) (Waterham et al. (1997) *Gene* 186(1):37-44); and FLD1 promoter (Shen et al. (1998) *Gene* 216(1):93-102). The GAP promoter is a strong constitutive promoter and the AOX and FLD1 promoters are inducible.

(43) Other yeast promoters include ADH1, alcohol dehydrogenase II, GAL4, PHO3, PHO5, Pyk, and chimeric promoters derived therefrom. Additionally, non-yeast promoters may be used in the invention such as mammalian, insect, plant, reptile, amphibian, viral, and avian promoters. Most typically the promoter will comprise a mammalian promoter (potentially endogenous to the expressed genes) or will comprise a yeast or viral promoter that provides for efficient transcription in yeast systems.

(44) Examples of mammalian promoters include cytomegalovirus (CMV) derived promoters, chicken 3-actin (CBM) derived promoters, adenomatous polyposis coli (APC) derived promoters, leucine-rich repeat containing G protein-coupled receptor 5 (LGR5) promoters, CAG promoter, Beta actin promoter, elongation factor-1 (EF1) promoter, early growth response 1 (EGR-1) promoter, eukaryotic initiation factor 4A (EIF4A1) promoter, simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter, among others. Combinations of two or more of the foregoing promoters may also be used. Further, inducible promoters may be used. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionine promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

(45) The polypeptides of interest may be produced recombinantly not only directly, but also as a fusion polypeptide with a heterologous polypeptide, e.g. a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. In general, the signal sequence may be a component of the vector, or it may be a part of the polypeptide coding sequence that is inserted into the vector. The heterologous signal sequence selected preferably is one that is recognized and processed through one of the standard pathways available within the host cell. The *S. cerevisiae* alpha factor pre-pro signal has proven effective in the secretion of a variety of recombinant proteins from *P. pastoris*. Other yeast signal sequences include the alpha mating factor signal sequence, the invertase signal sequence, and signal sequences derived from other secreted yeast polypeptides. Additionally, these signal peptide sequences may be engineered to provide for enhanced secretion in diploid yeast expression systems. Secretion signals for use in mammalian as well as yeast cells include mammalian signal sequences, which may be heterologous to the protein being secreted, or may be a native sequence for the protein being secreted. Signal sequences include pre-peptide sequences, and in some instances may include propeptide sequences. Many such signal sequences are known in the art, including the signal sequences found on immunoglobulin chains, e.g., K28 preprotoxin sequence, PHA-E, FACE, human MCP-1, human serum albumin signal sequences, human Ig heavy chain, human Ig light chain, and the like. For example, see Hashimoto et. al. *Protein Eng* 11(2) 75 (1998); and Kobayashi et. al. *Therapeutic Apheresis* 2(4) 257 (1998).

(46) Transcription may be increased by inserting a transcriptional activator sequence into the

vector. These activators are cis-acting elements of DNA, usually about from 10 to 300 bp, which act on a promoter to increase its transcription. Transcriptional enhancers are relatively orientation and position independent, having been found 5' and 3' to the transcription unit, within an intron, as well as within the coding sequence itself. The enhancer may be spliced into the expression vector at a position 5' or 3' to the coding sequence, but is preferably located at a site 5' from the promoter.

(47) Expression vectors used in eukaryotic host cells may also contain sequences necessary for the termination of transcription and for stabilizing the mRNA. Such sequences are commonly available from 3' to the translation termination codon, in untranslated regions of eukaryotic or viral DNAs or cDNAs. These regions contain nucleotide segments transcribed as polyadenylated fragments in the untranslated portion of the mRNA.

(48) Construction of suitable vectors containing one or more of the above-listed components employs standard ligation techniques or PCR/recombination methods. Isolated plasmids or DNA fragments are cleaved, tailored, and re-ligated in the form desired to generate the plasmids required or via recombination methods. For analysis to confirm correct sequences in plasmids constructed, the ligation mixtures are used to transform host cells, and successful transformants selected by antibiotic resistance (e.g. ampicillin or Zeocin) where appropriate. Plasmids from the transformants are prepared, analyzed by restriction endonuclease digestion and/or sequenced.

(49) As an alternative to restriction and ligation of fragments, recombination methods based on att sites and recombination enzymes may be used to insert DNA sequences into a vector. Such methods are described, for example, by Landy (1989) *Ann. Rev. Biochem.* 58:913-949; and are known to those of skill in the art. Such methods utilize intermolecular DNA recombination that is mediated by a mixture of lambda and *E. coli*-encoded recombination proteins. Recombination occurs between specific attachment (att) sites on the interacting DNA molecules. For a description of att sites see Weisberg and Landy (1983) Site-Specific Recombination in Phage Lambda, in *Lambda II*, Weisberg, ed. (Cold Spring Harbor, NY: Cold Spring Harbor Press), pp. 211-250. The DNA segments flanking the recombination sites are switched, such that after recombination, the att sites are hybrid sequences comprised of sequences donated by each parental vector. The recombination can occur between DNAs of any topology.

(50) Att sites may be introduced into a sequence of interest by ligating the sequence of interest into an appropriate vector; generating a PCR product containing att B sites through the use of specific primers; generating a cDNA library cloned into an appropriate vector containing att sites; and the like.

(51) Folding, as used herein, refers to the three-dimensional structure of polypeptides and proteins, where interactions between amino acid residues act to stabilize the structure. Proper folding is typically the arrangement of a polypeptide that results in optimal biological activity, and in the case of antibodies can conveniently be monitored by assays for activity, e.g. antigen binding.

(52) The expression host may be further modified by the introduction of sequences encoding one or more enzymes that enhance folding and disulfide bond formation, i.e. foldases, chaperonins, etc. Such sequences may be constitutively or inducibly expressed in the yeast host cell, using vectors, markers, etc. as known in the art. Preferably the sequences, including transcriptional regulatory elements sufficient for the desired pattern of expression, are stably integrated in the yeast genome through a targeted methodology.

(53) For example, the eukaryotic PDI is not only an efficient catalyst of protein cysteine oxidation and disulfide bond isomerization, but also exhibits chaperone activity. Co-expression of PDI can facilitate the production of active proteins having multiple disulfide bonds. Also of interest is the expression of BIP (immunoglobulin heavy chain binding protein); cyclophilin; and the like. In one embodiment of the invention, each of the haploid parental strains expresses a distinct folding enzyme, e.g. one strain may express BIP, and the other strain may express PDI or combinations thereof.

(54) The terms "desired protein" or "desired antibody" are used interchangeably and refer generally

to a parent antibody specific to a target, i.e., CGRP or a chimeric or humanized antibody or a binding portion thereof derived therefrom as described herein. The term “antibody” is intended to include any polypeptide chain-containing molecular structure with a specific shape that fits to and recognizes an epitope, where one or more non-covalent binding interactions stabilize the complex between the molecular structure and the epitope. The archetypal antibody molecule is the immunoglobulin, and in particular IgG, from all sources, e.g. human, rodent, rabbit, cow, sheep, pig, dog, other mammals, chicken, other avians, etc., are considered to be “antibodies.” A preferred source for producing antibodies useful as starting material according to the invention is rabbits. Numerous antibody coding sequences have been described; and others may be raised by methods well-known in the art. Examples thereof include chimeric antibodies, human antibodies and other non-human mammalian antibodies, humanized antibodies, single chain antibodies (such as scFvs), camelbodies, nanobodies, IgNAR (single-chain antibodies derived from sharks), small-modular immunopharmaceuticals (SMIPs), and antibody fragments such as Fabs, Fab', F(ab')₂ and the like. See Streltsov V A, et al., Structure of a shark IgNAR antibody variable domain and modeling of an early-developmental isotype, *Protein Sci.* 2005 November; 14(11):2901-9. Epub 2005 Sep. 30; Greenberg A S, et al., A new antigen receptor gene family that undergoes rearrangement and extensive somatic diversification in sharks, *Nature*. 1995 Mar. 9; 374(6518):168-73; Nuttall S D, et al., Isolation of the new antigen receptor from wobbegong sharks, and use as a scaffold for the display of protein loop libraries, *Mol Immunol.* 2001 August; 38(4):313-26; Hamers-Casterman C, et al., Naturally occurring antibodies devoid of light chains, *Nature*. 1993 Jun. 3; 363(6428):446-8; Gill D S, et al., Biopharmaceutical drug discovery using novel protein scaffolds, *Curr Opin Biotechnol.* 2006 December; 17(6):653-8. Epub 2006 Oct. 19.

(55) For example, antibodies or antigen binding fragments may be produced by genetic engineering. In this technique, as with other methods, antibody-producing cells are sensitized to the desired antigen or immunogen. The messenger RNA isolated from antibody producing cells is used as a template to make cDNA using PCR amplification. A library of vectors, each containing one heavy chain gene and one light chain gene retaining the initial antigen specificity, is produced by insertion of appropriate sections of the amplified immunoglobulin cDNA into the expression vectors. A combinatorial library is constructed by combining the heavy chain gene library with the light chain gene library. This results in a library of clones which co-express a heavy and light chain (resembling the Fab fragment or antigen binding fragment of an antibody molecule). The vectors that carry these genes are co-transfected into a host cell. When antibody gene synthesis is induced in the transfected host, the heavy and light chain proteins self-assemble to produce active antibodies that can be detected by screening with the antigen or immunogen.

(56) Antibody coding sequences of interest include those encoded by native sequences, as well as nucleic acids that, by virtue of the degeneracy of the genetic code, are not identical in sequence to the disclosed nucleic acids, and variants thereof. Variant polypeptides can include amino acid (aa) substitutions, additions or deletions. The amino acid substitutions can be conservative amino acid substitutions or substitutions to eliminate non-essential amino acids, such as to alter a glycosylation site, or to minimize misfolding by substitution or deletion of one or more cysteine residues that are not necessary for function. Variants can be designed so as to retain or have enhanced biological activity of a particular region of the protein (e.g., a functional domain, catalytic amino acid residues, etc). Variants also include fragments of the polypeptides disclosed herein, particularly biologically active fragments and/or fragments corresponding to functional domains. Techniques for in vitro mutagenesis of cloned genes are known. Also included in the subject invention are polypeptides that have been modified using ordinary molecular biological techniques so as to improve their resistance to proteolytic degradation or to optimize solubility properties or to render them more suitable as a therapeutic agent.

(57) Chimeric antibodies may be made by recombinant means by combining the variable light and heavy chain regions (V_L and V_H), obtained from antibody producing cells of one species

with the constant light and heavy chain regions from another. Typically chimeric antibodies utilize rodent or rabbit variable regions and human constant regions, in order to produce an antibody with predominantly human domains. The production of such chimeric antibodies is well known in the art, and may be achieved by standard means (as described, e.g., in U.S. Pat. No. 5,624,659, incorporated herein by reference in its entirety). It is further contemplated that the human constant regions of chimeric antibodies of the invention may be selected from IgG1, IgG2, IgG3, and IgG4 constant regions.

(58) Humanized antibodies are engineered to contain even more human-like immunoglobulin domains, and incorporate only the complementarity-determining regions of the animal-derived antibody. This is accomplished by carefully examining the sequence of the hyper-variable loops of the variable regions of the monoclonal antibody, and fitting them to the structure of the human antibody chains. Although facially complex, the process is straightforward in practice. See, e.g., U.S. Pat. No. 6,187,287, incorporated fully herein by reference.

(59) In addition to entire immunoglobulins (or their recombinant counterparts), immunoglobulin fragments comprising the epitope binding site (e.g., Fab', F(ab')₂, or other fragments) may be synthesized. "Fragment," or minimal immunoglobulins may be designed utilizing recombinant immunoglobulin techniques. For instance "Fv" immunoglobulins for use in the present invention may be produced by synthesizing a fused variable light chain region and a variable heavy chain region. Combinations of antibodies are also of interest, e.g. diabodies, which comprise two distinct Fv specificities. In another embodiment of the invention, SMIPs (small molecule immunopharmaceuticals), camelbodies, nanobodies, and IgNAR are encompassed by immunoglobulin fragments.

(60) Immunoglobulins and fragments thereof may be modified post-translationally, e.g. to add effector moieties such as chemical linkers, detectable moieties, such as fluorescent dyes, enzymes, toxins, substrates, bioluminescent materials, radioactive materials, chemiluminescent moieties and the like, or specific binding moieties, such as streptavidin, avidin, or biotin, and the like may be utilized in the methods and compositions of the present invention. Examples of additional effector molecules are provided infra.

(61) A polynucleotide sequence "corresponds" to a polypeptide sequence if translation of the polynucleotide sequence in accordance with the genetic code yields the polypeptide sequence (i.e., the polynucleotide sequence "encodes" the polypeptide sequence), one polynucleotide sequence "corresponds" to another polynucleotide sequence if the two sequences encode the same polypeptide sequence.

(62) A "heterologous" region or domain of a DNA construct is an identifiable segment of DNA within a larger DNA molecule that is not found in association with the larger molecule in nature. Thus, when the heterologous region encodes a mammalian gene, the gene will usually be flanked by DNA that does not flank the mammalian genomic DNA in the genome of the source organism. Another example of a heterologous region is a construct where the coding sequence itself is not found in nature (e.g., a cDNA where the genomic coding sequence contains introns, or synthetic sequences having codons different than the native gene). Allelic variations or naturally-occurring mutational events do not give rise to a heterologous region of DNA as defined herein.

(63) A "coding sequence" is an in-frame sequence of codons that (in view of the genetic code) correspond to or encode a protein or peptide sequence. Two coding sequences correspond to each other if the sequences or their complementary sequences encode the same amino acid sequences. A coding sequence in association with appropriate regulatory sequences may be transcribed and translated into a polypeptide. A polyadenylation signal and transcription termination sequence will usually be located 3' to the coding sequence. A "promoter sequence" is a DNA regulatory region capable of binding RNA polymerase in a cell and initiating transcription of a downstream (3' direction) coding sequence. Promoter sequences typically contain additional sites for binding of regulatory molecules (e.g., transcription factors) which affect the transcription of the coding

sequence. A coding sequence is “under the control” of the promoter sequence or “operatively linked” to the promoter when RNA polymerase binds the promoter sequence in a cell and transcribes the coding sequence into mRNA, which is then in turn translated into the protein encoded by the coding sequence.

(64) Vectors are used to introduce a foreign substance, such as DNA, RNA or protein, into an organism or host cell. Typical vectors include recombinant viruses (for polynucleotides) and liposomes (for polypeptides). A “DNA vector” is a replicon, such as plasmid, phage or cosmid, to which another polynucleotide segment may be attached so as to bring about the replication of the attached segment. An “expression vector” is a DNA vector which contains regulatory sequences which will direct polypeptide synthesis by an appropriate host cell. This usually means a promoter to bind RNA polymerase and initiate transcription of mRNA, as well as ribosome binding sites and initiation signals to direct translation of the mRNA into a polypeptide(s). Incorporation of a polynucleotide sequence into an expression vector at the proper site and in correct reading frame, followed by transformation of an appropriate host cell by the vector, enables the production of a polypeptide encoded by said polynucleotide sequence.

(65) “Amplification” of polynucleotide sequences is the in vitro production of multiple copies of a particular nucleic acid sequence. The amplified sequence is usually in the form of DNA. A variety of techniques for carrying out such amplification are described in a review article by Van Brunt (1990, *Bio Technol.*, 8(4):291-294). Polymerase chain reaction or PCR is a prototype of nucleic acid amplification, and use of PCR herein should be considered exemplary of other suitable amplification techniques.

(66) The general structure of antibodies in vertebrates now is well understood (Edelman, G. M., *Ann. N.Y. Acad. Sci.*, 190: 5 (1971)). Antibodies consist of two identical light polypeptide chains of molecular weight approximately 23,000 daltons (the “light chain”), and two identical heavy chains of molecular weight 53,000-70,000 (the “heavy chain”). The four chains are joined by disulfide bonds in a “Y” configuration wherein the light chains bracket the heavy chains starting at the mouth of the “Y” configuration. The “branch” portion of the “Y” configuration is designated the F.sub.ab region; the stem portion of the “Y” configuration is designated the Fc region. The amino acid sequence orientation runs from the N-terminal end at the top of the “Y” configuration to the C-terminal end at the bottom of each chain. The N-terminal end possesses the variable region having specificity for the antigen that elicited it, and is approximately 100 amino acids in length, there being slight variations between light and heavy chain and from antibody to antibody.

(67) The variable region is linked in each chain to a constant region that extends the remaining length of the chain and that within a particular class of antibody does not vary with the specificity of the antibody (i.e., the antigen eliciting it). There are five known major classes of constant regions that determine the class of the immunoglobulin molecule (IgG, IgM, IgA, IgD, and IgE corresponding to γ , μ , α , δ , and ϵ (gamma, mu, alpha, delta, or epsilon) heavy chain constant regions). The constant region or class determines subsequent effector function of the antibody, including activation of complement (Kabat, E. A., *Structural Concepts in Immunology and Immunochemistry*, 2nd Ed., p. 413-436, Holt, Rinehart, Winston (1976)), and other cellular responses (Andrews, D. W., et al., *Clinical Immunobiology*, pp 1-18, W. B. Sanders (1980); Kohl, S., et al., *Immunology*, 48: 187 (1983)); while the variable region determines the antigen with which it will react. Light chains are classified as either κ (kappa) or λ (lambda). Each heavy chain class can be prepared with either kappa or lambda light chain. The light and heavy chains are covalently bonded to each other, and the “tail” portions of the two heavy chains are bonded to each other by covalent disulfide linkages when the immunoglobulins are generated either by hybridomas or by B cells.

(68) The expression “variable region” or “VR” refers to the domains within each pair of light and heavy chains in an antibody that are involved directly in binding the antibody to the antigen. Each heavy chain has at one end a variable domain (V.sub.H) followed by a number of constant domains.

Each light chain has a variable domain (V.sub.L) at one end and a constant domain at its other end; the constant domain of the light chain is aligned with the first constant domain of the heavy chain, and the light chain variable domain is aligned with the variable domain of the heavy chain.

(69) The expressions “complementarity determining region,” “hypervariable region,” or “CDR” refer to one or more of the hyper-variable or complementarity determining regions (CDRs) found in the variable regions of light or heavy chains of an antibody (See Kabat, E. A. et al., Sequences of Proteins of Immunological Interest, National Institutes of Health, Bethesda, Md., (1987)). These expressions include the hypervariable regions as defined by Kabat et al. (“Sequences of Proteins of Immunological Interest,” Kabat E., et al., US Dept. of Health and Human Services, 1983) or the hypervariable loops in 3-dimensional structures of antibodies (Chothia and Lesk, *J Mol. Biol.* 196 901-917 (1987)). The CDRs in each chain are held in close proximity by framework regions and, with the CDRs from the other chain, contribute to the formation of the antigen binding site. Within the CDRs there are select amino acids that have been described as the selectivity determining regions (SDRs) which represent the critical contact residues used by the CDR in the antibody-antigen interaction (Kashmiri, S., *Methods*, 36:25-34 (2005)). In the present invention when specific antibody amino acid or nucleic acid residues are referenced by number this generally refers to its position within a specified amino acid or nucleic acid sequence (i.e., particular sequence identifier) and/or in accordance with Kabat et al numbering.

(70) The expressions “framework region” or “FR” refer to one or more of the framework regions within the variable regions of the light and heavy chains of an antibody (See Kabat, E. A. et al., Sequences of Proteins of Immunological Interest, National Institutes of Health, Bethesda, Md., (1987)). These expressions include those amino acid sequence regions interposed between the CDRs within the variable regions of the light and heavy chains of an antibody.

(71) “Cmax” refers to the maximum (or peak) concentration that an antibody or other compound achieves in tested area (e.g., in the serum or another compartment such as cerebrospinal fluid) after the drug has been administered. For example, serum Cmax may be measured from serum, e.g., prepared by collecting a blood sample, allowing it to clot and separating solid components by centrifugation or other means to yield serum (blood containing neither blood cells nor clotting factors), and then detecting the concentration of the analyte in the serum by ELISA or other means known in the art.

(72) “AUC” refers to the area under the concentration-time curve which is expressed in units of mg/mL*hr (or equivalently mg*hr/ml) unless otherwise specified. “AUC.sub.0-t” refers to the area under the concentration-time curve from time=0 to last quantifiable concentration. “AUC.sub.0-inf” refers to the area under the concentration-time curve from time=0 extrapolated to infinity.

(73) “I.sub.max” refers to the maximal pharmacodynamic response elicited by an anti-CGRP antibody dosage, preferably a dosage of 350 mg or more, more typically at least 750 or 1000 mg, as compared to the response elicited by a lower anti-CGRP antibody doses, e.g., wherein such response may be detected by the inhibition of vasodilation after topical application of capsaicin.

(74) Anti-CGRP Antibodies and Binding Fragments Thereof Having Binding Specificity for CGRP

(75) The invention specifically includes the use of Ab6, which is a specific anti-CGRP antibody or antibody fragment, which comprises or consists of the CDR, VL, VH, CL, CH polypeptides sequences identified in FIGS. 1-12. The polypeptides comprised in the anti-CGRP antibody, Ab6 is further described below.

(76) Antibody Ab6

(77) TABLE-US-00002 (SEQ ID NO: 222)

QVLTQSPSSLASVSGDRVTINCQASQSVYHNTYLAWYQQKPGKVPKQLI
YDASTLASGVPSRFSGSGSGTDFTLTISSLQPEDVATYYCLGSYDCTNG
DCFVFGGGTKVEIKR.

(78) The invention also includes humanized antibodies having binding specificity to CGRP and possessing a light chain sequence comprising the sequence set forth below:

(79) TABLE-US-00003 (SEQ ID NO: 221)

QVLTQSPSSLSASVGDRVTINCQASQSVYHNTYLAWYQQKPGKVPKQLI
YDASTLASGVPSRFSGSGSGTDFTLTISSLQPEDVATYYCLGSYDCTNG
DCFVFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNFPYPR
EAKVQWKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKV
YACEVTHQGLSSPVTKSFNRGEC.

(80) The invention further includes humanized antibodies having binding specificity to CGRP and possessing a variable heavy chain sequence comprising the sequence set forth below:

(81) TABLE-US-00004 (SEQ ID NO: 202)

EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNSTTVYLQMNSLRADTAVYFCARG
DIWGQGTLVTVSS.

(82) The invention also includes humanized antibodies having binding specificity to CGRP and possessing a heavy chain sequence comprising the sequence set forth below:

(83) TABLE-US-00005 (SEQ ID NO: 201)

EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNSTTVYLQMNSLRADTAVYFCARG
DIWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNV
NHHKPSNTKVDARVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVL
DSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK.

(84) Alternatively, the heavy chain of Ab6 may lack the C-terminal lysine of SEQ ID NO: 201, i.e., a heavy chain sequence comprising the sequence set forth below:

(85) TABLE-US-00006 (SEQ ID NO: 566)

EVQLVESGGGLVQPGGSLRLSCAVSGIDLSGYIMNWVRQAPGKGLEWVG
VIGINGATYYASWAKGRFTISRDNSTTVYLQMNSLRADTAVYFCARG
DIWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNV
NHHKPSNTKVDARVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVL
DSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPG.

(86) The invention further contemplates antibodies comprising one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221, and/or one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566, or combinations of these polypeptide sequences. In another embodiment of the invention, the antibodies of the invention or fragments thereof comprise, or alternatively consist of, combinations of one or more of the CDRs, the variable heavy and variable light chain sequences, and the heavy and light chain sequences set forth above, including all of them.

(87) The invention also contemplates fragments of the antibody having binding specificity to CGRP. In one embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 222 or SEQ ID NO: 221. In

another embodiment of the invention, antibody fragments of the invention comprise, or alternatively consist of, the polypeptide sequence of SEQ ID NO: 202 or SEQ ID NO: 201 or SEQ ID NO: 566.

(88) In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable light chain sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

(89) In a further embodiment of the invention, fragments of the antibody having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208 which correspond to the complementarity-determining regions (CDRs, or hypervariable regions) of the variable heavy chain sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

(90) The invention also contemplates antibody fragments which include one or more of the antibody fragments described herein. In one embodiment of the invention, fragments of the antibodies having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following antibody fragments: the variable light chain region of SEQ ID NO: 222; the variable heavy chain region of SEQ ID NO: 202; the complementarity-determining regions (SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228) of the variable light chain region of SEQ ID NO: 222; and the complementarity-determining regions (SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208) of the variable heavy chain region of SEQ ID NO: 202.

(91) In a particularly preferred embodiment of the invention, the humanized anti-CGRP antibody is Ab6, comprising, or alternatively consisting of, SEQ ID NO: 221 and SEQ ID NO: 201 or SEQ ID NO: 566, and having at least one of the biological activities set forth herein.

(92) In a further particularly preferred embodiment of the invention, antibody fragments comprise, or alternatively consist of, Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the Fab fragment includes the variable light chain sequence of SEQ ID NO: 222 and the variable heavy chain sequence of SEQ ID NO: 202. This embodiment of the invention further contemplates additions, deletions, and variants of SEQ ID NO: 222 and/or SEQ ID NO: 202 in said Fab while retaining binding specificity for CGRP.

(93) In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202, which polypeptides optionally are respectively linked to human light and heavy constant region polypeptides, e.g., human IgG1, IgG2, IgG3 or IgG4 constant regions, which constant regions optionally may be modified to alter glycosylation or proteolysis, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells.

(94) In another particularly preferred embodiment of the invention, said anti-CGRP antibody may comprise the antibody expression product isolated from recombinant cells which express nucleic acid sequences encoding the light chain of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566, wherein said recombinant cells optionally comprise yeast or mammalian cells, e.g., *Pichia pastoris* or CHO cells, wherein the constant regions thereof optionally may be modified to alter glycosylation or proteolysis or other effector functions.

(95) In another particularly preferred embodiment of the invention, any of the aforementioned anti-CGRP antibodies or antibody fragments may be optionally comprised in a formulation as disclosed herein, e.g., comprising histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg anti-CGRP antibody, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8.

(96) In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

(97) In another embodiment, antibody fragments may be present in one or more of the following non-limiting forms: Fab, Fab', F(ab')₂, Fv and single chain Fv antibody forms. In a preferred embodiment, the anti-CGRP antibodies described herein further comprises the kappa constant light chain sequence comprising the sequence set forth below:

(98) TABLE-US-00007 (SEQ ID NO: 563)

TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSG
NSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC.

(99) In another preferred embodiment, the anti-CGRP antibodies described herein further comprises the gamma-1 constant heavy chain polypeptide sequence comprising the sequence set forth below or the same sequence lacking the carboxy terminal lysine residue (SEQ ID NO: 564 and SEQ ID NO: 565, respectively):

(100) TABLE-US-00008 (SEQ ID NO: 564)

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVTV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLT
VDKSRWQQGNVVFSCSVMHEALHNHYTQKSLSLSPGK. (SEQ ID NO: 565)

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVTV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLT
VDKSRWQQGNVVFSCSVMHEALHNHYTQKSLSLSPG.

(101) For clarity, any antibody disclosed herein is intended to include any variant of the disclosed constant region variant sequences, e.g., Ab6 may comprise the constant region of SEQ ID NO: 564 containing the C-terminal lysine or may comprise the constant region of SEQ ID NO: 565 lacking the C-terminal lysine. Thus, every disclosure herein of the heavy chain of SEQ ID NO: 201 also includes a variant lacking the C-terminal lysine residue thereof, i.e., having the heavy chain variable region sequence of Ab6 (SEQ ID NO: 202) and the constant region sequence of SEQ ID NO: 565. For example, the sequence encoding an antibody comprising a C-terminal lysine in the heavy chain may, when expressed in cell lines such as CHO cells, produce an antibody lacking said C-terminal lysine due to proteolysis, or a mixture of heavy chains containing or lacking said C-terminal lysine.

(102) In one embodiment of the invention, the antibodies or V_H or V_L polypeptides originate or are selected from one or more rabbit B cell populations prior to initiation of the humanization process referenced herein.

(103) In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof do not have binding specificity for CGRP-R. In a further embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R. In another embodiment of the invention, the anti-CGRP antibodies and fragments thereof inhibit the association of CGRP with CGRP-R and/or additional proteins and/or multimers thereof, and/or

antagonizes the biological effects thereof.

(104) As stated herein, antibodies and fragments thereof may be modified post-translationally to add effector moieties such as chemical linkers, detectable moieties such as for example fluorescent dyes, enzymes, substrates, bioluminescent materials, radioactive materials, and chemiluminescent moieties, or functional moieties such as for example streptavidin, avidin, biotin, a cytotoxin, a cytotoxic agent, and radioactive materials.

(105) Antibodies or fragments thereof may also be chemically modified to provide additional advantages such as increased solubility, stability and circulating time (in vivo half-life) of the polypeptide, or decreased immunogenicity (See U.S. Pat. No. 4,179,337). The chemical moieties for derivatization may be selected from water soluble polymers such as polyethylene glycol, ethylene glycol/propylene glycol copolymers, carboxymethylcellulose, dextran, polyvinyl alcohol and the like. The antibodies and fragments thereof may be modified at random positions within the molecule, or at predetermined positions within the molecule and may include one, two, three or more attached chemical moieties.

(106) The polymer may be of any molecular weight, and may be branched or unbranched. For polyethylene glycol, the preferred molecular weight is between about 1 kDa and about 100 kDa (the term "about" indicating that in preparations of polyethylene glycol, some molecules will weigh more, some less, than the stated molecular weight) for ease in handling and manufacturing. Other sizes may be used, depending on the desired therapeutic profile (e.g., the duration of sustained release desired, the effects, if any on biological activity, the ease in handling, the degree or lack of antigenicity and other known effects of the polyethylene glycol to a therapeutic protein or analog). For example, the polyethylene glycol may have an average molecular weight of about 200, 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500, 10,000, 10,500, 11,000, 11,500, 12,000, 12,500, 13,000, 13,500, 14,000, 14,500, 15,000, 15,500, 16,000, 16,500, 17,000, 17,500, 18,000, 18,500, 19,000, 19,500, 20,000, 25,000, 30,000, 35,000, 40,000, 50,000, 55,000, 60,000, 65,000, 70,000, 75,000, 80,000, 85,000, 90,000, 95,000, or 100,000 kDa. Branched polyethylene glycols are described, for example, in U.S. Pat. No. 5,643,575; Morpurgo et al., *Appl. Biochem. Biotechnol.* 56:59-72 (1996); Vorobjev et al., *Nucleosides Nucleotides* 18:2745-2750 (1999); and Caliceti et al., *Bioconjug. Chem.* 10:638-646 (1999), the disclosures of each of which are incorporated herein by reference.

(107) There are a number of attachment methods available to those skilled in the art, See e.g., EP 0 401 384, herein incorporated by reference (coupling PEG to G-CSF), See also Malik et al., *Exp. Hematol.* 20:1028-1035 (1992) (reporting pegylation of GM-CSF using tresyl chloride). For example, polyethylene glycol may be covalently bound through amino acid residues via a reactive group, such as, a free amino or carboxyl group. Reactive groups are those to which an activated polyethylene glycol molecule may be bound. The amino acid residues having a free amino group may include lysine residues and the N-terminal amino acid residues; those having a free carboxyl group may include aspartic acid residues glutamic acid residues and the C-terminal amino acid residue. Sulfhydryl groups may also be used as a reactive group for attaching the polyethylene glycol molecules. Preferred for therapeutic purposes is attachment at an amino group, such as attachment at the N-terminus or lysine group.

(108) As suggested above, polyethylene glycol may be attached to proteins via linkage to any of a number of amino acid residues. For example, polyethylene glycol can be linked to polypeptides via covalent bonds to lysine, histidine, aspartic acid, glutamic acid, or cysteine residues. One or more reaction chemistries may be employed to attach polyethylene glycol to specific amino acid residues (e.g., lysine, histidine, aspartic acid, glutamic acid, or cysteine) or to more than one type of amino acid residue (e.g., lysine, histidine, aspartic acid, glutamic acid, cysteine and combinations thereof).

(109) Alternatively, antibodies or fragments thereof may have increased in vivo half-lives via fusion with albumin (including but not limited to recombinant human serum albumin or fragments or variants thereof (See, e.g., U.S. Pat. No. 5,876,969, issued Mar. 2, 1999, EP Patent 0 413 622,

and U.S. Pat. No. 5,766,883, issued Jun. 16, 1998, herein incorporated by reference in their entirety)) or other circulating blood proteins such as transferrin or ferritin. In a preferred embodiment, polypeptides and/or antibodies of the present invention (including fragments or variants thereof) are fused with the mature form of human serum albumin (i.e., amino acids 1-585 of human serum albumin as shown in FIGS. 1 and 2 of EP Patent 0 322 094) which is herein incorporated by reference in its entirety. Polynucleotides encoding fusion proteins of the invention are also encompassed by the invention.

(110) Regarding detectable moieties, further exemplary enzymes include, but are not limited to, horseradish peroxidase, acetylcholinesterase, alkaline phosphatase, beta-galactosidase and luciferase. Further exemplary fluorescent materials include, but are not limited to, rhodamine, fluorescein, fluorescein isothiocyanate, umbelliferone, dichlorotriazinylamine, phycoerythrin and dansyl chloride. Further exemplary chemiluminescent moieties include, but are not limited to, luminol. Further exemplary bioluminescent materials include, but are not limited to, luciferin and aequorin. Further exemplary radioactive materials include, but are not limited to, Iodine 125 (.sup.125I), Carbon 14 (.sup.14C), Sulfur 35 (.sup.35S), Tritium (H) and Phosphorus 32 (.sup.32P).

(111) Regarding functional moieties, exemplary cytotoxic agents include, but are not limited to, methotrexate, aminopterin, 6-mercaptopurine, 6-thioguanine, cytarabine, 5-fluorouracil decarbazine; alkylating agents such as mechlorethamine, thioepa chlorambucil, melphalan, carmustine (BSNU), mitomycin C, lomustine (CCNU), 1-methylnitrosourea, cyclophosphamide, mechlorethamine, busulfan, dibromomannitol, streptozotocin, mitomycin C, cis-dichlorodiamine platinum (II) (DDP) cisplatin and carboplatin (paraplatin); anthracyclines include daunorubicin (formerly daunomycin), doxorubicin (adriamycin), detorubicin, carminomycin, idarubicin, epirubicin, mitoxantrone and bisantrene; antibiotics include dactinomycin (actinomycin D), bleomycin, calicheamicin, mithramycin, and anthramycin (AMC); and antimetabolic agents such as the vinca alkaloids, vincristine and vinblastine. Other cytotoxic agents include paclitaxel (taxol), ricin, pseudomonas exotoxin, gemcitabine, cytochalasin B, gramicidin D, ethidium bromide, emetine, etoposide, teniposide, colchicin, dihydroxy anthracin dione, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, procarbazine, hydroxyurea, asparaginase, corticosteroids, mytotan (O,P'-(DDD)), interferons, and mixtures of these cytotoxic agents.

(112) Further cytotoxic agents include, but are not limited to, chemotherapeutic agents such as carboplatin, cisplatin, paclitaxel, gemcitabine, calicheamicin, doxorubicin, 5-fluorouracil, mitomycin C, actinomycin D, cyclophosphamide, vincristine and bleomycin. Toxic enzymes from plants and bacteria such as ricin, diphtheria toxin and *Pseudomonas* toxin may be conjugated to the humanized or chimeric antibodies, or binding fragments thereof, to generate cell-type-specific-killing reagents (Youle, et al., *Proc. Nat'l Acad. Sci. USA* 77:5483 (1980); Gilliland, et al., *Proc. Nat'l Acad. Sci. USA* 77:4539 (1980); Krolick, et al., *Proc. Nat'l Acad. Sci. USA* 77:5419 (1980)).

(113) Other cytotoxic agents include cytotoxic ribonucleases as described by Goldenberg in U.S. Pat. No. 6,653,104. Embodiments of the invention also relate to radioimmunoconjugates where a radionuclide that emits alpha or beta particles is stably coupled to the antibody, or binding fragments thereof, with or without the use of a complex-forming agent. Such radionuclides include beta-emitters such as Phosphorus-32 (.sup.32P), Scandium-47 (.sup.47Sc), Copper-67 (.sup.67Cu), Gallium-67 (.sup.67Ga), Yttrium-88 (.sup.88Y), Yttrium-90 (.sup.90Y), Iodine-125 (.sup.125I) Iodine-131 (.sup.131I), Samarium-153 (.sup.153Sm), Lutetium-177 (.sup.177Lu), Rhenium-186 (.sup.186Re) or Rhenium-188 (.sup.188Re), and alpha-emitters such as Astatine-211 (.sup.211At), Lead-212 (.sup.212Pb), Bismuth-212 (.sup.212Bi) or -213 (.sup.213Bi) or Actinium-225 (.sup.225Ac).

(114) Methods are known in the art for conjugating an antibody or binding fragment thereof to a detectable moiety and the like, such as for example those methods described by Hunter et al, *Nature* 144:945 (1962); David et al, *Biochemistry* 13:1014 (1974); Pain et al, *J Immunol. Meth.*

40:219 (1981); and Nygren, J., *Histochem. and Cytochem.* 30:407 (1982).

(115) Embodiments described herein further include variants and equivalents that are substantially homologous to the antibodies, antibody fragments, diabodies, SMIPs, camelbodies, nanobodies, IgNAR, polypeptides, variable regions and CDRs set forth herein. These may contain, e.g., conservative substitution mutations, (i.e., the substitution of one or more amino acids by similar amino acids). For example, conservative substitution refers to the substitution of an amino acid with another within the same general class, e.g., one acidic amino acid with another acidic amino acid, one basic amino acid with another basic amino acid, or one neutral amino acid by another neutral amino acid. What is intended by a conservative amino acid substitution is well known in the art.

(116) In another embodiment, the invention contemplates polypeptide sequences having at least 90% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. More preferably, the invention contemplates polypeptide sequences having at least 95% or greater sequence homology, even more preferably at least 98% or greater sequence homology, and still more preferably at least 99% or greater sequence homology to any one or more of the polypeptide sequences of antibody fragments, variable regions and CDRs set forth herein. Methods for determining homology between nucleic acid and amino acid sequences are well known to those of ordinary skill in the art.

(117) In another embodiment, the invention further contemplates the above-recited polypeptide homologs of the antibody fragments, variable regions and CDRs set forth herein further having anti-CGRP activity. Non-limiting examples of anti-CGRP activity are set forth herein.

(118) The invention further contemplates treatment methods wherein the one or more anti-human CGRP antibodies discussed above are aglycosylated or if glycosylated are only mannosylated; that contain an Fc region that has been modified to alter effector function, half-life, proteolysis, and/or glycosylation; are human, humanized, single chain or chimeric; and are a humanized antibody derived from a rabbit (parent) anti-human CGRP antibody. An exemplary mutation which impairs glycosylation comprises the mutation of the Asn residue at position 297 of an IgG heavy chain constant region such as IgG1 to another amino acid, such as Ala as described in U.S. Pat. No. 5,624,821, which is incorporated by reference in its entirety.

(119) The invention further contemplates one or more anti-human CGRP antibodies wherein the framework regions (FRs) in the variable light region and the variable heavy regions of said antibody respectively are human FRs which are unmodified or which have been modified by the substitution of one or more human FR residues in the variable light or heavy chain region with the corresponding FR residues of the parent rabbit antibody, and wherein said human FRs have been derived from human variable heavy and light chain antibody sequences which have been selected from a library of human germline antibody sequences based on their high level of homology to the corresponding rabbit variable heavy or light chain regions relative to other human germline antibody sequences contained in the library.

(120) The invention also contemplates that the treatment method may involve the administration of two or more anti-CGRP antibodies or fragments thereof and disclosed herein. If more than one antibody is administered to the patient, the multiple antibodies may be administered simultaneously or concurrently, or may be staggered in their administration. The anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, may also be described by their strength of binding or their affinity for CGRP. In one embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with a dissociation constant ($K_{\text{sub.D}}$) of less than or equal to 5×10^{-7} M, 10^{-7} M, 5×10^{-8} M, 10^{-9} M, 5×10^{-9} M, 10^{-9} M, 5×10^{-10} M, 10^{-11} M, 5×10^{-11} M, 10^{-12} M, 5×10^{-12} M, 10^{-12} M, 5×10^{-13} M, or 10^{-13} M. Preferably, the anti-CGRP antibodies and fragments thereof bind CGRP with a dissociation constant of less than or equal to 10^{-1} M,

5×10^{sup.}-2 M, or 10^{sup.}-12 M. In a specific embodiment of the invention the anti-CGRP antibody is Ab6 having a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM when measured using surface plasmon resonance (Misura, K et al, July 2019, Poster P220LB, AHS 61' annual scientific meeting). In another embodiment of the invention, the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to a linear or conformational CGRP epitope.

(121) In another embodiment of the invention, the anti-CGRP activity of the anti-CGRP antibodies of the present invention, and fragments thereof having binding specificity to CGRP, bind to CGRP with an off-rate of less than or equal to 10^{sup.}-4 S^{sup.}-1, 5×10^{sup.}-5 S^{sup.}-1, 10^{sup.}-5 S^{sup.}-1, 5×10^{sup.}-6 S^{sup.}-1, 10^{sup.}-6 S^{sup.}-1, 5×10^{sup.}-7 S^{sup.}-1, or 10^{sup.}-7 S^{sup.}-1. In a specific embodiment of the invention the anti-CGRP antibody is Ab6 having an off-rate of less than or equal to 5×10^{sup.}-6 S^{sup.}-1, such as less than or equal to 4×10^{sup.}-6 S^{sup.}-1, such as less than or equal to 3×10^{sup.}-6 S^{sup.}-1, such as less than or equal to 2×10^{sup.}-6 S^{sup.}-1, such as less than or equal to 1×10^{sup.}-6 S^{sup.}-1 when measured using surface plasmon resonance.

(122) Polynucleotides Encoding Anti-CGRP Antibody Polypeptides

(123) As aforementioned the invention specifically includes the use of specific anti-CGRP antibody or antibody fragment referred to herein as Ab6, which comprises or consists of the CDR, VL, VH, CL, and CH polypeptides having the sequences identified in FIGS. 1-12. The nucleic acid sequences encoding the foregoing VL, VH, CL, and CH polypeptides comprised in Ab6 are also comprised in FIGS. 1-12. The nucleic acid sequences which encode the CDR, VL, VH, CL, and CH polypeptides of an especially preferred anti-CGRP antibody, Ab6, are further described below.

(124) Polynucleotides Encoding Antibody Ab6

(125) The invention is further directed to polynucleotides encoding antibody polypeptides having binding specificity to CGRP. In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable light chain polypeptide sequence of SEQ ID NO: 222:

(126) TABLE-US-00009 (SEQ ID NO: 232)

CAAGTGCTGaccagtcctccatcctccctgtctgcatctgtaggagaca
gagtcaccatcAATtgcCAGGCCAGTCAGAGTGTTTATCATAACACCTA
CCTGGCCtggtatcagcagaaaccagggaaagttcctaagCAActgatc
tatGATGCATCCACTCTGGCATCTgggggtcccatctcgtttcagtggca
gtggatctgggacagattcactctcaccatcagcagcctgcagcctga
agatgttgcaactattactgtCTGGGCAGTTATGATTGTACTAATGGT
GATTGTTTTGTTttcggcgaggaggaaccaaggtggaaatcaaactgt.

(127) In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the light chain polypeptide sequence of SEQ ID NO: 221:

(128) TABLE-US-00010 (SEQ ID NO: 231)

CAAGTGCTGaccagtcctccatcctccctgtctgcatctgtaggagaca
gagtcaccatcAATtgcCAGGCCAGTCAGAGTGTTTATCATAACACCTA
CCTGGCCtggtatcagcagaaaccagggaaagttcctaagCAActgatc
tatGATGCATCCACTCTGGCATCTgggggtcccatctcgtttcagtggca
gtggatctgggacagattcactctcaccatcagcagcctgcagcctga
agatgttgcaactattactgtCTGGGCAGTTATGATTGTACTAATGGT
GATTGTTTTGTTttcggcgaggaggaaccaaggtggaaatcaaactgtACGG
TGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTTGAA
ATCTGGAAGTGCCTCTGTTGTGTGCTGCTGAATAACTTCTATCCCAGA
GAGGCCAAAGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAAGT
CCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGCCT

CAGCAGCCTGACGCTGAGCAGAAACACAAAGTC
TACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGA
GCTTCAACAGGGGAGAGTGTTAG.

(129) In another embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the variable heavy chain polypeptide sequence of SEQ ID NO: 202:

(130) TABLE-US-00011 (SEQ ID NO: 212)

gaggtgcagctTgtggagtctgggggaggcttggtccagcctgggggggt
ccctgagactctcctgtgcaGTCtctggaATCGACCTCagtGGCTACTA
CATGAACtgggtccgtcaggctccagggaaggggctggagtgggtcGGA
GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCC
gattcaccatctccagagacaattccaagACCACGGTGtatcttcaa
gaacagcctgagagctgaggacactgctgtgtatTTCtgtGCTAGAGGG
GACATCtggggccaagggaccctcgtcaccgtcTCGAGC.

(131) In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 201:

(132) TABLE-US-00012 (SEQ ID NO: 211)

gaggtgcagctTgtggagtctgggggaggcttggtccagcctgggggggt
ccctgagactctcctgtgcaGTCtctggaATCGACCTCagtGGCTACTA
CATGAACtgggtccgtcaggctccagggaaggggctggagtgggtcGGA
GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCC
gattcaccatctccagagacaattccaagACCACGGTGtatcttcaa
gaacagcctgagagctgaggacactgctgtgtatTTCtgtGCTAGAGGG
GACATCtggggccaagggaccctcgtcaccgtcTCGAGCGCCTCCACCA
AGGGCCCATCGGTCTTCCCCCTGGCAcCCTCCTCCaAGAGCACCTCTGG
GGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCG
GTGACGGTGTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGACACCT
TCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGT
GACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG
AATCACAAGCCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCAAAT
CTTGTGACAAAACCTCACACATGCCACCGTGCCACGACCTGAACTCCT
GGGGGGACCGTCAGTCTTCTCTTCCCCCCTAAAACCCAAGGACACCCTC
ATGaTCTCCCgGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCC
ACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGAGGT
GCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACGCCAGCACGTAC
CGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCA
AGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGA
GAAAACCATCTCCAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTAC
ACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGA
CCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGA
GAGCAATGGGCAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCTG
GACTCCGACGGCTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGA
GCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGC
TCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA TGA.

(133) In one embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, the following polynucleotide sequence encoding the heavy chain polypeptide sequence of SEQ ID NO: 566:

(134) TABLE-US-00013 (SEQ ID NO: 567)

gaggtgcagctTgtggagtctgggggaggcttggtccagcctgggggggt

ccctgagactctgtgcaGTctctggaATCGACCTCagtGGCTACTA
CATGAACtgggtccgtcaggctccaggggaaggggtggagtgggtcGGA
GTCATTGGTATTAATGGTGCCACATACTACGCGAGCTGGGCGAAAGGCc
gattcaccatctccagagacaattccaagACCACGGTGtatcttcaa
gaacagcctgagagctgaggacactgctgtgtatTTCtgtGCTAGAGGG
GACATCtggggccaagggaccctcgtcaccgtcTCGAGCGCCTCCACCA
AGGGCCCATCGGTCTTCCCCCTGGCAcCCTCCTCCaAGAGCACCTCTGG
GGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCG
GTGACGGTGTTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGACACCT
TCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGT
GACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG
AATCACAAGCCCAGCAACACCAAGGTGGACGCGAGAGTTGAGCCCCAAAT
CTTGTGACAAAACCTCACACATGCCCACCGTGCCAGCACCTGAACTCCT
GGGGGGACCGTCAGTCTTCTTCTTCCCCCCTAAAACCCAAGGACACCTC
ATGaTCTCCCgGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCC
ACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGAGGT
GCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACGCCAGCACGTAC
CGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCA
AGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGA
GAAAACCATCTCCAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTAC
ACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGA
CCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGA
GAGCAATGGGCAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCTG
GACTCCGACGGCTCCTTCTTCTTCTTCTACAGCAAGCTCACCGTGGACAAGA
GCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGC
TCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTTGA.

(135) In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221.

(136) In a further embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one or more of the polynucleotide sequences of SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218 which correspond to polynucleotides encoding the complementarity-determining regions (CDRs, or hypervariable regions) of the heavy chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

(137) The invention also contemplates polynucleotide sequences including one or more of the polynucleotide sequences encoding antibody fragments described herein. In one embodiment of the invention, polynucleotides encoding antibody fragments having binding specificity to CGRP comprise, or alternatively consist of, one, two, three or more, including all of the following polynucleotides encoding antibody fragments: the polynucleotide SEQ ID NO: 232 encoding the light chain variable sequence of SEQ ID NO: 222; the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221; the polynucleotide SEQ ID NO: 212 encoding the heavy chain variable sequence of SEQ ID NO: 202; the polynucleotide SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201; the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566; polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238) of the light chain variable sequence of SEQ ID NO: 222 or the light chain sequence of SEQ ID NO: 221; and

polynucleotides encoding the complementarity-determining regions (SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218) of the heavy chain variable sequence of SEQ ID NO: 202 or the heavy chain sequence of SEQ ID NO: 201 or SEQ ID NO: 566.

(138) In a preferred embodiment of the invention, polynucleotides of the invention comprise, or alternatively consist of, polynucleotides encoding Fab (fragment antigen binding) fragments having binding specificity for CGRP. With respect to antibody Ab6, the polynucleotides encoding the full length Ab6 antibody comprise, or alternatively consist of, the polynucleotide SEQ ID NO: 231 encoding the light chain sequence of SEQ ID NO: 221 and the polynucleotide SEQ ID NO: 211 encoding the heavy chain sequence of SEQ ID NO: 201 or the polynucleotide SEQ ID NO: 567 encoding the heavy chain sequence of SEQ ID NO: 566.

(139) Another embodiment of the invention contemplates these polynucleotides incorporated into an expression vector for expression in mammalian cells such as CHO, NSO, HEK-293, or in fungal, insect, or microbial systems such as yeast cells such as the yeast *Pichia*. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*. In one embodiment of the invention described herein (infra), Fab fragments may be produced by enzymatic digestion (e.g., papain) of Ab6 following expression of the full-length polynucleotides in a suitable host. In another embodiment of the invention, anti-CGRP antibodies such as Ab6 or Fab fragments thereof may be produced via expression of Ab6 polynucleotides in mammalian cells such as CHO, NSO or HEK 293 cells, fungal, insect, or microbial systems such as yeast cells (for example diploid yeast such as diploid *Pichia*) and other yeast strains. Suitable *Pichia* species include, but are not limited to, *Pichia pastoris*.

(140) Host cells and vectors comprising said polynucleotides are also contemplated.

(141) The invention further contemplates vectors comprising the polynucleotide sequences encoding the variable heavy and light chain polypeptide sequences, as well as the individual complementarity-determining regions (CDRs, or hypervariable regions), as set forth herein, as well as host cells comprising said vector sequences. In one embodiment of the invention, the host cell is a yeast cell. In another embodiment of the invention, the yeast host cell belongs to the genus *Pichia*.

(142) Methods of Producing Antibodies and Fragments Thereof

(143) In another embodiment, the present invention contemplates methods for producing anti-CGRP antibodies and fragments thereof. Methods for producing antibodies and fragments thereof secreted from polyploid, preferably diploid or tetraploid strains of mating competent yeast are taught, for example, in U.S. patent application publication no. US 2009/0022659 to Olson et al., and in U.S. Pat. No. 7,935,340 to Garcia-Martinez et al., the disclosures of each of which are herein incorporated by reference in their entireties. Methods for producing antibodies and fragments thereof in mammalian cells, e.g., CHO cells are further well known in the art.

(144) Other methods of producing antibodies are also well known to those of ordinary skill in the art. For example, methods of producing chimeric antibodies are now well known in the art (See, for example, U.S. Pat. No. 4,816,567 to Cabilly et al.; Morrison et al., *P.N.A.S. USA*, 81:8651-55 (1984); Neuberger, M. S. et al., *Nature*, 314:268-270 (1985); Boulianne, G. L. et al., *Nature*, 312:643-46 (1984), the disclosures of each of which are herein incorporated by reference in their entireties).

(145) Likewise, other methods of producing humanized antibodies are now well known in the art (See, for example, U.S. Pat. Nos. 5,530,101, 5,585,089, 5,693,762, and 6,180,370 to Queen et al; U.S. Pat. Nos. 5,225,539 and 6,548,640 to Winter; U.S. Pat. Nos. 6,054,297, 6,407,213 and 6,639,055 to Carter et al; U.S. Pat. No. 6,632,927 to Adair; Jones, P. T. et al, *Nature*, 321:522-525 (1986); Reichmann, L., et al, *Nature*, 332:323-327 (1988); Verhoeyen, M, et al, *Science*, 239:1534-36 (1988), the disclosures of each of which are herein incorporated by reference in their entireties).

(146) The present invention further includes the use of any of the pharmaceutical formulations disclosed herein in the manufacture of a medicament for the treatment, prevention and/or

amelioration of most bothersome symptom associated with migraine.

(147) Administration

(148) In one embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of between about 0.1 and 100.0 mg/kg of body weight of recipient subject. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject at a concentration of about 0.4 mg/kg of body weight of recipient subject and/or at a dosage of 100 or 300 mg. In a preferred embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a recipient subject with a frequency of once every twenty-six weeks or six months or less, such as once every sixteen weeks or four months or less, once every eight weeks or two months or less, once every four weeks or monthly or less, once every two weeks or bimonthly or less, once every week or less, or once daily or less. In general the administration of sequential doses may vary by plus or minus a few days from the aforementioned schedule, e.g., administration every 3 months or every 12 weeks includes administration of a dose varying from the schedule day by plus or minus 1, 2, 3, 4, 5, 5, or 7 days.

(149) Fab fragments may be administered every two weeks or less, every week or less, once daily or less, multiple times per day, and/or every few hours. In one embodiment of the invention, a patient receives Fab fragments of 0.1 mg/kg to 40 mg/kg per day given in divided doses of 1 to 6 times a day, or in a sustained release form, effective to obtain desired results.

(150) It is to be understood that the concentration of the antibody or Fab administered to a given patient may be greater or lower than the exemplary administration concentrations set forth above.

(151) A person of skill in the art would be able to determine an effective dosage and frequency of administration through routine experimentation, for example guided by the disclosure herein and the teachings in Goodman, L. S., Gilman, A., Brunton, L. L., Lazo, J. S., & Parker, K. L. (2006). Goodman & Gilman's the pharmacological basis of therapeutics. New York: McGraw-Hill; Howland, R. D., Mycek, M. J., Harvey, R. A., Champe, P. C., & Mycek, M. J. (2006). Pharmacology. Lippincott's illustrated reviews. Philadelphia: Lippincott Williams & Wilkins; and Golan, D. E. (2008). Principles of pharmacology: the pathophysiologic basis of drug therapy. Philadelphia, Pa., [etc.]: Lippincott Williams & Wilkins.

(152) In another embodiment of the invention, the anti-CGRP antibodies described herein, or CGRP binding fragments thereof, as well as combinations of said antibodies or antibody fragments, are administered to a subject in a pharmaceutical formulation.

(153) A "pharmaceutical composition" refers to a chemical or biological composition suitable for administration to a mammal. Such compositions may be specifically formulated for administration via one or more of a number of routes, including but not limited to buccal, epicutaneous, epidural, inhalation, intraarterial, intracardial, intracerebroventricular, intradermal, intramuscular, intranasal, intraocular, intraperitoneal, intraspinal, intrathecal, intravenous, oral, parenteral, rectally via an enema or suppository, subcutaneous, subdermal, sublingual, transdermal, and transmucosal, preferably intravenous. In addition, administration can occur by means of injection, powder, liquid, gel, drops, or other means of administration.

(154) A "pharmaceutical excipient" or a "pharmaceutically acceptable excipient" is a carrier, usually a liquid, in which an active therapeutic agent is formulated. In one embodiment of the invention, the active therapeutic agent is a humanized antibody described herein, or one or more fragments thereof. The excipient generally does not provide any pharmacological activity to the formulation, though it may provide chemical and/or biological stability, and release characteristics. Exemplary formulations can be found, for example, in Remington's Pharmaceutical Sciences, 19th Ed., Grennaro, A., Ed., 1995 which is incorporated by reference.

(155) As used herein “pharmaceutically acceptable carrier” or “excipient” includes any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents that are physiologically compatible. In one embodiment, the carrier is suitable for parenteral administration. Alternatively, the carrier can be suitable for intravenous, intraperitoneal, intramuscular, or sublingual administration. Pharmaceutically acceptable carriers include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The use of such media and agents for pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the pharmaceutical compositions of the invention is contemplated. Supplementary active compounds can also be incorporated into the compositions.

(156) Pharmaceutical compositions typically must be sterile and stable under the conditions of manufacture and storage. The invention contemplates that the pharmaceutical composition is present in lyophilized form. The composition can be formulated as a solution, microemulsion, liposome, or other ordered structure suitable to high drug concentration. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof. The invention further contemplates the inclusion of a stabilizer in the pharmaceutical composition. The proper fluidity can be maintained, for example, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants.

(157) In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, or sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent which delays absorption, for example, monostearate salts and gelatin. Moreover, the alkaline polypeptide can be formulated in a time release formulation, for example in a composition which includes a slow release polymer. The active compounds can be prepared with carriers that will protect the compound against rapid release, such as a controlled release formulation, including implants and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, polylactic acid and polylactic, polyglycolic copolymers (PLG). Many methods for the preparation of such formulations are known to those skilled in the art.

(158) An exemplary composition comprises, consists essentially of Ab6, an excipient such as histidine, an isotonic agent such as sorbitol, and a surfactant such as polysorbate 80 in an aqueous solution. For example, the composition may comprise, consist essentially of, or consist of histidine (L-histidine), sorbitol, polysorbate 80, such as, per 1 mL volume, about 100 mg Ab6, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, having a pH of about 5.8, or approximately that constitution, e.g., within 10% of those values, within 5% of those values, within 1% of those values, within 0.5% of those values, or within 0.1% of those values, and water. For example, the pH value may be within 10% of 5.8, i.e., between 5.22 and 6.38. The Ab6 antibody may comprise or consist of the variable light and heavy chain polypeptides of SEQ ID NO: 222 and SEQ ID NO: 202 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, or the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively. The composition may be in the form of an aqueous solution, or a concentrate (e.g., lyophilized) which when reconstituted, e.g., by addition of water, yields the aforementioned constitution. An exemplary composition consists of, per mL, 100 mg of the light and heavy chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 201 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q.S, or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of those quantities, within 0.5% of those quantities, or within 0.10% of those quantities. Another exemplary composition consists of, per mL, 100 mg of the light and heavy

chain polypeptides of SEQ ID NO: 221 and SEQ ID NO: 566 respectively, about 3.1 mg L-Histidine, about 40.5 mg Sorbitol, and about 0.15 mg Polysorbate 80, and water Q.S, or approximately that constitution, e.g., within 10% of those quantities, within 5% of those quantities, within 1% of those quantities, within 0.5% of those quantities, or within 0.10% of those quantities. The composition may be suitable for intravenous or subcutaneous administration, preferably intravenous administration. For example, the composition may be suitable for mixing with an intravenous solution (such as 0.9% sodium chloride) at an amount of between about 100 mg and about 300 mg antibody added to 100 mL of intravenous solution. Preferably the composition may be shelf-stable for at least 1, 3, 6, 12, 18, or 24 months, e.g., showing formation of aggregates of no more than 5% or no more than 10% of the antibody or fragment after storage at room temperature or when refrigerated at 4° C. for the specified duration, or in an accelerated aging test that simulates storage for that duration.

(159) For each of the recited embodiments, the compounds can be administered by a variety of dosage forms. Any biologically-acceptable dosage form known to persons of ordinary skill in the art, and combinations thereof, are contemplated. Examples of such dosage forms include, without limitation, reconstitutable powders, elixirs, liquids, solutions, suspensions, emulsions, powders, granules, particles, microparticles, dispersible granules, cachets, inhalants, aerosol inhalants, patches, particle inhalants, implants, depot implants, injectables (including subcutaneous, intramuscular, intravenous, and intradermal, preferably intravenous), infusions, and combinations thereof.

(160) The above description of various illustrated embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. The teachings provided herein of the invention can be applied to other purposes, other than the examples described above.

(161) These and other changes can be made to the invention in light of the above detailed description. In general, in the following claims, the terms used should not be construed to limit the invention to the specific embodiments disclosed in the specification and the claims. Accordingly, the invention is not limited by the disclosure, but instead the scope of the invention is to be determined entirely by the following claims.

(162) The invention may be practiced in ways other than those particularly described in the foregoing description and examples. Numerous modifications and variations of the invention are possible in light of the above teachings and, therefore, are within the scope of the appended claims.

(163) Certain CGRP antibody polynucleotides and polypeptides are disclosed in the sequence listing accompanying this patent application filing, and the disclosure of said sequence listing is herein incorporated by reference in its entirety.

(164) The entire disclosure of each document cited (including patents, patent applications, journal articles, abstracts, manuals, books, or other disclosures) in the Background of the Invention, Detailed Description, and Examples is herein incorporated by reference in their entireties.

(165) The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the subject invention, and are not intended to limit the scope of what is regarded as the invention. Efforts have been made to ensure accuracy with respect to the numbers used (e.g. amounts, temperature, concentrations, etc.) but some experimental errors and deviations should be allowed for. Unless otherwise indicated, parts are parts by weight, molecular weight is average molecular weight, temperature is in degrees centigrade; and pressure is at or near atmospheric.

Additional Exemplary Embodiments

(166) Additional exemplary embodiments of the invention are provided as follows:

(167) S1. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most

bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody.

(168) S2. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said migraine patient suffers from chronic migraine.

(169) S3. Use of an anti-CGRP antibody for the manufacturing of a medicament for treating most bothersome symptom (MBS) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said patient suffers from episodic migraine.

(170) S4. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved at 1-12 hours post-completion of administration or infusion, such as 1-5 hours post-completion of administration or infusion, 1-2 hours post-completion of administration or infusion, or about 2 hours post-completion of administration or infusion.

(171) S5. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 1 month from the first dosing with said anti-CGRP antibody.

(172) S6. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 3 month from the first dosing with said anti-CGRP antibody.

(173) S7. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 6 month from the first dosing with said anti-CGRP antibody.

(174) S8. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

(175) S9. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

(176) S10. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty.

(177) S11. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes and Sensitivity to smell (osmophobia or olfactophobia).

(178) S12. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody.

(179) S13. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said migraine patient suffers from chronic migraine.

(180) S14. Use of an anti-CGRP antibody for the manufacturing of a medicament for improving patient global impression of change (PGIC) associated with migraine, comprising administering to a migraine patient an anti-CGRP antibody, wherein said patient suffers from episodic migraine.

(181) S15. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine within 1 month from the first dosing with said anti-CGRP antibody.

(182) S16. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine within 3 month from the first dosing with said anti-CGRP antibody.

(183) S17. Use of the anti-CGRP antibody of any one of embodiments S12-S14, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine within 6 month from the first dosing with said anti-CGRP antibody.

(184) S18. Use of the anti-CGRP antibody of any one of embodiments S12-S17, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine, and wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

(185) S19. Use of the anti-CGRP antibody of any one of embodiments S12-S18, wherein the administration of said medicament improves patient global impression of change (PGIC) associated with migraine, and wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

(186) S20. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous or subcutaneous infusion.

(187) S21. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous infusion.

(188) S22. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of administration or infusion.

(189) S23. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises Ab6.

(190) S24. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively.

(191) S25. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively.

(192) S26. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

(193) S27. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

(194) S28. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

(195) S29. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

(196) S30. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

(197) S31. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

(198) S32. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

(199) S33. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

(200) S34. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

(201) S35. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

(202) S36. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

(203) S37. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

(204) S38. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

(205) S39. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(206) S40. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

(207) S41. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(208) S42. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is between about 100 mg and about 300 mg, or is about 100 mg, or is about 300 mg.

(209) S43. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is 100 mg.

(210) S44. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is for intravenous administration in a dosage of 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

(211) S45. Use of the anti-CGRP antibody of any one of embodiments S1-S42, wherein said medicament is for intravenous administration in a dosage of 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.

(212) S46. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 1-10 migraine attacks per month in the month or in the 3 months prior to administration.

(213) S47. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 2-8 migraine attacks per month in the month or in the 3 months prior to administration.

(214) S48. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits between 3-7 migraine attacks per month in the month or in the 3 months prior to administration.

(215) S49. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 25 headache days per month in the month or in the 3 months prior to administration.

(216) S50. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 20 headache days per month in the month or in the 3 months prior to administration.

(217) S51. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 15 headache days per month in the month or in the 3 months prior to administration.

(218) S52. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein, prior to administration of said medicament, the patient exhibits less than 10 headache days per month in the month or in the 3 months prior to administration.

(219) S53. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 10 years prior to administration of said medicament.

(220) S54. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 15 years prior to administration of said medicament.

(221) S55. Use of the anti-CGRP antibody of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 18 or at least 19 years prior to administration of said medicament.

(222) S56. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 50% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(223) S57. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(224) S58. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(225) S59. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(226) S60. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(227) S61. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said patient further has a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to administration of said medicament.

(228) S62. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament is further for administration in a second dose of said anti-CGRP antibody about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months after administration of said medicament.

(229) S63. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said medicament comprises about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

(230) S64. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody is aglycosylated or if glycosylated only contains only mannose residues.

(231) S65. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

(232) S66. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said

anti-CGRP antibody consists of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(233) S67. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said headache or said migraine is diagnosed according to the third edition of the International Classification of Headache Disorders.

(234) S68. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

(235) S69. Use of the anti-CGRP antibody of any of any one of embodiments S1-S67, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

(236) S70. Use of the anti-CGRP antibody of any one of the foregoing embodiments, wherein said anti-CGRP antibody or anti-CGRP antibody fragment is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

(237) S71. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

(238) S72. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

(239) S73. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

(240) S74. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

(241) S75. Use of the anti-CGRP antibody of embodiment S70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

(242) S76. Use of the anti-CGRP antibody of any of any one of the foregoing embodiments, wherein the anti-CGRP antibody has a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM.

Further Exemplary Embodiments

(243) Further exemplary embodiments of the invention are provided as follows:

(244) E1. An anti-CGRP antibody for use in treating most bothersome symptom (MBS) associated with migraine in a patient suffering from migraine.

(245) E2. The anti-CGRP antibody for use of embodiment E1, wherein the patient suffers from chronic migraine.

(246) E3. The anti-CGRP antibody for use of embodiment E1, wherein the patient suffers from episodic migraine.

(247) E4. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved at 1-12 hours post-completion of administration or infusion, such as 1-5 hours post-completion of administration or infusion, 1-2 hours post-completion of administration or infusion, or about 2 hours post-completion of administration or infusion.

(248) E5. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved

within 1 month from the first dosing with said anti-CGRP antibody.

(249) E6. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 3 month from the first dosing with said anti-CGRP antibody.

(250) E7. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the patients most bothersome symptom (MBS) associated with migraine is improved within 6 month from the first dosing with said anti-CGRP antibody.

(251) E8. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the improvement is sustained for at least 3 months from the first dosing with said anti-CGRP antibody.

(252) E9. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the improvement is sustained for at least 6 months from the first dosing with said anti-CGRP antibody.

(253) E10. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes, Sensitivity to smell (osmophobia or olfactophobia), Visual impact, Pressure/tightness, Pain (anatomical), Eye pain, Neck pain, Dizziness, Allodynia, Inactivity, Sensory disturbance, Sleep disturbance and Speech difficulty.

(254) E11. The anti-CGRP antibody for use according to any of the foregoing embodiments, wherein the MBS is selected from the group consisting of: Sensitivity to light (photophobia), Nausea/vomiting, Headache, Sensitivity to sound (phonophobia), Aura, Pain with activity, Pain, Throbbing/pulsation, Cognitive disruption, Fatigue, Mood changes and Sensitivity to smell (osmophobia or olfactophobia).

(255) E12. An anti-CGRP antibody for use in improving patient global impression of change (PGIC) associated with migraine in a patient suffering from migraine.

(256) E13. The anti-CGRP antibody for use of embodiment E12, wherein the patient suffers from chronic migraine.

(257) E14. The anti-CGRP antibody for use of embodiment E12, wherein the patient suffers from episodic migraine.

(258) E15. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 1 month from the first dosing with said anti-CGRP antibody.

(259) E16. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 3 month from the first dosing with said anti-CGRP antibody.

(260) E17. The anti-CGRP antibody for use according to any of embodiments E12-E14, wherein the improvement of patient global impression of change (PGIC) associated with migraine is observed within 6 month from the first dosing with said anti-CGRP antibody.

(261) E18. The anti-CGRP antibody for use according to any of embodiments E12-E17, wherein the improvement of patient global impression of change (PGIC) associated with migraine is sustained for 3 months from the first dosing with said anti-CGRP antibody.

(262) E19. The anti-CGRP antibody for use according to any of embodiments E12-E18, wherein the improvement of patient global impression of change (PGIC) associated with migraine is sustained for 6 months from the first dosing with said anti-CGRP antibody.

(263) E20. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous or subcutaneous infusion.

(264) E21. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous infusion.

(265) E22. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient is headache free 2 hours post-completion of administration or infusion.

(266) E23. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises Ab6.

(267) E24. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively.

(268) E25. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively.

(269) E26. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

(270) E27. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

(271) E28. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224; SEQ ID NO: 226; and SEQ ID NO: 228, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 204; SEQ ID NO: 206; and SEQ ID NO: 208, respectively.

(272) E29. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 234; SEQ ID NO: 236; and SEQ ID NO: 238, respectively and heavy chain CDR 1, 2, and 3 polypeptide sequences encoded by SEQ ID NO: 214; SEQ ID NO: 216; and SEQ ID NO: 218, respectively.

(273) E30. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222.

(274) E31. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232.

(275) E32. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide of SEQ ID NO: 202.

(276) E33. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

(277) E34. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide of SEQ ID NO: 222 and the variable heavy chain polypeptide of SEQ ID NO: 202.

(278) E35. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the variable light chain polypeptide encoded by SEQ ID NO: 232 and the variable heavy chain polypeptide encoded by SEQ ID NO: 212.

(279) E36. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221.

(280) E37. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231.

(281) E38. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

(282) E39. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

- (283) E40. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.
- (284) E41. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody comprises the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.
- (285) E42. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is between about 100 mg and about 300 mg, or is about 100 mg, or is about 300 mg.
- (286) E43. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the administered amount of said anti-CGRP antibody is 100 mg.
- (287) E44. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is for intravenous administration in a dosage of 100 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.
- (288) E45. The anti-CGRP antibody for use of any one of embodiments E1-E42, wherein said anti-CGRP antibody is for intravenous administration in a dosage of administering 300 mg of said anti-CGRP antibody every 10-14 weeks, preferably every 11-13 weeks, more preferably every 12 weeks.
- (289) E46. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 1-10 migraine attacks per month in the month or in the 3 months prior to administration.
- (290) E47. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 2-8 migraine attacks per month in the month or in the 3 months prior to administration.
- (291) E48. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits between 3-7 migraine attacks per month in the month or in the 3 months prior to administration.
- (292) E49. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 25 headache days per month in the month or in the 3 months prior to administration.
- (293) E50. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 20 headache days per month in the month or in the 3 months prior to administration.
- (294) E51. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 15 headache days per month in the month or in the 3 months prior to administration.
- (295) E52. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein, prior to administration of said anti-CGRP antibody, the patient exhibits less than 10 headache days per month in the month or in the 3 months prior to administration.
- (296) E53. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 10 years prior to the administration of said anti-CGRP antibody.
- (297) E54. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 15 years prior to the administration of said anti-CGRP antibody.
- (298) E55. The anti-CGRP antibody for use of any one of the foregoing embodiments wherein said patient was diagnosed with migraine at least 18 or at least 19 years prior to the administration of said anti-CGRP antibody.
- (299) E56. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 50% in the one month period

after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(300) E57. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 75% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(301) E58. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by 100% in the one month period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(302) E59. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 50% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(303) E60. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by at least 75% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(304) E61. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said patient has a reduction in the number of migraine days by 100% in the 12 week period after being administered said antibody relative to the baseline number of migraine days experienced by that patient prior to the administration of said anti-CGRP antibody.

(305) E62. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said use comprises administering a second dose of said anti-CGRP antibody to said patient about 10-14 weeks, preferably 11-13 weeks, more preferably about 12 weeks or about 3 months after the administration of said anti-CGRP antibody.

(306) E63. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said use comprises administering about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg of said anti-CGRP antibody.

(307) E64. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody is aglycosylated or if glycosylated only contains only mannose residues.

(308) E65. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide of SEQ ID NO: 221 and the heavy chain polypeptide of SEQ ID NO: 201 or SEQ ID NO: 566.

(309) E66. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody consists of the light chain polypeptide encoded by SEQ ID NO: 231 and the heavy chain polypeptide encoded by SEQ ID NO: 211 or SEQ ID NO: 567.

(310) E67. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said headache or said migraine is diagnosed according to the third edition of the International Classification of Headache Disorders.

(311) E68. The anti-CGRP antibody for use of any of any one of the foregoing embodiments, wherein said anti-CGRP antibody is expressed in or obtained by expression in *Pichia pastoris*.

(312) E69. The anti-CGRP antibody for use of any of any one of embodiments E1-E67, wherein said anti-CGRP antibody is expressed in or obtained by expression in CHO cells.

(313) E70. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein said anti-CGRP antibody or anti-CGRP antibody fragment is comprised in a formulation comprising or consisting of histidine (L-histidine), sorbitol, polysorbate 80, and water.

(314) E71. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 10\%$ of

said values, and having a pH of 5.8 or within $\pm 10\%$ of said value.

(315) E72. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 5\%$ of said values, and/or having a pH of 5.8 or within $\pm 5\%$ of said value.

(316) E73. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 1\%$ of said values, and/or having a pH of 5.8 or within $\pm 1\%$ of said value.

(317) E74. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.5\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.5\%$ of said value.

(318) E75. The anti-CGRP antibody for use of embodiment E70, wherein said formulation comprises or consists of, per 1 mL volume, 100 mg anti-CGRP antibody, 3.1 mg L-Histidine, 40.5 mg Sorbitol, and 0.15 mg Polysorbate 80, or having amounts of each constituent within $\pm 0.1\%$ of said values, and/or having a pH of 5.8 or within $\pm 0.1\%$ of said value.

(319) E76. The anti-CGRP antibody for use of any one of the foregoing embodiments, wherein the anti-CGRP antibody has a dissociation constant of less than or equal to 10 pM, such as 2-8 pM, such as 3-6 pM, such as less than or equal to about 5 pM.

EXAMPLES

(320) The following examples are provided in order to illustrate the invention, but are not to be construed as limiting the scope of the claims in any way.

Example 1

(321) Preparation of Antibodies that Bind CGRP

(322) The preparation of exemplary anti-CGRP antibody Ab6 having the sequences in FIGS. 1-12 is disclosed in commonly owned PCT Application WO/2012/162243, published on Nov. 29, 2012, the contents of which are incorporated by reference herein. This application exemplifies synthesis of these antibodies in *Pichia pastoris* cells. The present Applicant further contemplates synthesis of anti-CGRP antibody Ab6 particularly in CHO cells.

Example 2

(323) Human Clinical Study Evaluating the Safety and Efficacy of an Anti-CGRP Antibody in Chronic Migraine Patients

(324) This example describes a randomized, double-blind, placebo-controlled clinical trial evaluating the safety and efficacy of Ab6 for chronic migraine prevention. In the study, 1,072 patients were randomized to receive Ab6 (300 mg or 100 mg), or placebo administered by infusion once every 12 weeks. The study design is depicted in FIG. 13. To be eligible for the trial, patients must have experienced at least 15 headache days per month, of which at least eight met criteria for migraine. Patients that participated in the trial had an average of 16.1 migraine days per month at baseline. The primary endpoint of the present study was the change from baseline in mean monthly migraine days (MMDs) over weeks 1-12 following the first infusion of Ab6. The change from baseline in mean monthly migraine days (MMDs) following the second infusion at week 12 was also assessed and the results are shown in FIG. 14.

(325) Study endpoints further included patient-identified MBS as part of the predefined key secondary endpoints. At screening, patients verbally identified the MBS associated with their migraine, which was pooled across treatment arms for this analysis. The change from baseline of these symptoms were then rated by the patient every month of the study beginning from Day 0.

(326) In the present study, patients verbally identified the most bothersome symptom (MBS) associated with their migraine at screening. The MBS associated with their migraine was then categorized by the investigator into a predefined list of 8 symptoms or an "other" option. The

predefined list included the terms nausea, vomiting, sensitivity to light, sensitivity to sound, mental cloudiness, fatigue, pain with activity, and mood changes. The “other” option provided investigators the opportunity to identify any migraine-associated symptom without limitation described by the patient as most bothersome but did not easily fit into the check list of symptoms included in the work study checklist. For those patients who selected the “other” category for their MBS, their “write-in” responses were re-coded post hoc and re-classified to the predefined list or to new symptom classes. At subsequent visits, patients were asked to rate the change from the screening visit in their self-reported MBS on a 7-point scale, which is shown below:

(327) TABLE-US-00014 Very Much Much Minimally No Minimally Much Very Improved
Improved Improved Change Worse Worse Much Worse

(328) In addition to MBS, the patients were also requested to evaluate the efficacy of the treatment on patient global impression of change (PGIC), which is a parameter comprising a single question assessing the patient's own impression of the overall change in their disease status since the start of the study. This parameter was also rated by the patients at a 7-point scale identical to the one used to assess change in MBS as displayed above and at the same time points in the study. In FIGS. 16-22 the “worse” category includes “minimally worse”, “much worse”, and “very much worse”.

(329) At the screening visit in, patients indicated a wide range of symptoms as their MBS, with the “other” category being the most frequent response (40%-42% across the 3 treatment groups). The patients who selected the “other” category generally provided more details and/or had more than 1 symptom as their MBS, allowing for these symptoms to be recoded. The overall list of MBS is summarized in Table 1 below.

(330) TABLE-US-00015 TABLE 1 Summary of patient-identified MBS in the present study as described in Example 2

	Eptinezumab 100 mg	Eptinezumab 300 mg	Placebo	Total Symptom, n (%)
ICHD-3 Symptoms				
Sensitivity to light	67 (18.8)	64 (18.3)	69 (18.9)	200 (18.7)
Nausea/vomiting	55 (15.4)	46 (13.1)	61 (16.7)	162 (15.1)
Headache	45 (12.6)	43 (12.3)	32 (8.7)	120 (11.2)
Sensitivity to sound	22 (6.2)	28 (8.0)	28 (7.7)	78 (7.3)
Aura	4 (1.1)	1 (<1)	2 (<1)	7 (0.7)
Additional Symptoms				
Pain with activity	53 (14.9)	45 (12.9)	49 (13.4)	147 (13.7)
Pain	35 (9.8)	45 (12.9)	53 (14.5)	133 (12.4)
Throbbing/pulsation	18 (5.1)	17 (4.9)	15 (4.1)	50 (4.7)
Cognitive disruption	17 (4.8)	14 (4.0)	13 (3.6)	44 (4.1)
Fatigue	7 (2.0)	11 (3.1)	8 (2.2)	26 (2.4)
Mood changes	8 (2.2)	4 (1.1)	4 (1.1)	16 (1.5)
Sensitivity to smell	1 (<1)	1 (<1)	8 (2.2)	10 (0.9)
Visual impact	2 (<1)	3 (<1)	3 (<1)	8 (0.7)
Pressure/tightness	2 (<1)	2 (<1)	3 (<1)	7 (0.7)
Pain, anatomical	3 (<1)	3 (<1)	0	6 (0.6)
Eye pain	4 (1.1)	1 (<1)	1 (<1)	6 (0.6)
Neck pain	1 (<1)	1 (<1)	3 (<1)	5 (0.5)
Dizziness	2 (<1)	2 (<1)	1 (<1)	5 (0.5)
Allodynia	1 (<1)	1 (<1)	1 (<1)	3 (0.3)
Inactivity	0	1 (<1)	1 (<1)	2 (0.2)
Sensory disturbance	1 (<1)	0	0	1 (0.1)
Sleep disturbance	0	0	1 (<1)	1 (0.1)
Speech difficulty	0	0	1 (<1)	1 (0.1)
Multiple*	7 (2.0)	12 (3.4)	8 (2.2)	27 (2.5)
Other	1 (<1)	5 (1.4)	1 (<1)	7 (0.7)

*Patient's most bothersome symptom included more than 1 symptom type. ICHD-3 = International Classification of Headache Disorders 3rd edition.

(331) The most commonly reported symptoms were light sensitivity, nausea/vomiting, pain with activity, pain, headache, sound sensitivity, throbbing/pulsation, cognitive disruption, fatigue, mood changes, and sensitivity to smell, with each category having at least 10 patients reporting these events as their MBS. At the end of the 28-day screening period (i.e, before dosing at the baseline visit), patients were asked to rate the change in their identified MBS from very much worse to very much improved, with >90% reporting no change in their MBS, which is illustrated in FIG. 16. This suggests that the bothersomeness of patient-identified MBS was quite stable among this cohort with chronic migraine during the screening period.

(332) Infusion of Ab6 in doses of 100 mg and 300 mg provided significantly reduction in mean MMDs across months 1-3 of the study, with further reduction after an additional infusion at week 12 of the study. This effect is shown in FIG. 14.

(333) The efficacy of Ab6 on the MBS was demonstrated at 1 month (FIG. 17), 3 months (FIG. 19), and 6 months (FIG. 21), following the first infusion of Ab6 in doses of 100 mg and 300 mg.

The efficacy of Ab6 on the PGIC was demonstrated at 1 month (FIG. 18), 3 months (FIG. 20), and 6 months (FIG. 22), following the first infusion of Ab6 in doses of 100 mg and 300 mg. The efficacy on these parameters were sustained or increased through 2 doses of Ab6 over 6 months; at Month 1, 75-82% of Ab6-treated patients indicated some level of improvement compared to 56-59% for the placebo-treated patients; at Month 3 ratings of improvement were similar to those of month 1; at Month 6, ~80% of Ab6-treated patients indicated ≥ 1 categorical level of improvement in MBS and PGIC. The distribution of ratings for MBS improvement and PGIC were similar across time points, suggesting that the 2 identically rated measures in patients with chronic migraine move in parallel. These data suggest that improvements in patient-identified most bothersome migraine-associated symptoms are highly correlated with the patient's perception of an improved disease status in patients with chronic migraine.

(334) The administered antibody, Ab6, is an anti-CGRP antibody consisting of the light chain polypeptide of SEQ ID NO: 221 and heavy chain polypeptide of SEQ ID NO: 201.

(335) Patient characteristics are summarized in Table 2 below, with separate columns for patients receiving placebo, 100 mg of the antibody, or 300 mg of the antibody. Patients had a mean number of years from migraine diagnosis of between 17.0 and 19.0 years, a mean duration of suffering from chronic migraine of between 11.5 and 12.4 years, and between 44.3% and 45.2% of patients utilized at least one prophylactic medication. In addition, patients with a dual diagnosis of chronic migraine and medications overuse excluding opioid and butalbital over were included in this study. At baseline, in both antibody treatment groups the mean number of migraine days per month was 16.1, while for the placebo group, the mean number of migraine days per month was 16.2.

(336) TABLE-US-00016 TABLE 2 Summarizes the characteristics of patients in each treatment group in the clinical trials described in Example 2.

	Placebo	Eptinezumab 100 mg	Eptinezumab 300 mg
n	356	350	366
Age (years), mean (SD)	41.0 (11.72)	41.0 (10.36)	39.6 (11.28)
Sex, n (%)			
Male	49 (13.8%)	36 (10.3%)	41 (11.2%)
Female	307 (86.2%)	314 (89.7%)	325 (88.8%)
Race, n (%)			
White	332 (93.3%)	322 (92.0%)	321 (87.7%)
Black or African American	21 (5.9%)	23 (6.6%)	38 (10.4%)
American Indian or Alaska Native	3 (0.8%)	5 (1.4%)	7 (1.9%)
Other*	3 (0.8%)	5 (1.4%)	7 (1.9%)
BMI (kg/m ²), mean (SD)	26.4 (4.98)	26.3 (7.14)	27.0 (5.56)
Age at migraine diagnosis (years), mean (SD)	22.8 (10.64)	22.0 (9.30)	22.6 (9.98)
Duration of migraine diagnosis (years), mean (SD)	18.3 (12.22)	19.0 (11.50)	17.0 (11.63)
Duration of chronic migraine (years), mean (SD)	11.6 (11.72)	12.3 (11.15)	11.6 (10.90)
Number of migraine days, mean (SD)	16.1 (4.61)	16.1 (4.77)	16.2 (4.55)
Medication-overuse [†] , n (%)	139 (39.0%)	147 (42.0%)	145 (39.6%)

headache diagnosis, n (%).^{sup.‡} BMI, body mass index; SD, standard deviation, *Other includes Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, multiple races, and other. ^{sup.†} As reported by the eDiary in the 28-day screening period. ^{sup.§} Based on 3rd edition of the International Classification of Headache Disorders (beta).

Claims

1. A method of individual therapy in a patient suffering from migraine, comprising: (a) identifying a most bothersome symptom (MBS) of the patient, wherein the MBS is a symptom which is associated with the migraine of the patient and is most bothersome to the patient; (b) intravenously administering to the patient an effective amount of an anti-calcitonin gene related peptide (CGRP) antibody; (c) assessing changes in the MBS after the administration of (a); and (d) if the MBS is improved in the assessment of (c), further intravenously administering to the patient an effective amount of the anti-CGRP antibody, wherein the anti-CGRP antibody comprises: (A) a heavy chain variable domain (VH) comprising heavy chain complementarity-determining region (CDR) 1, 2, and 3 polypeptide sequences of SEQ ID NOS: 204, 206, and 208, respectively; and (B) a light chain variable domain (VL) comprising light chain CDR 1, 2, and 3 polypeptide sequences of SEQ ID NO: 224, 226, and 228, respectively.

2. The method of claim 1, wherein the assessing in (c) is performed at 1-12 hours, within 1 month, within 3 months, or within 6 months from the administering in (a).
 3. The method of claim 1, wherein: (A) the amino acid sequence of the VH comprises SEQ ID NO: 202; and/or (B) the amino acid sequence of the VL comprises SEQ ID NO: 222.
 4. The method of claim 1, wherein the anti-CGRP comprises: (A) a heavy chain polypeptide comprising SEQ ID NO: 201 or SEQ ID NO: 566; and/or (B) a light chain polypeptide comprising SEQ ID NO: 221.
 5. The method of claim 1, wherein the MBS identified in (a) is not headache and not pain.
 6. The method of claim 1, wherein the MBS identified in (a) is selected from the group consisting of: sensitivity to light (photophobia); nausea and/or vomiting; sensitivity to sound (phonophobia); aura; throbbing and/or pulsation; cognitive disruption; fatigue; mood changes; sensitivity to smell (osmophobia or olfactophobia); visual impact; pressure and/or tightness; dizziness; inactivity; sensory disturbance; sleep disturbance; and speech difficulty.
 7. The method of claim 1, wherein the MBS identified in (a) is selected from the group consisting of: vomiting; throbbing and/or pulsation; cognitive disruption; fatigue; mood changes; sensitivity to smell (osmophobia or olfactophobia); visual impact; pressure and/or tightness; dizziness; inactivity; sensory disturbance; sleep disturbance; and speech difficulty.
 8. The method of claim 1, wherein the effective amount in (b) and/or (d) is between about 100 mg and about 300 mg, optionally about 100 mg, about 125 mg, about 150 mg, about 175 mg, about 200 mg, about 225 mg, about 250 mg, about 275 mg, or about 300 mg.
 9. The method of claim 1, wherein the administering in (d) is performed about 10-14 weeks, optionally 11-13 weeks, further optionally about 12 weeks or about 3 months, after the administering in (b).
 10. The method of claim 1, wherein the administering in (d) is performed every 10-14 weeks, optionally every 11-13 weeks, further optionally every 12 weeks.
 11. The method of claim 1, wherein the patient has chronic migraine when the administering in (b) is performed.
 12. The method of claim 1, wherein the patient has episodic migraine when the administering in (b) is performed.
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